

**The Spectral Theorem for unbounded
self-adjoint operators**
Semester Project

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Abstract

In this project, we generalize the functional calculus for bounded self-adjoint operators, which is obtained by the mean of the Weierstrass Approximation Theorem, to unbounded operators and then prove the so-called Spectral Theorem. This generalization is achieved with abstract integration and tools from complex analysis. We then provide applications of the theorem to two other seemingly unrelated problems, namely the Hamburger Moment Problem and the characterization of operator monotone functions.

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Notation

1. $\mathbb{K}$ denotes $\mathbb{R}$ or $\mathbb{C}$. If not specified otherwise, the vector spaces we consider are always over $\mathbb{K}$.
2. $\mathbb{H}$ denotes the upper half-complex plane, namely $\mathbb{H} = \{z \in \mathbb{C} \mid \Im z > 0\}$.
3. Let V_1, V_2 be two vector spaces over $\mathbb{K}$. Let $T : D(T) \subset V_1 \rightarrow V_2$, $D(T) \subset V_1$ a subspace, be a linear map. We write
 - (a) Kernel of T : $N(T) = \{x \in D(T) \mid Tx = 0\} \subset D(T) \subset V_1$
 - (b) Image of T : $R(T) = \{y \in V_2 \mid \exists x \in D(T) : y = Tx\} \subset V_2$
 - (c) Graph of T : $G(T) = \{(x, y) \in V_1 \times V_2 \mid x \in D(T), y = Tx\} \subset V_1 \times V_2$
 Since T is linear and $D(T)$ is a subspace, $N(T)$, $R(T)$, $G(T)$ are subspaces as well. If not specified otherwise, $D(T)$ will always be a subspace.
4. Let V_1, V_2, V_3 , be three vector spaces over $\mathbb{K}$.
 - (a) Let $T : D(T) \subset V_1 \rightarrow V_2$ be linear and $a \in \mathbb{K} \setminus \{0\}$. We define the linear operator $aT : D(T) \rightarrow V_2$ by $(aT)(f) = aT(f)$.
 - (b) Let $T : D(T) \subset V_1 \rightarrow V_2$, $S : D(S) \subset V_1 \rightarrow V_2$ be linear. We define the linear operator $T+S : D(T+S) := D(T) \cap D(S) \rightarrow V_2$ by $(T+S)(f) = T(f) + S(f)$.
 - (c) Let $T : D(T) \subset V_1 \rightarrow V_2$, $S : D(S) \subset V_2 \rightarrow V_3$ be linear. We define the linear operator $ST : D(ST) := \{f \in D(T) \mid Tf \in D(S)\} \rightarrow V_3$ by $ST(f) = S \circ T(f)$.
5. Let $f : \mathbb{R} \rightarrow \mathbb{C}$ be any function. For $x_0 \in \mathbb{R}$, we write $f(x_0-)$ the limit $\lim_{x \rightarrow x_0^-} f(x)$ and $f(x_0+)$ the limit $\lim_{x \rightarrow x_0^+} f(x)$ (if they exist).
6. Let X be a topological space. We denote by
 - (a) $C^0(X)$ the set of continuous functions from X to $\mathbb{R}$.
 - (b) $C_b(X)$ the set of continuous and bounded functions from X to $\mathbb{R}$.
 - (c) $C_c(X)$ the set of continuous and compactly supported functions from X to $\mathbb{R}$.
 - (d) $C_0(X)$ the closure of $C_c(X)$ in the normed space $(C_b(X), \|\cdot\|_\infty)$.

Chapter 1

Generalities about bounded linear operators

We begin by recalling some basic definitions and results about linear operators between normed spaces.

Definition 1.1 (Bounded linear operator)

Let $(X, \|\cdot\|_X)$, $(Y, \|\cdot\|_Y)$ be two normed spaces over $\mathbb{K}$.

Let $D(T)$ be a subspace of X . A linear operator $T : D(T) \subset X \rightarrow Y$ is called *bounded* if

$$\exists M > 0 \forall x \in D(T) : \|Tx\|_Y \leq M\|x\|_X$$

The smallest such M is called the operator norm of T and is denoted $\|T\|$.

We denote by $B(X, Y)$ the set of all bounded linear operators from X to Y .

Proposition 1.1 (Characterization of bounded linear operators)

Let $X, Y, D(T)$ be as in the above definition. Let $T : D(T) \subset X \rightarrow Y$ be a linear operator.

The following are equivalent :

1. T is bounded,
2. $\sup\{\|Tx\|_Y \mid x \in D(T), \|x\|_X \leq 1\} < +\infty$
3. T is continuous at 0,
4. T is continuous on $D(T)$.

If we further assume that $D(T) \neq \{0\}$, the above conditions are equivalent to

5. $\sup\{\|Tx\|_Y \mid x \in D(T), \|x\|_X = 1\} < +\infty$.

Proposition 1.2 (About norm of a bounded linear operator)

Let $X, Y, D(T)$ be as in the above definition. Let $T : D(T) \subset X \rightarrow Y$ be a bounded linear operator. Then

$$\|T\| = \sup\{\|Tx\|_Y \mid x \in D(T), \|x\|_X \leq 1\}.$$

If we further assume that $D(T) \neq \{0\}$, we also have

$$\|T\| = \sup\{\|Tx\|_Y \mid x \in D(T), \|x\|_X = 1\}.$$

Definition 1.2 (Sesquilinear form)

Let X be a vector space over $\mathbb{K}$. A *sesquilinear form* on X is a mapping $s : X \times X \rightarrow \mathbb{K}$ satisfying :

1. $\langle x, \lambda y + \mu z \rangle = \lambda \langle x, y \rangle + \mu \langle x, z \rangle \quad \forall x, y, z \in X, \quad \forall \lambda, \mu \in \mathbb{K},$
2. $\langle \lambda x + \mu y, z \rangle = \bar{\lambda} \langle x, z \rangle + \bar{\mu} \langle y, z \rangle \quad \forall x, y, z \in X, \quad \forall \lambda, \mu \in \mathbb{K}$

The associated map $q : X \rightarrow \mathbb{K}$, $q(x) = s(x, x)$, is called the *quadratic form* induced by s .

Proposition 1.3

Let X be a vector space over $\mathbb{K}$. Let s be a sesquilinear form on X with associated quadratic form q . The following identities hold :

1. $\forall x, y \in X : 2q(x) + 2q(y) = q(x + y) + q(x - y)$ (Parallelogram law)
2. If $\mathbb{K} = \mathbb{R}$, $\forall x, y \in X : s(x, y) = \frac{1}{4} (q(x + y) - q(x - y))$ (Polarization identity)
3. If $\mathbb{K} = \mathbb{C}$, $\forall x, y \in X : s(x, y) = \frac{1}{4} (q(x + y) - q(x - y) + iq(x - iy) - iq(x + iy))$ (Polarization identity)

Definition 1.3 (Scalar Product)

Let X be a vector space over $\mathbb{K}$. A *semi-scalar product* on X is a mapping $\langle \cdot, \cdot \rangle : X \times X \rightarrow \mathbb{K}$ satisfying :

1. $\langle x, y + z \rangle = \langle x, y \rangle + \langle x, z \rangle \quad \forall x, y, z \in X,$
2. $\langle x, \lambda y \rangle = \lambda \langle x, y \rangle \quad \forall x, y \in X, \quad \forall \lambda \in \mathbb{K},$
3. $\langle x, y \rangle = \overline{\langle y, x \rangle} \quad \forall x, y \in X,$
4. $\langle x, x \rangle \geq 0 \quad \forall x \in X$

In particular, a semi-scalar product is a sesquilinear form.

If, in addition, $\langle \cdot, \cdot \rangle$ satisfies $\langle x, x \rangle = 0 \iff x = 0$, then $\langle \cdot, \cdot \rangle$ is called a *scalar product*.

One can verify that $x \mapsto \|x\| := \sqrt{\langle x, x \rangle}$ defines a (semi-)norm over X , which we say is *induced* by the (semi-)scalar product $\langle \cdot, \cdot \rangle$.

Definition 1.4 ((Pre-)Hilbert space)

Let X be a vector space over $\mathbb{K}$ with a scalar product $\langle \cdot, \cdot \rangle$. The space $(X, \langle \cdot, \cdot \rangle)$ is called a *Pre-Hilbert space*.

Let $\|\cdot\|$ be the norm induced by $\langle \cdot, \cdot \rangle$. If the normed space $(X, \|\cdot\|)$ is complete, we say that $(X, \langle \cdot, \cdot \rangle)$ is a *Hilbert space*.

Proposition 1.4 (Cauchy-Schwarz Inequality)

Let X be a vector space over $\mathbb{K}$ with a semi-scalar product $\langle \cdot, \cdot \rangle$. The Cauchy-Schwarz inequality holds :

$$\forall x, y \in X : |\langle x, y \rangle| \leq \|x\| \cdot \|y\|$$

Proposition 1.5

Let $(X, \langle \cdot, \cdot \rangle)$ be a Pre-Hilbert space on $\mathbb{K}$ and let $T : D(T) \subset X \rightarrow X$ be a linear map. The mapping $(x, y) \mapsto \langle x, Ty \rangle$ defines a sesquilinear form on the Pre-Hilbert space $(D(T), \langle \cdot, \cdot \rangle_{|D(T) \times D(T)})$.

Theorem 1.6

Let $(X, \|\cdot\|_X)$ be a normed space, $(Y, \langle \cdot, \cdot \rangle)$ be a Pre-Hilbert space. Let $T : D(T) \subset X \rightarrow Y$ be a linear map, where $D(T), R(T) \neq \{0\}$. Assume $M \subset Y$ is a subspace such that

$R(T) \subset \overline{M}$. Then T is bounded if and only if

$$\sup\{|\langle Tx, y \rangle| \mid x \in D(T), y \in M, \|x\|_X = \|y\|_Y = 1\} < +\infty$$

In that case, $\|T\| = \sup\{|\langle Tx, y \rangle| \mid x \in D(T), y \in M, \|x\|_X = \|y\|_Y = 1\}$.

Proof.

Write $s(T) = \sup\{|\langle Tx, y \rangle| \mid x \in D(T), y \in M, \|x\|_X = \|y\|_Y = 1\} \in [0, +\infty]$.

Assume T is bounded. Then by the Cauchy-Schwarz inequality, $|\langle Tx, y \rangle| \leq \|Tx\|_Y \cdot \|y\|_Y \forall x \in D(T), y \in M$, and we get $s(T) \leq \|T\| < +\infty$ by taking the supremum over $\{x \in D(T), y \in M \mid \|x\|_X = \|y\|_Y = 1\}$.

Assume $s(T) < +\infty$. Since $R(T) \subset \overline{M}$, for every $x \in D(T)$ such that $Tx \neq 0$ and $\|x\|_X = 1$, there exists a sequence $(y_n)_{n=1}^\infty \subset M$ such that $y_n \xrightarrow{n \rightarrow +\infty} Tx$ and $y_n \neq 0 \forall n \in \mathbb{N}$. Then

$$|\langle Tx, \frac{y_n}{\|y_n\|_Y} \rangle| \leq s(T) \quad \forall n \in \mathbb{N}$$

and by taking the limit $n \rightarrow +\infty$, we get $\|Tx\|_Y \leq s(T)$. This inequality also holds if $Tx = 0$ and $\|x\|_X = 1$. Finally, we have $\|T\| \leq s(T) < +\infty$ by taking the supremum over $\{x \in D(T) \mid \|x\|_X = 1\}$. $\square$

Remark 1.7

In particular, $\|T\| = \sup\{|\langle Tx, y \rangle| \mid x \in D(T), y \in M, \|x\|_X = \|y\|_Y = 1\}$ for any dense subspace $M \subset Y$.

Theorem 1.8 (Extension of bounded linear operators)

Let $(X, \|\cdot\|_X)$ be a normed space, $(Y, \|\cdot\|_Y)$ be a Banach space and $T : D(T) \subset X \rightarrow Y$ be a bounded linear operator. Then there exists a unique bounded extension $S : \overline{D(T)} \rightarrow Y$ of T . Furthermore, $\|S\| = \|T\|$.

Proof.

Let $x \in \overline{D(T)}$ and let $(x_n)_{n=1}^\infty \subset D(T)$ be a sequence such that $x_n \xrightarrow{n \rightarrow +\infty} x$. In particular, $(x_n)_{n=1}^\infty$ is Cauchy. Then $(Tx_n)_{n=1}^\infty \subset Y$ is also a Cauchy sequence.

Indeed, for $n, m \in \mathbb{N}$, we have

$$\|Tx_n - Tx_m\| = \|T(x_n - x_m)\| \leq \|T\| \cdot \|x_n - x_m\| \xrightarrow{n, m \rightarrow +\infty} 0.$$

Since Y is Banach, $(Tx_n)_{n=1}^\infty$ converges to a limit we will call Sx . The limit Sx does not depend on the choice of (x_n) . Indeed, if $(x_n)_{n=1}^\infty, (y_n)_{n=1}^\infty$ both converges to x , then

$$\|Tx_n - Ty_n\| \leq \|T\| \cdot \|x_n - y_n\| \leq \|T\| \cdot (\|x_n - x\| + \|x - y_n\|) \xrightarrow{n \rightarrow +\infty} 0,$$

so they both have the same limit Sx .

Thus, we have defined a map $x \mapsto Sx$ from $\overline{D(T)}$ to Y , which coincides with T on $D(T)$ (consider the constant sequence $x_n = x \forall n$). It is easy to check that S is linear and we immediately have that $\|T\| \leq \|S\|$.

Now let $x \in \overline{D(T)}$ and $(x_n)_{n=1}^\infty \subset D(T)$ be such that $x_n \xrightarrow{n \rightarrow +\infty} x$. Then

$$\|Sx\| = \lim_{n \rightarrow +\infty} \|Tx_n\|_Y \leq \lim_{n \rightarrow +\infty} \|T\| \cdot \|x_n\|_X = \|T\| \cdot \|x\|_X,$$

so we get $\|S\| \leq \|T\|$.

Finally, assume that S' is another bounded extension of T on $\overline{D(T)}$. For $x \in \overline{D(T)}$, let $(x_n)_{n=1}^\infty \subset D(T)$ be such that $x_n \xrightarrow{n \rightarrow +\infty} x$. By continuity,

$$S'x = \lim_{n \rightarrow +\infty} S'x_n = \lim_{n \rightarrow +\infty} Tx_n = Sx.$$

□

Notation (Extension of an operator)

If S is an extension of T , we write $T \subset S$.

Chapter 2

The Adjoint Operator and Basic Spectral Theory

In this chapter, we will define all the functional analysis-related notions needed to prove the Spectral Theorem. We need to introduce the notions of the adjoint of an operator, of an orthogonal projector and study basic properties of the resolvent of a linear operator. The Spectral Theorem can then be used to give more information about the resolvent of a self-adjoint operator or prove existence of some useful decompositions of self-adjoint operators.

2.1 Preliminary Theorems

In this section, we study basic properties of the orthogonal complement of a set and we prove the Riesz Representation Theorem for bounded linear operators. This theorem is essential if we want to have a nice characterization of the domain of an adjoint. It will also be a powerful tool in the proof of the Spectral Theorem.

Theorem 2.1 (Distance from a point to a closed convex set)

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and $C \subset H$ be a closed convex set. We define the distance from a point $x \in H$ to C as

$$d(x, C) = \inf\{\|x - c\| \mid c \in C\}.$$

Then $\forall x \in H, \exists! c_x \in C$ such that $d(x, C) = \|x - c_x\|$.

Proof.

Let $x \in H$. Consider a sequence $(c_n)_{n=1}^\infty \subset C$ such that $\|x - c_n\| \xrightarrow{n \rightarrow +\infty} d(x, C)$. Then $(c_n)_{n=1}^\infty$ is a Cauchy sequence. Indeed, by the Parallelogram Law :

$$\begin{aligned} \|c_n - c_m\|^2 &= \|(c_n - x) + (x - c_m)\|^2 = 2\|c_n - x\|^2 + 2\|x - c_m\|^2 - \|c_n + c_m - 2x\|^2 \\ &= 2\|c_n - x\|^2 + 2\|x - c_m\|^2 - 4\|\frac{1}{2}(c_n + c_m) - x\|^2 \\ &\leq 2\|c_n - x\|^2 + 2\|x - c_m\|^2 - 4d(x, C)^2 \xrightarrow{n, m \rightarrow +\infty} 0 \end{aligned}$$

where we used $\frac{1}{2}(c_n + c_m) \in C$ by convexity.

Since H is complete and C is closed, the sequence $(c_n)_{n=1}^\infty$ converges to a limit $c_x \in C$ and we have $\|x - c_n\| \xrightarrow{n \rightarrow +\infty} \|x - c_x\| = d(x, C)$.

Finally, if $c_1, c_2 \in C$ are such that $\|x - c_1\| = \|x - c_2\| = d(x, C)$, then by the parallelogram law :

$$\begin{aligned}\|c_1 - c_2\|^2 &= \|(c_1 - x) + (x - c_2)\|^2 = 2\|c_1 - x\|^2 + 2\|x - c_2\|^2 - \|c_1 + c_2 - 2x\|^2 \\ &= 2d(x, C)^2 + 2d(x, C)^2 - 4\left\|\frac{1}{2}(c_1 + c_2) - x\right\|^2 \\ &\leq 2d(x, C)^2 + 2d(x, C)^2 - 4d(x, C)^2 = 0\end{aligned}$$

and thus $c_1 = c_2$. □

Definition 2.1 (Orthogonal Complement)

Let $(X, \langle \cdot, \cdot \rangle)$ be a Pre-Hilbert space and $S \subset X$ be an arbitrary subset. The orthogonal complement of S is the closed subspace

$$S^\perp = \{x \in X \mid \langle x, s \rangle = 0 \ \forall s \in S\}.$$

Theorem 2.2 (Projection Theorem for closed subspaces)

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and $M \subset H$ be a closed subspace. Then $H = M \oplus M^\perp$.

Proof.

Let $x \in H$. Since M is a subspace, it is convex. By Theorem 2.1, there exists a unique $y \in M$ such that $d(x, M) = \|x - y\| = \inf\{\|x - m\| \mid m \in M\}$.

We write $x = y + (x - y)$ and we show that $x - y \in M^\perp$. Indeed, let $m \in M \setminus \{0\}$. For $\alpha \in \mathbb{K}$, we have

$$\|x - y\|^2 \leq \|x - (y + \alpha m)\|^2 = \|x - y\|^2 - 2\Re(\langle x - y, \alpha m \rangle) + |\alpha|^2 \|m\|^2$$

Set $\alpha = \frac{\overline{\langle x - y, m \rangle}}{\|m\|^2}$. Then

$$0 \leq -2 \frac{|\langle x - y, m \rangle|^2}{\|m\|^2} + \frac{|\langle x - y, m \rangle|^2}{\|m\|^2} = -\frac{|\langle x - y, m \rangle|^2}{\|m\|^2}$$

Thus, $\langle x - y, m \rangle = 0$ and we have $H = M + M^\perp$.

Finally, assume that $x \in M \cap M^\perp$. This implies $\langle x, x \rangle = 0$ and so $x = 0$. Therefore, $M \cap M^\perp = \{0\}$. □

Corollary 2.3

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and $M \subset H$ be a closed subspace. Then $M = (M^\perp)^\perp$.

Proof.

By Theorem 2.2, $H = M \oplus M^\perp = M^\perp \oplus (M^\perp)^\perp$ since $M^\perp$ is a closed subspace. Since the sum is direct, we get $M = (M^\perp)^\perp$. □

Lemma 2.4 (Closure of a subspace is a subspace)

Let $(X, \|\cdot\|)$ be a normed space and let $M \subset X$ be a subspace of X . The topological closure $\overline{M}$ of M is a subspace of X .

Proposition 2.5

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and $M \subset H$ be a subspace. Then $(M^\perp)^\perp = \overline{M}$.

Proof.

Clearly, $(\overline{M})^\perp = M^\perp$. By Corollary 2.3, $(M^\perp)^\perp = ((\overline{M})^\perp)^\perp = \overline{M}$. $\square$

Proposition 2.6

Let $(X, \langle \cdot, \cdot \rangle)$ be a Pre-Hilbert space. For all $g \in X$, the map $T_g : X \rightarrow \mathbb{K}$, $T_g(x) = \langle g, x \rangle$ is linear and bounded with norm $\|T_g\| = \|g\|_X$.

Proof.

If $g = 0$, this is trivial. Assume $g \neq 0$.

We immediately deduce $\|T_g\| \leq \|g\|_X$ using the Cauchy-Schwarz inequality. To get $\|T_g\| \geq \|g\|_X$, we just evaluate T_g at $\frac{g}{\|g\|_X}$. $\square$

Theorem 2.7 (Riesz Representation Theorem)

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and $T : H \rightarrow \mathbb{K}$ be a bounded linear map. Then there exists a unique $g \in H$ such that $T(x) = T_g(x) = \langle g, x \rangle \forall x \in H$. Furthermore, $\|T\| = \|g\|_H$.

Proof.

If $N(T) = H$, then $T = 0$ on H and we can choose $g = 0$.

Otherwise, since $N(T)$ is closed by continuity of T , we have $N(T)^\perp \neq \{0\}$ by Theorem 2.2. Let $a \in N(T)^\perp$ be an element with $\|a\| = 1$. Notice that $\forall x \in H$, $T(x)a - T(a)x \in N(T)$ by linearity of T . Since $a \in N(T)^\perp$, we get

$$0 = \langle a, T(x)a - T(a)x \rangle = T(x)\|a\|^2 - \langle \overline{T(a)}a, x \rangle \forall x \in H$$

Thus, $T(x) = \langle \overline{T(a)}a, x \rangle = T_g(x) \forall x \in H$, where $g = \overline{T(a)}a$.

Finally, assume that $T(x) = \langle g, x \rangle = \langle h, x \rangle \forall x \in H$ for some $g, h \in H$. Then $\langle g - h, x \rangle = 0 \forall x \in H$ and we can take $x = g - h$ to conclude that $g = h$. $\square$

Remark 2.8

The map $T \mapsto g$ such that $T(x) = T_g(x) = \langle g, x \rangle \forall x \in X$ is antilinear.

Definition 2.2 (Densely defined linear operators)

Let $(X, \|\cdot\|_X)$, $(Y, \|\cdot\|_Y)$ be two normed spaces and let $T : D(T) \subset X \rightarrow Y$ be a linear map. We say that T is *densely defined* if $D(T) \subset X$ is a dense subset.

Lemma 2.9

Let $(X, \langle \cdot, \cdot \rangle)$ be a Pre-Hilbert space and $D \subset X$ be a dense subset. Then $D^\perp = \{0\}$.

Proof. Let $x \in D^\perp$ and let $(x_n)_{n=1}^\infty \subset D$ be a sequence such that $x_n \xrightarrow{n \rightarrow +\infty} x$. Then

$$\|x\|^2 = \langle x, x \rangle = \lim_{n \rightarrow \infty} \langle x, x_n \rangle = \lim_{n \rightarrow \infty} 0 = 0.$$

$\square$

Corollary 2.10 (Riesz Representation Theorem for densely defined operators)

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and $T : D(T) \rightarrow \mathbb{K}$ be a densely defined, bounded, linear map. Then there exists a unique $g \in H$ such that $T(x) = T_g(x) = \langle g, x \rangle \forall x \in D(T)$.

Proof.

By Theorem 1.8, T has a bounded extension on the whole H . Denote $\hat{T}$ this extension. By Riesz Representation Theorem, there exists $g \in H$ such that $\hat{T}(x) = \langle g, x \rangle \forall x \in H$.

In particular, $\hat{T}(x) = T(x) = \langle g, x \rangle \forall x \in D(T)$.

Now, assume that there exists $g, h \in H$ such that $T(x) = \langle g, x \rangle = \langle h, x \rangle \forall x \in D(T)$. Then $g - h \in D(T)^\perp = \{0\}$ by density of $D(T)$, which allows us to conclude. $\square$

2.2 The Adjoint Operator

Definition 2.3 (Formal adjoint)

Let $(H_1, \langle \cdot, \cdot \rangle_1)$, $(H_2, \langle \cdot, \cdot \rangle_2)$ be two Hilbert spaces and $T : D(T) \subset H_1 \rightarrow H_2$, $S : D(S) \subset H_2 \rightarrow H_1$ be two linear operators. We say that S is a *formal adjoint* of T if

$$\langle g, Tf \rangle_2 = \langle Sg, f \rangle_1 \quad \forall f \in D(T), g \in D(S)$$

Remark 2.11

1. The linear operator $S : \{0\} \subset H_2 \rightarrow H_1$ is a formal adjoint of any $T : D(T) \subset H_1 \rightarrow H_2$.
2. S is a formal adjoint of T if and only if T is a formal adjoint of S .

Proposition 2.12

Let $(H_1, \langle \cdot, \cdot \rangle_1)$, $(H_2, \langle \cdot, \cdot \rangle_2)$ be two Hilbert spaces and $T : D(T) \subset H_1 \rightarrow H_2$, $S : D(S) \subset H_2 \rightarrow H_1$ be two linear operators.

If S is a formal adjoint of T , then $\forall g \in D(S)$, the operator

$$\begin{aligned} L_g : D(T) &\rightarrow \mathbb{K} \\ f &\mapsto \langle g, Tf \rangle_2 \end{aligned}$$

is linear and bounded.

Proof. By the Cauchy-Schwarz inequality, we have

$$|L_g(f)| = |\langle g, Tf \rangle_2| = |\langle Sg, f \rangle_1| \leq \|Sg\|_1 \cdot \|f\|_1.$$

$\square$

Definition 2.4 (Adjoint of a densely defined operator)

Let $(H_1, \langle \cdot, \cdot \rangle_1)$, $(H_2, \langle \cdot, \cdot \rangle_2)$ be two Hilbert spaces and $T : D(T) \subset H_1 \rightarrow H_2$ be a densely defined linear operator.

We define the set

$$\begin{aligned} D^* &= \{g \in H_2 \mid L_g : D(T) \rightarrow \mathbb{K}, L_g(f) = \langle g, Tf \rangle_2 \text{ is continuous}\} \\ &= \{g \in H_2 \mid \exists h_g \in H_1 : L_g(f) = \langle g, Tf \rangle_2 = \langle h_g, f \rangle_1 \quad \forall f \in D(T)\} \end{aligned}$$

where the equality follows from Riesz Representation Theorem for densely defined operators (Corollary 2.10).

On D^* , we define the operator

$$\begin{aligned} T^* : D(T^*) = D^* \subset H_2 &\rightarrow H_1 \\ g &\mapsto T^*g := h_g \end{aligned}$$

This operator, called the *adjoint* of T , is well-defined (because the element h_g is unique as seen in Corollary 2.10) and linear.

Proposition 2.13 (On the relation between the adjoint and formal adjoints)

Let $(H_1, \langle \cdot, \cdot \rangle_1)$, $(H_2, \langle \cdot, \cdot \rangle_2)$ be two Hilbert spaces and $T : D(T) \subset H_1 \rightarrow H_2$ be a densely defined linear operator.

1. The adjoint T^* of T is a formal adjoint of T ;
2. The adjoint T^* of T is an extension of any formal adjoint S of T ;
3. If T^* is densely defined, then $T^{**} = (T^*)^*$ is an extension of T ;
4. $N(T^*) = R(T)^\perp$;
5. $N(T^*)^\perp = \overline{R(T)}$.

Proof.

1. This follows from the definition of T^* .
2. Let $S : D(S) \subset H_2 \rightarrow H_1$ be a formal adjoint of T . Let $g \in D(S)$. Then

$$\langle g, Tf \rangle_2 = \langle Sg, f \rangle_1 =: \langle h_g, f \rangle \quad \forall f \in D(T)$$

so $g \in D(T^*)$ and $T^*g = h_g = Sg$.

3. By 2., we just need to show that T is a formal adjoint of T^* , which follows from 1.
4. If $g \in N(T^*)$, $0 = \langle T^*g, f \rangle_1 = \langle g, Tf \rangle_2 \quad \forall f \in D(T)$, i.e. $g \in R(T)^\perp$.

Conversely, let $g \in R(T)^\perp$, i.e. $\langle g, Tf \rangle_2 = 0 \quad \forall f \in D(T)$. In particular, $\langle g, Tf \rangle_2 = \langle 0, f \rangle_1 \quad \forall f \in D(T)$, so $g \in D(T^*)$ and $T^*g = 0$.

5. This follows from point 4. and Proposition 2.5.

□

Example 2.1 (Adjoint of the differentiation operator)

Consider the real Hilbert space

$$H = \mathcal{L}^2([0, 1]) = \{f : [0, 1] \rightarrow \mathbb{R} \mid |f|^2 \text{ is Lebesgue integrable on } [0, 1]\}$$

(where f and g are identified if $f = g$ almost everywhere) endowed with the $\mathcal{L}^2$ -scalar product

$$\langle g, f \rangle = \int_0^1 g(x)f(x)dx.$$

A function $f : [0, 1] \rightarrow \mathbb{R}$ is called *absolutely continuous* if it is differentiable almost everywhere on $[0, 1]$, its derivative f' is Lebesgue integrable on $[0, 1]$ and

$$f(x) = f(0) + \int_0^x f'(t)dt \text{ almost everywhere on } [0, 1].$$

Consider the differentiation operator

$$T : D(T) \subset \mathcal{L}^2([0, 1]) \rightarrow \mathcal{L}^2([0, 1])$$

$$f(x) \mapsto f'(x)$$

where $D(T) \subset H$ is the set of classes for which we can find an absolutely continuous representative $f : [0, 1] \rightarrow \mathbb{R}$ such that $f, f' \in \mathcal{L}^2([0, 1])$. The domain of T is densely defined since the subset of C^∞ and compactly supported functions $C_c^\infty([0, 1]) \subset D(T)$ is dense in $\mathcal{L}^2([0, 1])$.

We want to find the adjoint of T , i.e. we want to find the elements $g \in \mathcal{L}^2([0, 1])$ such that there exists $h = h_g \in \mathcal{L}^2([0, 1])$ satisfying

$$\langle g, Tf \rangle = \langle h, f \rangle \quad \forall f \in D(T)$$

i.e.

$$\int_0^1 g(x)f'(x)dx = \int_0^1 h(x)f(x)dx \quad \forall f \in D(T)$$

Integrating by parts the right-hand side, we find

$$\int_0^1 h(x)f(x)dx = H(1)f(1) - H(0)f(0) - \int_0^1 H(x)f'(x)dx \quad \forall f \in D(T).$$

where $H(x) = \int_0^x h(t)dt$, $H(0) = 0$, $H(1) = \int_0^1 h(t)dt$, i.e. g and h must satisfy

$$\int_0^1 (g(x) + H(x))f'(x)dx = \left(\int_0^1 h(t)dt \right) f(1) \quad \forall f \in D(T).$$

Setting $f(x) = -\int_x^1 v(t)dt + d$, $v \in C_0^\infty([0, 1])$, $d \in \mathbb{R}$, we get

$$\int_0^1 (g(x) + H(x))v(x)dx = d \left(\int_0^1 h(t)dt \right) \quad \forall v \in C_0^\infty([0, 1]), \quad \forall d \in \mathbb{R}.$$

Evaluating this at $d \in \{0, 1\}$, we deduce $H(1) = \int_0^1 h(t)dt = 0$.

By density of $C_c^\infty([0, 1])$, evaluating this at a sequence $(v_n) \subset C_c^\infty([0, 1])$ converging to $g(x) + H(x)$ in $\mathcal{L}^2([0, 1])$, we deduce $g(x) = -H(x) = -\int_0^x h(t)dt$.

We conclude that $D(T^*) \subset H$ is the set of classes for which we can find an absolutely continuous representative $g : [0, 1] \rightarrow \mathbb{R}$ such that $g, g' \in \mathcal{L}^2([0, 1])$ and $g(0) = g(1) = 0$. Furthermore, $T^*g(x) = -g'(x) \quad \forall g \in D(T^*)$.

Proposition 2.14 (On the adjoint of bounded operators)

Let $(H_1, \langle \cdot, \cdot \rangle_1)$, $(H_2, \langle \cdot, \cdot \rangle_2)$ be two Hilbert spaces and $T : D(T) \subset H_1 \rightarrow H_2$ be a densely defined linear operator. Then

1. T is bounded if and only if $T^* \in B(H_2, H_1)$.
2. If T is bounded, $\|T\| = \|T^*\|$.
3. If T is bounded, T^{**} is the unique bounded extension on H_1 from Theorem 1.8. If $T \in B(H_1, H_2)$, then $T^{**} = T$.

Proof.

We only prove the first part. The two other parts follow from it.

If $D(T) = \{0\}$ or $R(T) = \{0\}$, the proposition is trivial. Assume $D(T), R(T) \neq \{0\}$.

$\Rightarrow$: If T is bounded, then for all $g \in H_2$, $L_g(f) = \langle g, Tf \rangle_2$ is bounded since

$$|L_g(f)| = |\langle g, Tf \rangle_2| \leq \|g\|_2 \cdot \|T\| \cdot \|f\|_1,$$

so we have $D(T^*) = H_2$. By Theorem 1.6,

$$\begin{aligned} \|T^*\| &= \sup\{|\langle T^*g, f \rangle_1| \mid g \in H_2, f \in D(T), \|g\|_2 = \|f\|_1 = 1\} \\ &= \sup\{|\langle g, Tf \rangle_2| \mid g \in H_2, f \in D(T), \|g\|_2 = \|f\|_1 = 1\} \\ &= \sup\{|\langle Tf, g \rangle_2| \mid g \in H_2, f \in D(T), \|g\|_2 = \|f\|_1 = 1\} = \|T\| \end{aligned}$$

In particular, we have proved that if T is bounded, then $\|T^*\| = \|T\|$.

$\Leftarrow$: If $T^* \in B(H_2, H_1)$, then $T^{**} \in B(H_1, H_2)$ is a bounded extension of T on $H_1 = \overline{D(T)}$ by $\Rightarrow$ and Proposition 2.13. In particular, T is bounded. $\square$

Proposition 2.15

Let $(H_1, \langle \cdot, \cdot \rangle_1)$, $(H_2, \langle \cdot, \cdot \rangle_2)$ be two Hilbert spaces and $T : D(T) \subset H_1 \rightarrow H_2$, $S : D(S) \subset H_1 \rightarrow H_2$ be two densely defined linear operators. We have the following properties for the adjoint :

1. If $S \subset T$, then $T^* \subset S^*$.
2. If $T+S$ is densely defined, then $T^* + S^* \subset (T+S)^*$. Furthermore, if $S \in B(H_1, H_2)$, then $T^* + S^* = (T+S)^*$.

Proof.

1. It suffices to prove that T^* is a formal adjoint of S by Proposition 2.13. Let $f \in D(T^*) \subset H_2$, $g \in D(S) \subset D(T) \subset H_1$. We have

$$\langle T^*f, g \rangle_1 = \langle f, Tg \rangle_2 = \langle f, Sg \rangle_2$$

2. Again, it suffices to prove that $T^* + S^*$ is a formal adjoint of $T + S$. Let $f \in D(T^* + S^*) = D(T^*) \cap D(S^*) \subset H_2$, $g \in D(T + S) = D(T) \cap D(S) \subset H_1$. We have

$$\langle (T^* + S^*)f, g \rangle_1 = \langle T^*f, g \rangle_1 + \langle S^*f, g \rangle_1 = \langle f, Tg \rangle_2 + \langle f, Sg \rangle_2 = \langle f, (T + S)g \rangle_2$$

Now, assume that $S \in B(H_1, H_2)$. Then $S^* \in B(H_2, H_1)$ by Proposition 2.13. We show that $D((T + S)^*) \subset D(T^* + S^*) = D(T^*)$. Let $g \in D((T + S)^*)$. We have

$$\begin{aligned} \langle g, Tf \rangle_2 &= \langle g, (T + S)f \rangle_2 - \langle g, Sf \rangle_2 = \langle (T + S)^*g, f \rangle_1 - \langle S^*g, f \rangle_1 \\ &= \langle ((T + S)^* - S^*)g, f \rangle_1 = \langle h_g, f \rangle_1 \quad \forall f \in D(T) \subset D(S) \end{aligned}$$

$\square$

Proposition 2.16

Let $(H_i, \langle \cdot, \cdot \rangle_i)$, $i = 1, 2, 3$, be three Hilbert spaces and $T : D(T) \subset H_1 \rightarrow H_2$, $S : D(S) \subset H_2 \rightarrow H_3$ be two densely defined linear operators.

1. If ST is densely defined, then $T^*S^* \subset (ST)^*$. Furthermore, if $S \in B(H_2, H_3)$, then $T^*S^* = (ST)^*$.
2. $(aT)^* = \bar{a}T^* \forall a \in \mathbb{K} \setminus \{0\}$

Proof.

1. It suffices to prove that T^*S^* is a formal adjoint of ST . Let $f \in D(T^*S^*) \subset H_3$, $g \in D(ST) \subset H_1$. We have

$$\langle (T^*S^*)f, g \rangle_1 = \langle T^*(S^*f), g \rangle_1 = \langle S^*f, Tg \rangle_2 = \langle f, STg \rangle_3$$

Now, assume that $S \in B(H_2, H_3)$. Then $S^* \in B(H_2, H_3)$ by Proposition 2.13. We show that $D((ST)^*) \subset D(T^*S^*) = \{g \in D(S^*) = H_3 \mid S^*g \in D(T^*)\}$. Let $g \in D((ST)^*)$. We have

$$\langle S^*g, Tf \rangle_2 = \langle g, STf \rangle_3 = \langle (ST)^*g, f \rangle_1 = \langle h_g, f \rangle_1 \quad \forall f \in D(T)$$

2. Follows from 1. for $S = aI$ and $H_2 = H_3$.

□

Definition 2.5

Let $(H_1, \langle \cdot, \cdot \rangle_1)$, $(H_2, \langle \cdot, \cdot \rangle_2)$ be two Hilbert spaces. We define the following scalar product on $H_1 \times H_2$: $\langle (x_1, y_1), (x_2, y_2) \rangle = \langle x_1, x_2 \rangle_1 + \langle y_1, y_2 \rangle_2$ and we have a similar scalar product on $H_2 \times H_1$.

Additionally, we define the two following maps :

$$\begin{aligned} U : H_1 \times H_2 &\rightarrow H_2 \times H_1, & V : H_1 \times H_2 &\rightarrow H_2 \times H_1 \\ (x, y) &\mapsto (y, -x) & (x, y) &\mapsto (y, x) \end{aligned}$$

$$\begin{aligned} U^{-1} : H_2 \times H_1 &\rightarrow H_1 \times H_2, & V^{-1} : H_2 \times H_1 &\rightarrow H_1 \times H_2 \\ (y, x) &\mapsto (-x, y) & (y, x) &\mapsto (x, y) \end{aligned}$$

Notice that for $(v_1, v_2) \in H_2 \times H_1$, $(w_1, w_2) \in H_1 \times H_2$, we have

$$\begin{aligned} \langle (v_1, v_2), U(w_1, w_2) \rangle &= 0 \iff \langle U^{-1}(v_1, v_2), (w_1, w_2) \rangle = 0 \text{ and} \\ \langle (v_1, v_2), V(w_1, w_2) \rangle &= 0 \iff \langle V^{-1}(v_1, v_2), (w_1, w_2) \rangle = 0 \end{aligned}$$

so that $(UM)^\perp = U(M^\perp)$ and $(VM)^\perp = V(M^\perp)$ for any subspace $M \subset H_1 \times H_2$.

Lemma 2.17 (On the graph of the adjoint)

Let $(H_1, \langle \cdot, \cdot \rangle_1)$, $(H_2, \langle \cdot, \cdot \rangle_2)$ be two Hilbert spaces and $T : D(T) \subset H_1 \rightarrow H_2$ be a densely defined linear operator. We have the following characterization for the graph of T^* :

$$G(T^*) = U(G(T)^\perp) = (UG(T))^\perp$$

where U is defined as in Definition 2.5 on $H_1 \times H_2$.

Proof.

$$\begin{aligned} G(T^*) &= \{(g, T^*g) \mid g \in D(T^*)\} = \{(g, h) \in H_2 \times H_1 \mid \langle h, f \rangle_1 = \langle g, Tf \rangle_2 \ \forall f \in D(T)\} \\ &= \{(g, h) \mid \langle (g, h), (Tf, -f) \rangle = 0 \ \forall f \in D(T)\} \\ &= (UG(T))^\perp \end{aligned}$$

The second equality follows from Definition 2.5 □

Theorem 2.18 (On the adjoint of an inverse)

Let $(H_1, \langle \cdot, \cdot \rangle_1)$, $(H_2, \langle \cdot, \cdot \rangle_2)$ be two Hilbert spaces and $T : D(T) \subset H_1 \rightarrow H_2$ be an injective and densely defined linear operator. If $R(T)$ is dense, then T^* is injective and $(T^*)^{-1} = (T^{-1})^*$.

Proof.

Assume $T^*g = T^*h$ for some $g, h \in D(T^*)$. Then

$$\langle g, Tf \rangle = \langle T^*g, f \rangle = \langle T^*h, f \rangle = \langle h, Tf \rangle \ \forall f \in D(T),$$

i.e. $g - h \in R(T)^\perp = \{0\}$.

Finally,

$$\begin{aligned} G((T^{-1})^*) &= (UG(T^{-1}))^\perp = (UV^{-1}G(T))^\perp = (VU^{-1}G(T))^\perp = V(U^{-1}(G(T)^\perp)) \\ &= V(G(T^*)) = G((T^*)^{-1}) \end{aligned}$$

using Lemma 2.17, where U and V are defined as in Definition 2.5 on $H_2 \times H_1$ (instead of $H_1 \times H_2$). □

Proposition 2.19 (On the graph of the double adjoint)

Let $(H_1, \langle \cdot, \cdot \rangle_1)$, $(H_2, \langle \cdot, \cdot \rangle_2)$ be two Hilbert spaces and let $T : D(T) \subset H_1 \rightarrow H_2$ be a densely defined linear operator such that T^* is densely defined. Then $G(T^{**}) = \overline{G(T)}$. In particular, if T has a closed graph, then $T = T^{**}$.

Proof.

$$G(T^{**}) = (U_{2,1}G(T^*))^\perp = (U_{2,1}U_{1,2}(G(T)^\perp))^\perp = (G(T)^\perp)^\perp = \overline{G(T)}$$

using Lemma 2.17, where $U_{1,2}$ and $U_{2,1}$ are defined as in Definition 2.5 on $H_1 \times H_2$ and $H_2 \times H_1$ respectively. □

Definition 2.6 (Self-adjoint)

Let $(H, \langle \cdot, \cdot \rangle)$ be an Hilbert space and $T : D(T) \subset H \rightarrow H$ be a linear operator. We say that T is self-adjoint if T is densely defined and $T = T^*$.

Theorem 2.20

Let $(H, \langle \cdot, \cdot \rangle)$ be an Hilbert space and $T : D(T) \subset H \rightarrow H$ be an injective, self-adjoint, linear operator. Then T^{-1} is self-adjoint.

Proof.

By Theorem 2.18, it is enough to prove that $R(T)$ is dense. We have $\{0\} = N(T) = N(T^*) = R(T)^\perp$ by Proposition 2.13. In particular, $\overline{R(T)}$ is a closed subspace such that $(\overline{R(T)})^\perp = \{0\}$. By Theorem 2.2, $\overline{R(T)} = H$. $\square$

Definition 2.7 (Closed operator)

Let $(X, \|\cdot\|_X)$, $(Y, \|\cdot\|_Y)$ be two Banach spaces and $T : D(T) \subset X \rightarrow Y$ be a linear operator. We say that T is *closed* if the graph $G(T)$ of T is closed in $X \times Y$ (endowed with the norm $\|(x, y)\|_{X \times Y} = \|x\|_X + \|y\|_Y$).

Theorem 2.21 (Closed Graph Theorem)

Let $(X, \|\cdot\|_X)$, $(Y, \|\cdot\|_Y)$ be two Banach spaces and $T : D(T) \subset X \rightarrow Y$ be a linear operator. The following are equivalent :

1. $D(T)$ is closed and T is closed;
2. $D(T)$ is closed and T is bounded;
3. T is closed and bounded.

Proposition 2.22 (Adjoint is closed)

Let $(H_1, \langle \cdot, \cdot \rangle_1)$, $(H_2, \langle \cdot, \cdot \rangle_2)$ be two Hilbert spaces and let $T : D(T) \subset H_1 \rightarrow H_2$ be a densely defined linear operator. Then T^* is closed.

Proof.

Let $(g_n, f_n)_{n=1}^\infty \subset G(T^*)$ be a sequence converging to some (g, f) in $H_2 \times H_1$.

Then for all $x \in D(T)$:

$$\langle g, Tx \rangle = \lim_{n \rightarrow \infty} \langle g_n, Tx \rangle = \lim_{n \rightarrow \infty} \langle T^* g_n, x \rangle = \lim_{n \rightarrow \infty} \langle f_n, x \rangle = \langle f, x \rangle,$$

so $g \in D(T^*)$ and $T^* g = f$, i.e. $(g, f) \in G(T^*)$. $\square$

Corollary 2.23

Let $(H_1, \langle \cdot, \cdot \rangle_1)$, $(H_2, \langle \cdot, \cdot \rangle_2)$ be two Hilbert spaces and let $T : D(T) \subset H_1 \rightarrow H_2$ be a densely defined linear operator such that T^* is densely defined. Then $T^* = (T^*)^{**} = (T^{**})^*$.

Proof. Follows from Proposition 2.19 and Proposition 2.22. $\square$

Proposition 2.24 (On the range of the double adjoint)

Let $(H_1, \langle \cdot, \cdot \rangle_1)$, $(H_2, \langle \cdot, \cdot \rangle_2)$ be two Hilbert spaces and let $T : D(T) \subset H_1 \rightarrow H_2$ be a densely defined, injective, linear operator such that T^* is densely defined and T^{**} is injective with bounded inverse. Then $R(T^{**}) = \overline{R(T)}$.

Proof.

This is equivalent to proving $D((T^{**})^{-1}) = \overline{D(T^{-1})}$. We already know that $D((T^{**})^{-1})$ is closed by the Closed Graph Theorem and $D(T^{-1}) \subset D((T^{**})^{-1})$ since $T \subset T^{**}$. It remains to prove that $D((T^{**})^{-1}) \subset \overline{D(T^{-1})}$.

Let $x \in D((T^{**})^{-1})$. Since $G((T^{**})^{-1}) = \overline{G(T^{-1})}$ by Proposition 2.19, we can find a sequence $((x_n, T^{-1}x_n))_{n=1}^\infty \subset G(T^{-1})$ converging to $(x, (T^{**})^{-1}x)$ in $H_2 \times H_1$. In particular, $(x_n)_{n=1}^\infty \subset D(T^{-1})$ converges to x in H_2 , so $x \in \overline{D(T^{-1})}$. $\square$

2.3 Orthogonal Projectors

Definition 2.8 (Symmetric operator)

Let $(X, \langle \cdot, \cdot \rangle)$ be a Pre-Hilbert space and $S : X \rightarrow X$ be a linear operator. We say that S is *symmetric* if

$$\langle Sx, y \rangle = \langle x, Sy \rangle \quad \forall x, y \in X$$

Definition 2.9

Let $(X, \langle \cdot, \cdot \rangle)$ be a Pre-Hilbert space and $P, S : X \rightarrow X$ be two symmetric linear operators. We write $P \leq S$ (and similarly for $<, \geq, >$) if

$$\langle Px, x \rangle \leq \langle Sx, x \rangle \quad \forall x \in X$$

Note that since S is symmetric, $\langle Sx, x \rangle$ is real for all $x \in X$.

Definition 2.10 (Orthogonal Projector)

Let $(X, \langle \cdot, \cdot \rangle)$ be a Pre-Hilbert space and $V, W \subset X$ two subspaces such that

1. $V \perp W$
2. $X = V \oplus W$

By hypothesis 2., $\forall x \in X, \exists! v \in V, w \in W$ such that $x = v + w$. Therefore, we have two well-defined maps

$$\begin{array}{ll} P : X \rightarrow X, & Q : X \rightarrow X \\ x \mapsto v & x \mapsto w \end{array}$$

P is called the *orthogonal projection* onto V and $Q (= I - P)$ is called the *orthogonal projection* onto W .

Example 2.2

Consider the Pre-Hilbert space $X = C^0([-1, 1])$ on $\mathbb{R}$ endowed with the $\mathcal{L}^2$ -scalar product

$$\langle g, f \rangle = \int_{-1}^1 g(x)f(x)dx.$$

Let $V \subset X$ be the subspace of even functions and $W \subset X$ be the subspace of odd functions. Then X is the orthogonal direct sum of V and W and the orthogonal projections P, Q are respectively given by

$$Pf(x) = \frac{f(x) + f(-x)}{2}, \quad Qf(x) = \frac{f(x) - f(-x)}{2}$$

We only check that $V \perp W$. Let $g(x) \in V$ be even, $f(x) \in W$ be odd. Then $g(x)f(x)$ is odd and

$$\begin{aligned} \langle g, f \rangle &= \int_{-1}^1 g(x)f(x)dx = \int_{-1}^0 g(x)f(x)dx + \int_0^1 g(x)f(x)dx \\ &= \int_0^1 g(-x)f(-x)dx + \int_0^1 g(x)f(x)dx \\ &= 0 \end{aligned}$$

Proposition 2.25 (Properties of an orthogonal projection)

Let X, V, W, P, Q be as in the above definition.

Then P is a bounded linear operator such that $\|P\| \leq 1$ (and $\|P\| = 1$ if $V \neq \{0\}$), $P^2 = P \circ P = P$ (i.e. P is idempotent), $N(P) = W$, $R(P) = V$, P is symmetric and $0 \leq P \leq I$.

Proof. We write $x = v + w$, $Px = v \in V$, $w \in W$.

1. Linearity : Easy to check.

2. By Pythagoras' Theorem, $\|x\|^2 = \|Px\|^2 + \|w\|^2 \ \forall x \in X$, so $\|Px\|^2 \leq \|x\|^2 \ \forall x \in X$.

This proves $\|P\| \leq 1$. Assume now that $V \neq \{0\}$. Since $Pv = v \ \forall v \in V$, we have $\|P\| = 1$ (take $v \in V$ such that $\|v\| = 1$).

3. Since $Px \in V \ \forall x \in X$, we have $P(Px) = Px \ \forall x \in X$, i.e. $P^2 = P$.

4. $N(P) = W$, $R(P) = V$ follow from the definition of P .

5. Let $x, y \in X$. Write $x = v_x + w_x$, $y = v_y + w_y$, $Px = v_x$, $Py = v_y \in V$, $w_x, w_y \in W$. Then

$$\begin{aligned}\langle Px, y \rangle &= \langle Px, Py + w_y \rangle = \langle Px, Py \rangle \\ \langle x, Py \rangle &= \langle Px + w_x, Py \rangle = \langle Px, Py \rangle\end{aligned}$$

where we used $V \perp W$.

6. For all $x \in X$:

$$\begin{aligned}\langle Px, x \rangle &= \langle P^2x, x \rangle = \langle Px, Px \rangle \geq 0 = \langle 0x, x \rangle \\ \langle Px, x \rangle &\leq \|Px\| \cdot \|x\| \leq \|P\| \cdot \|x\|^2 \leq 1 \cdot \|x\|^2 = \langle Ix, x \rangle\end{aligned}$$

□

Proposition 2.26

Let $(X, \langle \cdot, \cdot \rangle)$ be a Pre-Hilbert space and $P : X \rightarrow X$ be a linear operator. The following are equivalent :

1. P is an orthogonal projection;
2. $I - P$ is an orthogonal projection;
3. P is idempotent and symmetric.

Proof.

(i) $\iff$ (ii) : By definition, P is an orthogonal projection if and only if $I - P$ is an orthogonal projection.

(i) $\implies$ (iii) : Follows from Proposition 2.25.

(iii) $\implies$ (i) : We shall prove that P is the orthogonal projection onto $R(P)$. First, we prove that $R(P) \perp R(I - P)$. For all $x, y \in X$, we have

$$\langle Px, (I - P)y \rangle = \langle x, P(I - P)y \rangle = \langle x, (P - P^2)y \rangle = \langle x, 0 \rangle = 0.$$

Furthermore, every $x \in X$ can be written as $x = Px + (I - P)x = v + w$, $v \in R(P)$, $w \in R(I - P)$, so $X = R(P) + R(I - P)$.

It only remains to prove that $R(P) \cap R(I - P) = \{0\}$ (i.e. the sum is direct) and (i) will follow. Assume that $x \in R(P) \cap R(I - P)$. We can write $x = Py = (I - P)z$ for some $y, z \in X$. Then

$$\|x\|^2 = \langle x, x \rangle = \langle Py, (I - P)z \rangle = \langle y, P(I - P)z \rangle = \langle y, 0 \rangle = 0.$$

□

Proposition 2.27

Let $(X, \langle \cdot, \cdot \rangle)$ be a Pre-Hilbert space and $P_1, P_2 : X \rightarrow X$ be two orthogonal projectors. The following are equivalent :

1. $P_1 \leq P_2$
2. $P_2 P_1 = P_1$
3. $P_1 P_2 = P_1$
4. $R(P_1) \subset R(P_2)$

Theorem 2.28

Let $(H, \langle \cdot, \cdot \rangle)$ be an Hilbert space and $(S_n)_{n=1}^\infty \subset B(H, H)$ be a decreasing sequence of symmetric operators (i.e. $0 \leq S_{n+1} \leq S_n \forall n \in \mathbb{N}$).

Then there exists $S \in B(H, H)$ symmetric, $S \geq 0$, $\|S\| \leq \|S_1\|$, such that for all $x \in H$:

$$\lim_{n \rightarrow +\infty} S_n x = Sx,$$

i.e. S_n converges strongly (pointwisely) to S . We write $S_n \xrightarrow{s} S$.

2.4 Basic Spectral Properties

Definition 2.11 (Eigenvalue, Spectrum, Resolvent Set)

Let $(H, \langle \cdot, \cdot \rangle)$ be an Hilbert space on $\mathbb{K}$ and $T : D(T) \rightarrow H$ be a linear operator.

1. An element $z \in \mathbb{K}$ is called an *eigenvalue* of T if $N(zI - T) \neq \{0\}$.
2. If $z \in \mathbb{K}$ is an eigenvalue of T , $N(zI - T)$ is called the *eigenspace* of z and the elements of $N(zI - T)$ are called *eigenvectors*. The dimension of $N(zI - T)$ is called the *multiplicity* of the eigenvalue.
3. The set $\rho(T) = \{z \in \mathbb{K} \mid zI - T \text{ is injective, } R(zI - T) = H \text{ and } (zI - T)^{-1} \text{ is bounded}\}$ is called the *resolvent set* of T . If $z \in \rho(T)$, $R(z, T) = (zI - T)^{-1}$ is called the *resolvent of T at point z* .
4. The set $\sigma(T) = \mathbb{K} \setminus \rho(T)$ is called the *spectrum* of T . Notice that any eigenvalue of T lies in the spectrum.

Remark 2.29 (About the resolvent set)

Suppose that T is not closed. Then $\rho(T) = \emptyset$.

Assume not and let $z \in \rho(T)$. Since T is not closed, $zI - T$ is not closed and so $R(z, T)$ is not closed as well. But $R(z, T) : H \rightarrow H$ is bounded, which then contradicts the Closed Graph Theorem.

Note that by Proposition 2.22, a self-adjoint operator is closed.

Example 2.3 (Spectrum of the differentiation operator)

Consider the real Hilbert space

$$H = \mathcal{L}^2([0, 1]) = \{f : [0, 1] \rightarrow \mathbb{R} \mid |f|^2 \text{ is Lebesgue integrable on } [0, 1]\}$$

endowed with the $\mathcal{L}^2$ -scalar product

$$\langle g, f \rangle = \int_0^1 g(x)f(x)dx.$$

Consider the differentiation operator

$$\begin{aligned} T : D(T) \subset \mathcal{L}^2([0, 1]) &\rightarrow \mathcal{L}^2([0, 1]) \\ f(x) &\mapsto f'(x) \end{aligned}$$

where $D(T) \subset H$ is the set of classes for which we can find an absolutely continuous representative $f : [0, 1] \rightarrow \mathbb{R}$ such that $f, f' \in \mathcal{L}^2([0, 1])$ and $f(0) = 0$. Then $\rho(T) = \mathbb{R}$ and $\sigma(T) = \emptyset$.

Indeed, let $\lambda \in \mathbb{R}$. First, we show that $\lambda I - T$ is injective and $R(\lambda I - T) = H$. Let $g \in \mathcal{L}^2([0, 1])$. We try to find $f \in \mathcal{L}^2([0, 1])$ satisfying

$$\lambda f(x) - f'(x) = g(x), \quad f(0) = 0$$

Multiplying by $-e^{-\lambda x}$, we get

$$\frac{d}{dx} [-e^{-\lambda x} f] = e^{-\lambda x} \lambda f(x) - e^{-\lambda x} f'(x) = -e^{-\lambda x} g(x).$$

Integrating this, we find

$$f(x) = e^{\lambda x} \int_0^x e^{-\lambda t} g(t) dt,$$

where there is no constant since $f(0) = 0$. We can check that this is indeed a solution in $D(T)$ to our equation (which is unique from the way we derived it).

Now, it remains to show that the operator

$$\begin{aligned} S : \mathcal{L}^2([0, 1]) &\rightarrow D(T) \\ g(x) &\mapsto e^{\lambda x} \int_0^x e^{-\lambda t} g(t) dt \end{aligned}$$

is bounded. This is clear since

$$\begin{aligned} \|Sg\|_{\mathcal{L}^2}^2 &= \int_0^1 e^{2\lambda x} \left(\int_0^x e^{-\lambda t} g(t) dt \right)^2 dx \\ &\leq (1 + e^{2\lambda}) \int_0^1 \left(\int_0^x e^{-2\lambda t} dt \right) \left(\int_0^x g^2(t) dt \right) dx \\ &\leq (1 + e^{2\lambda}) \cdot \|e^{-2\lambda x}\|_{\mathcal{L}^2}^2 \cdot \|g\|_{\mathcal{L}^2}^2 \end{aligned}$$

where we used the Cauchy-Schwarz Inequality for the second inequality.

We saw in Example 2.1 that T is not a self-adjoint operator. An easy consequence of the Spectral Theorem is that a self-adjoint operator always has non-empty spectrum, so that would be another way to prove that T is not self-adjoint.

Theorem 2.30 (First Resolvent Identity)

Let $(H, \langle \cdot, \cdot \rangle)$ be an Hilbert space and $T : D(T) \rightarrow H$ be a linear operator. For all $z, z' \in \rho(T)$, the first resolvent identity holds :

$$R(z, T) - R(z', T) = (z' - z)R(z, T)R(z', T) = (z' - z)R(z', T)R(z, T).$$

Proposition 2.31

Let $(H, \langle \cdot, \cdot \rangle)$ be an Hilbert space and $T : D(T) \rightarrow H$ be a linear operator. For all $z, z' \in \rho(T)$, $z \neq z'$, the following identity holds :

$$zI - T = ((z' - z)^{-1}I - R(z', T))(z'I - T)(z' - z)I$$

Corollary 2.32 (On the spectrum of the resolvent)

Let $(H, \langle \cdot, \cdot \rangle)$ be an Hilbert space and $T : D(T) \rightarrow H$ be a linear operator such that $\rho(T) \neq \emptyset$. Let $z' \in \rho(T)$ be fixed.

Then $z \in \rho(T)$, $z \neq z'$, if and only if $(z' - z)^{-1} \in \rho(R(z', T))$, $z \neq z'$.

In particular, $\sigma(R(z', T)) \setminus \{0\} = \{(z' - z)^{-1} \mid z \in \sigma(T)\}$.

Proposition 2.33 (On the spectrum of a complex self-adjoint operator)

Let $(H, \langle \cdot, \cdot \rangle)$ be an Hilbert space on $\mathbb{C}$ and $T : D(T) \rightarrow H$ be a self-adjoint operator.

If $z \in \mathbb{C}$, $\Im z \neq 0$, we have $\|(zI - T)x\| \geq |\Im z| \cdot \|x\| > 0 \ \forall x \in H$.

In particular, $z \in \rho(T)$, $\|R(z, T)\| \leq \frac{1}{|\Im z|}$ and the spectrum of T is real.

Proof.

Since T is self-adjoint, we have $\langle x, Tx \rangle = \langle Tx, x \rangle = \overline{\langle x, Tx \rangle} \in \mathbb{R} \ \forall x \in D(T)$.

By the Cauchy-Schwarz inequality,

$$\|(zI - T)x\| \cdot \|x\| \geq |\langle x, (zI - T)x \rangle| \geq |\Im \langle x, (zI - T)x \rangle| = |\Im z| \cdot \|x\|^2 \ \forall x \in D(T).$$

This implies $\|(zI - T)x\| \geq |\Im z| \cdot \|x\| \ \forall x \in D(T)$ (*) and so the first part of the statement follows.

This inequality also implies that $zI - T$ is injective. If $(zI - T)x_1 = (zI - T)x_2$, then $(zI - T)(x_1 - x_2) = 0$ and $0 = \|(zI - T)(x_1 - x_2)\| \geq |\Im z| \cdot \|x_1 - x_2\| \geq 0$, so $x_1 = x_2$.

By Theorem 2.20, $R(zI - T)$ is dense. Since T is self-adjoint, $G(T)$ and $G(zI - T)$ are closed. Thus, $R(z, T)$ is a densely defined and closed inverse of $zI - T$. Furthermore, it is bounded by $1/|\Im z|$ because of (*). By the Closed Graph Theorem, $D(R(z, T))$ is closed, i.e. $D(R(z, T)) = H$.

□

Theorem 2.34 (von Neumann Series)

Let $(X, \|\cdot\|_X)$ be a Banach space and $T : X \rightarrow X$ be a bounded linear operator such that $\|T\| < 1$.

Then the sequence $(\sum_{i=0}^n T^i)_{n=1}^\infty$ converges in $(B(X, X), \|\cdot\|)$. Additionally, its limit, denoted by S , satisfies

$$(I - T) \circ S = S \circ (I - T) = I,$$

i.e. S is invertible in $B(X, X)$ with inverse $I - T$ and $\|S\| \leq \frac{1}{1 - \|T\|}$.

Corollary 2.35 (Resolvent set is open)

Let $(H, \langle \cdot, \cdot \rangle)$ be an Hilbert space and $T : D(T) \rightarrow H$ be a linear operator. Then $\rho(T)$ is open and $\sigma(T)$ is closed.

More precisely, if $\rho(T) \neq \emptyset$, then for any $z_0 \in \rho(T)$ and $z \in \mathbb{K}$ such that $|z - z_0| < \|R(z_0, T)\|^{-1}$, we have $z \in \rho(T)$ and

$$R(z, T) = \sum_{n=0}^{\infty} (z_0 - z)^n R(z_0, T)^{n+1}.$$

Proof.

If $\rho(T) = \emptyset$, $\rho(T)$ is trivially open. Assume not and let $z_0 \in \rho(T)$.

Let $z \in \mathbb{K}$ be such that $|z - z_0| < \|R(z_0, T)\|^{-1}$, $z \neq z_0$. Apply Theorem 2.34 to $(z_0 - z)R(z_0, T) \in B(H, H)$. Thus, the operator

$$S = \sum_{n=0}^{\infty} (z_0 - z)^n R(z_0, T)^n \in B(H, H)$$

is well-defined. Consider the operator

$$\hat{S} = R(z_0, T)S = \sum_{n=0}^{\infty} (z_0 - z)^n R(z_0, T)^{n+1} \in B(H, H)$$

Denote by $\hat{S}_p$ the p -th partial sum associated to $\hat{S}$. Then

$$\begin{aligned} (zI - T)\hat{S}_p &= (-(z_0 - z)I + (z_0I - T))\hat{S}_p \\ &= -\sum_{n=0}^p (z_0 - z)^{n+1} R(z_0, T)^{n+1} + \sum_{n=0}^p (z_0 - z)^n R(z_0, T)^n \\ &= I - \sum_{n=0}^{p+1} (z_0 - z)^n R(z_0, T)^n + \sum_{n=0}^p (z_0 - z)^n R(z_0, T)^n \\ &\xrightarrow{p \rightarrow +\infty} I - S + S = I \end{aligned}$$

i.e. $(zI - T)\hat{S} = I$.

Similarly, we can show that $\hat{S}(zI - T) = I$. Therefore, $(zI - T)$ has an inverse $\hat{S} = R(z, T) \in B(H, H)$. $\square$

Corollary 2.36

Let $(H, \langle \cdot, \cdot \rangle)$ be an Hilbert space and $T : D(T) \rightarrow H$ be a linear operator.

If $\sigma(T) \neq \emptyset$, for any $z_0 \in \rho(T)$, we have $d(z_0, \sigma(T)) \geq \|R(z_0, T)\|^{-1}$.

Proof.

If $z_0 \in \rho(T)$, for any $z \in \mathbb{K}$ such that $|z - z_0| < \|R(z_0, T)\|^{-1}$, we have $z \in \rho(T)$. Therefore, $|z - z_0| \geq \|R(z_0, T)\|^{-1}$ for all $z \in \sigma(T)$, so $d(z_0, \sigma(T)) \geq \|R(z_0, T)\|^{-1}$ by taking the infimum over all $z \in \sigma(T)$. $\square$

Proposition 2.37 (Spectrum of a bounded operator is compact)

Let $(H, \langle \cdot, \cdot \rangle)$ be an Hilbert space and $T : H \rightarrow H$ be a bounded operator. Then $\sigma(T)$ is compact (closed and bounded).

More precisely, if $\lambda \in \sigma(T)$, then $|\lambda| \leq \|T\|$. Equivalently, if $\lambda \in \mathbb{K}$ is such that $|\lambda| > \|T\|$, then $\lambda \in \rho(T)$.

Proof.

Let $\lambda \in \mathbb{K}$ with $|\lambda| > \|T\|$. Then $\|\frac{1}{\lambda}T\| < 1$ so $I - \frac{1}{\lambda}T$ has a bounded inverse defined on the whole H by Theorem 2.34. Therefore, $\lambda(I - \frac{1}{\lambda}T) = \lambda I - T$ has a bounded inverse defined on the whole H and so $\lambda \in \rho(T)$. $\square$

Definition 2.12 (Analyticity)

Let U be a non-empty open subset of $\mathbb{C}$ and let $(X, \|\cdot\|_X)$ be a Banach space. A function $F : U \rightarrow X$ is called *complex analytic* if for every $z_0 \in U$, there exists a $r > 0$ and a sequence $(f_n)_{n=1}^\infty \subset X$ such that

$$F(z) = \sum_{n=0}^{\infty} (z - z_0)^n f_n \quad \forall z \in U \text{ s.t. } |z - z_0| < r$$

i.e.

$$\forall z \in U \text{ s.t. } |z - z_0| < r : \|F(z) - \sum_{k=0}^n (z - z_0)^k f_k\|_X \xrightarrow{n \rightarrow +\infty} 0.$$

The quantity $R = R(z_0) = \left(\limsup_{n \rightarrow +\infty} \|f_n\|_X^{1/n} \right)^{-1}$ is called the *radius of convergence* of the series around z_0 . One can show that the series converges uniformly in $\{z \in U \mid |z - z_0| < R\}$ and diverges if $z \in U$ is such that $|z - z_0| > R$.

Theorem 2.38 (Resolvent is analytic)

Let $(H, \langle \cdot, \cdot \rangle)$ be an Hilbert space on $\mathbb{C}$, $T : D(T) \rightarrow H$ be a linear operator such that $\rho(T) \neq \emptyset$ and let $x, y \in H$ be fixed. Then the following functions are analytic :

$$\begin{aligned} R(\cdot, T) : \rho(T) &\rightarrow B(H, H), & z &\mapsto R(z, T) \\ R(\cdot, T)x : \rho(T) &\rightarrow H, & z &\mapsto R(z, T)x \\ \langle y, R(\cdot, T)x \rangle : \rho(T) &\rightarrow \mathbb{C}, & z &\mapsto \langle y, R(z, T)x \rangle \end{aligned}$$

In particular, $\langle y, R(\cdot, T)x \rangle$ is holomorphic in the usual sense.

Proof. This follows from Corollary 2.35. $\square$

Chapter 3

Abstract Measure and Integration Theory

Our version of the Spectral Theorem will not assume boundedness of the operator. Therefore, defining a functional calculus on the set $C^0(\sigma(T))$ by the mean of the Weierstrass Approximation Theorem do not work anymore ($\sigma(T)$ is not necessarily compact). The functional calculus for unbounded operators will be defined using abstract integration.

For this purpose, we recall the fundamental theory of Measure and Integration. See [Sha05] (Chapter 6) for more details.

3.1 Abstract Measure Theory

Notation (Extended Real Numbers)

We denote by $\mathbb{R} = \mathbb{R} \cup \{\pm\infty\}$ the extended real numbers with the usual arithmetic operations and the convention $0 \times \pm\infty = 0$.

Definition 3.1 (Outer Measure and Measure)

An *outer measure* on a topological space X is a function $\mu_* : \mathcal{P}(X) \rightarrow \mathbb{R}_{\geq 0} \cup \{+\infty\}$ satisfying :

1. $\mu_*(\emptyset) = 0$
2. If $A_1 \subset A_2$, then $\mu_*(A_1) \leq \mu_*(A_2)$
- 3.

$$\mu_*\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu_*(A_i)$$

A set $A \subset X$ is said to be *Carathéodory measurable* with respect to μ if

$$\mu_*(S) = \mu_*(A \cap S) + \mu_*(A^c \cap S) \quad \forall S \subset X.$$

Denote by $\mathcal{M}$ the set of μ -measurable subsets of X . Then $\mathcal{M}$ is a σ -algebra (i.e. it is non empty, closed by complementary and closed by countable union) and the function $\mu = \mu_{*|\mathcal{M}} : \mathcal{M} \rightarrow \mathbb{R}_{\geq 0} \cup \{+\infty\}$ is called a *measure* on $\mathcal{M}$. This function satisfies

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

for any countable family of disjoint sets of $\mathcal{M}$.

We will use the following terminology :

1. A triple $(X, \mathcal{M}', \mu|_{\mathcal{M}'})$, where $\mathcal{M}' \subset \mathcal{M}$ is a σ -algebra, is called a *measure space*.
2. The space is called *complete* if $\forall E \in \mathcal{M}'$ such that $\mu(E) = 0 : A \subset E \Rightarrow A \in \mathcal{M}'$.
3. The space is called *Borel* if all Borel sets of X (i.e. the family of countable union, countable intersection and complement of open subsets of X) are in $\mathcal{M}'$.
4. The space is called *σ -finite* if X is the countable union of measurable sets with finite measure.
5. The measure is called *finite* if $\mu(A) < +\infty \forall A \in \mathcal{M}'$.
6. A subset $A \subset X$ will be called *μ -measurable* if $A \in \mathcal{M}'$.

Proposition 3.1

Let $(X, \mathcal{M}, \mu)$ be a measure space. If A and B are measurable, then

$$\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B).$$

Proposition 3.2 (Continuity of the measure)

Let $(X, \mathcal{M}, \mu)$ be a measure space.

1. If $(A_n)_{n=1}^{\infty} \subset \mathcal{M}$ is an increasing sequence of measurable sets (i.e. $A_n \subset A_{n+1} \forall n \in \mathbb{N}$), then

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow +\infty} \mu(A_n).$$

2. If $(A_n)_{n=1}^{\infty} \subset \mathcal{M}$ is a decreasing sequence of measurable sets (i.e. $A_{n+1} \subset A_n \forall n \in \mathbb{N}$) such that $\mu(A_1) < +\infty$, then

$$\mu\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow +\infty} \mu(A_n).$$

Definition 3.2 (Measurability)

Let $(X, \mathcal{M}, \mu)$ be a measure space.

A function $f : X \rightarrow \overline{\mathbb{R}}$ is said to be *μ -measurable* if the sets $\{x \in X \mid f(x) < s\}$ are μ -measurable $\forall s \in \overline{\mathbb{R}}$.

A function $f : X \rightarrow \mathbb{C}$ is said to be *μ -measurable* if $\Re f$ and $\Im f$ are μ -measurable.

An assertion is said to be true *μ -almost everywhere* (abbreviated a.e.) if there exists a μ -measurable set $N \subset X$ such that $\mu(N) = 0$ and the assertion is true $\forall x \in X \setminus N$.

Definition 3.3 (Stieltjes Measure)

Let $\rho : \mathbb{R} \rightarrow \mathbb{R}$ be a non-decreasing and right-continuous function.

Then ρ defines a unique σ -finite Borel measure on $\mathbb{R}$ (denoted by ρ as well) such that

$$\rho(J) = \begin{cases} \rho(b) - \rho(a) & \text{for } J = (a, b) \\ \rho(b) - \rho(a-) & \text{for } J = [a, b) \\ \rho(b-) - \rho(a) & \text{for } J = (a, b] \\ \rho(b-) - \rho(a-) & \text{for } J = [a, b] \end{cases}$$

where $\rho(a-) = \lim_{t \rightarrow a^-} \rho(t)$. This measure, called a *Lebesgue-Stieltjes measure*, is induced by the outer measure

$$\rho_*(A) = \inf \left\{ \sum_{j=1}^{\infty} (\rho(b_j) - \rho(a_j)) \mid A \subset \bigcup_{j=1}^{\infty} (a_j, b_j], a_j < b_j \in \mathbb{R} \right\}.$$

The standard *Lebesgue measure* is the Stieltjes measure associated to $\rho(x) = x$.

Definition 3.4 (Cumulative Distribution Function)

Let $(\mathbb{R}, \mathcal{M}, \mu)$ be a measure space such that μ is Borel and finite. We define the *Cumulative Distribution Function* of μ as the function $F : \mathbb{R} \rightarrow \mathbb{R}$, $F(x) = \mu((-\infty, x])$.

Then F is a non-negative, non-decreasing, bounded and right-continuous function such that $\lim_{x \rightarrow -\infty} F(x) = 0$. Furthermore, the Stieltjes measure associated to F coincide with μ .

3.2 Abstract Integration Theory

Definition 3.5 (Integral of simple and non-negative functions)

Let $(X, \mathcal{M}, \mu)$ be a σ -finite measure space.

A *simple function* is a function $\phi : X \rightarrow \mathbb{R}$ of the form

$$\phi = \sum_{i=1}^n a_i \chi_{A_i},$$

where $A_1, \dots, A_n$ is a sequence of disjoint measurable subsets of X and $a_i \in \mathbb{R} \forall i = 1, \dots, n$. If $X = \mathbb{R}$ and the A_i 's are disjoint bounded intervals, ϕ is called a *step function*.

The *integral* of a simple function with respect to the measure μ is defined as

$$\int_X \phi d\mu = \sum_{i=1}^n a_i \mu(A_i)$$

The integral of a measurable, non-negative, function $f : X \rightarrow \overline{\mathbb{R}}$ is defined as

$$\int_X f d\mu = \sup \left\{ \int_X \phi d\mu \mid 0 \leq \phi \leq f, \phi \text{ is a simple function} \right\}.$$

The integral of a measurable, non-negative, function $f : X \rightarrow \overline{\mathbb{R}}$ on a measurable subset $A \subset X$ is defined as

$$\int_A f d\mu = \int_X f \cdot \chi_A d\mu.$$

Definition 3.6 (Integral and Integrability)

Let $(X, \mathcal{M}, \mu)$ be a σ -finite measure space. Let A be measurable subset of X .

We say that a measurable function $f : X \rightarrow \overline{\mathbb{R}}$ is μ -*integrable* on A if the functions $f^+ = \max\{0, f(x)\}$ and $f^- = \max\{0, -f(x)\}$ satisfy

$$\int_A f^+ d\mu < +\infty, \int_A f^- d\mu < +\infty.$$

In this case, the *integral* of f on A with respect to the measure μ is defined as

$$\int_A f d\mu = \int_A f^+ d\mu - \int_A f^- d\mu$$

The integral of a complex-valued function is defined in a similar fashion.

Theorem 3.3 (Monotone Convergence Theorem)

Let $(\mathbb{R}, \mathcal{M}, \mu)$ be a measure space.

1. Let $(f_n)_{n=1}^\infty$, $f_n : \mathbb{R} \rightarrow \overline{\mathbb{R}}_{\geq 0}$, be a non-decreasing sequence of non-negative measurable functions converging pointwisely almost everywhere to $f : \mathbb{R} \rightarrow \overline{\mathbb{R}}_{\geq 0}$. Then f is measurable and

$$\lim_{n \rightarrow +\infty} \int f_n d\mu = \int \lim_{n \rightarrow +\infty} f_n d\mu = \int f d\mu$$

2. Let $(f_n)_{n=1}^\infty$, $f_n : \mathbb{R} \rightarrow \overline{\mathbb{R}}$, be a non-increasing sequence of real-valued measurable functions such that f_1 is integrable and the sequence (f_n) converges pointwisely almost everywhere to $f : \mathbb{R} \rightarrow \overline{\mathbb{R}}$. Then f is measurable and

$$\lim_{n \rightarrow +\infty} \int f_n d\mu = \int \lim_{n \rightarrow +\infty} f_n d\mu = \int f d\mu$$

Theorem 3.4 (Fatou's Lemma)

Let $(\mathbb{R}, \mathcal{M}, \mu)$ be a measure space. Let $(f_n)_{n=1}^\infty$, $f_n : \mathbb{R} \rightarrow \overline{\mathbb{R}}_{\geq 0}$, be a sequence of non-negative measurable functions. Define the function $f : \mathbb{R} \rightarrow \overline{\mathbb{R}}_{\geq 0}$ by

$$f(x) = \liminf_{n \rightarrow +\infty} f_n(x), \quad x \in \mathbb{R}.$$

Then f is measurable and

$$\int f d\mu = \int \liminf_{n \rightarrow +\infty} f_n d\mu \leq \liminf_{n \rightarrow +\infty} \int f_n d\mu.$$

Theorem 3.5 (Dominated Convergence Theorem)

Let $(\mathbb{R}, \mathcal{M}, \mu)$ be a measure space. Let $(f_n)_{n=1}^\infty$, $f_n : \mathbb{R} \rightarrow \overline{\mathbb{R}}$, be a sequence of measurable functions converging pointwisely almost everywhere to $f : \mathbb{R} \rightarrow \overline{\mathbb{R}}$. Assume there exists an integrable function $g : \mathbb{R} \rightarrow \overline{\mathbb{R}}_{\geq 0}$ such that $0 \leq |f_n| \leq g$ almost everywhere for all $n \in \mathbb{N}$. Then f_n is integrable for all $n \in \mathbb{N}$, f is integrable and

$$\lim_{n \rightarrow +\infty} \int f_n d\mu = \int \lim_{n \rightarrow +\infty} f_n d\mu = \int f d\mu.$$

Theorem 3.6 (Relation between Lebesgue and Riemann Integral)

Consider the standard Lebesgue measure μ .

Let $-\infty < a < b < +\infty$ and let $f : [a, b] \rightarrow \mathbb{R}$ be Riemann integrable. Then f is Lebesgue integrable and

$$\int_{[a,b]}^{\mathcal{R}} f(x) dx = \int_{[a,b]}^{\mathcal{L}} f d\mu,$$

where the integral on the left-hand side is the standard Riemann integral and the integral on the right-hand side is the standard Lebesgue integral.

Proof. [Sha05] (Theorem 1.5 p.57) □

Corollary 3.7

Let $f : [a, +\infty) \rightarrow \mathbb{R}$ be a function that is Riemann integrable on any closed interval $[a, b] \subset [a, +\infty)$ and Lebesgue integrable on $[a, +\infty)$. Then the improper Riemann integral

$$\lim_{t \rightarrow +\infty} \int_{[a, t]}^R f(x) dx$$

exists and

$$\lim_{t \rightarrow +\infty} \int_{[a, t]}^R f(x) dx = \int_{[a, +\infty)}^L f d\mu$$

Remark 3.8

There exists functions that are improperly Riemann integrable, but not Lebesgue integrable. For example, the function $f(x) = \sin(x)/x$ is improperly Riemann integrable on $[0, +\infty)$ (using, for example, the Abel-Dirichlet Test for convergence of improper Riemann integrals) but not Lebesgue integrable.

Definition 3.7 (Pushforward measure)

Let $(X_1, \mathcal{M}_1, \mu)$ be a measure space with associated outer measure μ_* , X_2 be a topological space and $F : X_1 \rightarrow X_2$ be any function.

Let $\mathcal{M}_2 = F(\mathcal{M}_1) = \{B \subset X_2 \mid F^{-1}(B) \in \mathcal{M}_1\}$ be the *pushforward σ -algebra* on X_2 . We define an outer measure $F\#\mu_*$ on X_2 as

$$F\#\mu_*(B) = \mu_*(F^{-1}(B)), \quad B \subset X_2$$

Then $(X_2, \mathcal{M}_2, (F\#\mu_*)|_{\mathcal{M}_2})$ defines a measure space on X_2 , i.e. every element of $\mathcal{M}_2$ is Carathéodory measurable with respect to $F\#\mu_*$, and $F\#\mu := (F\#\mu_*)|_{\mathcal{M}_2}$ is called the *pushforward measure* on $(X_2, \mathcal{M}_2)$.

Theorem 3.9 (Change of variables)

Let $(X_1, \mathcal{M}_1, \mu)$ be a measure space, X_2 be a topological space and $F : X_1 \rightarrow X_2$ be any function. Let $g : X_2 \rightarrow \mathbb{R}$ be a $(F\#\mu)$ -integrable function.

Then $g \circ F$ is μ -integrable and

$$\int_{X_2} g d(F\#\mu) = \int_{X_1} g \circ F d\mu.$$

Definition 3.8

Let $(X, \mathcal{M}, \mu)$ be a measure space with associated outer measure μ_* and $f : X \rightarrow \mathbb{R}$ be a μ -measurable non-negative function.

We define an outer measure $f\mu_*$ on X as

$$f\mu_*(B) = \inf \left\{ \sum_{i=1}^{\infty} \int_{A_i} f d\mu \mid (A_i)_{i=1}^{\infty} \subset \mathcal{M}, B \subset \bigcup_{i=1}^{\infty} A_i \right\}, \quad B \subset X$$

Then $(X, \mathcal{M}, (f\mu_*)|_{\mathcal{M}})$ defines a measure space on X , i.e. every element of $\mathcal{M}$ is Carathéodory measurable with respect to $f\mu_*$. Furthermore,

$$f\mu_*(B) = \int_B f d\mu \quad \forall B \in \mathcal{M}.$$

We write $f\mu := (f\mu_*)|_{\mathcal{M}}$.

Proposition 3.10

Let $(X, \mathcal{M}, \mu)$ be a measure space and $f : X \rightarrow \mathbb{R}$ be a μ -measurable non-negative function. For any μ -measurable function $g : X \rightarrow \overline{\mathbb{R}}$, we have

$$\int_X gf \, d\mu = \int_X g \, d(f\mu).$$

3.3 $\mathcal{L}^p$ spaces

We need the four additional results related to $\mathcal{L}^p$ spaces, whose proofs can be found in [Rud86] (pp.41-47, 65-70).

Theorem 3.11 ($\mathcal{L}^p$ is a complete space)

Let $(X, \mathcal{M}, \mu)$ be a measure space. For $1 \leq p < +\infty$, we denote by $\mathcal{L}_{\mathbb{K}}^p(X, \mathcal{M}, \mu)$ the set $\{f : X \rightarrow \mathbb{K} \mid |f|^p \text{ is integrable}\}$ (where f and g are identified if $f = g$ almost everywhere). Then $\mathcal{L}_{\mathbb{K}}^p(X, \mathcal{M}, \mu)$ is a $\mathbb{K}$ -vector space.

If we equip $\mathcal{L}_{\mathbb{K}}^p(X, \mathcal{M}, \mu)$ with the norm

$$\|f\|_{\mathcal{L}^p} = \left(\int |f(t)|^p d\mu \right)^{1/p},$$

then $\mathcal{L}_{\mathbb{K}}^p(X, \mathcal{M}, \mu)$ is complete.

Notation

If the measure space $(X, \mathcal{M}, \mu)$ and $\mathbb{K}$ are clear from the context, we write $\mathcal{L}^p(\mu) := \mathcal{L}_{\mathbb{K}}^p(X, \mathcal{M}, \mu)$.

Theorem 3.12

Let $(X, \mathcal{M}, \mu)$ be a measure space and let $1 \leq p < +\infty$. If $(f_n)_{n=1}^{\infty} \subset \mathcal{L}_{\mathbb{K}}^p(X, \mathcal{M}, \mu)$ converges to f in $\mathcal{L}_{\mathbb{K}}^p(X, \mathcal{M}, \mu)$, then there exists a subsequence $(f_{n_k})_{k=1}^{\infty}$ of (f_n) such that (f_{n_k}) converges pointwisely to f almost everywhere.

Definition 3.9 (Radon measure)

Let $(X, \mathcal{M}, \mu)$ be a measure space, where X is a locally compact Hausdorff space. The measure μ is called a *Radon measure* if it satisfies

1. $\mu(K) < +\infty$ for all compact subsets $K \subset X$;
2. $\mu(E) = \inf\{\mu(V) \mid E \subset V, V \text{ open}\} \forall E \in \mathcal{M}$;
3. $\mu_*(E) = \sup\{\mu_*(K) \mid K \subset E, K \text{ compact}\} \forall E \in \mathcal{M}$ such that $\mu(E) < +\infty$ and for all open subsets $E \subset X$.

Remark 3.13

If X is a compact topological space, a Radon measure μ on X is necessarily finite.

Theorem 3.14 (About density of step functions)

Let $(\mathbb{R}, \mathcal{M}, \mu)$ be a complete measure space and μ be a Radon measure. Then

1. Continuous compactly supported functions are dense in $\mathcal{L}_{\mathbb{K}}^p(\mathbb{R}, \mathcal{M}, \mu) \forall 1 \leq p < +\infty$.

2. Step functions are dense in $\mathcal{L}_{\mathbb{R}}^p(\mathbb{R}, \mathcal{M}, \mu) \forall 1 \leq p < +\infty$.

Remark 3.15

Any Stieltjes measure on $\mathbb{R}$ satisfies the assumptions of Theorem 3.14.

Theorem 3.16 (Riesz-Markov-Kakutani Representation Theorem)

Let X be a locally compact Hausdorff space and let $C_c(X)$ be the vector space of continuous and compactly supported functions from X to $\mathbb{R}$ endowed with the sup norm.

Let $\Lambda : C_c(X) \rightarrow \mathbb{R}$ be a linear functional satisfying $\Lambda(f) \geq 0 \forall f \in C_c(X)$ such that $f(X) \subset [0, +\infty)$. Then there exists a complete Borel measure space $(X, \mathcal{M}, \mu)$, where μ is a Radon measure, such that

$$\Lambda(f) = \int_X f d\mu \quad \forall f \in C_c(X).$$

Furthermore, μ is the only Radon measure on $\mathcal{M}$ with this property.

3.4 Weak convergence of measures

The following results about sequences of measures will be helpful to prove some representation theorems for analytic functions.

Theorem 3.17 (Layer-Cake Representation)

Let $(\mathbb{R}, \mathcal{M}, \mu)$ be a measure space. Let $f \in \mathcal{L}^1(\mu)$ be a positive and finite-valued function. Then

$$\int_{\mathbb{R}} f d\mu = \int_{(0, +\infty)} \mu(\{x \in \mathbb{R} \mid f(x) > t\}) dt$$

where the integral on the right-hand side is to be interpreted as a standard Lebesgue integral.

Proof.

For $n, k \in \mathbb{N}$, define $E_{n,k} = \{x \in \mathbb{R} \mid f(x) > \frac{k}{2^n}\}$. Since f is measurable, each $E_{n,k}$ is measurable. Notice that if $k = 2j$, then $E_{n+1,k} = E_{n,j}$ and if $k = 2j - 1$, then $E_{n+1,k} \supset E_{n,j}$.

For $n \in \mathbb{N}$, set

$$f_n = \frac{1}{2^n} \sum_{k=1}^{4^n} \chi_{E_{n,k}}.$$

Then $(f_n)_{n=1}^{\infty}$ is an increasing sequence because

$$\begin{aligned} f_{n+1} &= \frac{1}{2^{n+1}} \sum_{k=1}^{4^{n+1}} \chi_{E_{n+1,k}} = \frac{1}{2^{n+1}} \left(\sum_{j=1}^{2 \cdot 4^n} \chi_{E_{n+1,2j}} + \sum_{j=1}^{2 \cdot 4^n} \chi_{E_{n+1,2j-1}} \right) \\ &\geq \frac{1}{2^{n+1}} \left(\sum_{j=1}^{2 \cdot 4^n} \chi_{E_{n,j}} + \sum_{j=1}^{2 \cdot 4^n} \chi_{E_{n,j}} \right) \geq f_n \quad \forall n \in \mathbb{N}. \end{aligned}$$

Furthermore, (f_n) converges pointwisely to f . Let $x \in \mathbb{R}$. If $f(x) = 0$, this is clear. Otherwise, let $n \in \mathbb{N}$ be large enough so that $\frac{1}{2^n} < f(x) \leq 2^n$. Then there exists a unique

$k = k(n) \in \mathbb{N}$, $1 \leq k \leq 4^n$, such that $\frac{k}{2^n} < f(x) \leq \frac{k+1}{2^n}$. For such a n and $k(n)$, we have

$$f_n(x) = \frac{1}{2^n} \sum_{j=1}^{k(n)} \chi_{E_{n,j}} = \frac{k(n)}{2^n} < f(x) \leq \frac{k(n)}{2^n} + \frac{1}{2^n} = f_n(x) + \frac{1}{2^n},$$

so the pointwise convergence follows.

By the Monotone Convergence Theorem,

$$\lim_{n \rightarrow +\infty} \int f_n d\mu = \int f d\mu.$$

Since (f_n) is an increasing sequence converging pointwisely to f , we have

$$\{x \in \mathbb{R} \mid f(x) > t\} = \bigcup_{n=1}^{\infty} \{x \in \mathbb{R} \mid f_n(x) > t\}$$

and $\{x \in \mathbb{R} \mid f_n(x) > t\}, n \in \mathbb{N}$, defines an increasing sequence of measurable sets. Therefore,

$$\mu(\{x \in \mathbb{R} \mid f(x) > t\}) = \lim_{k \rightarrow +\infty} \mu(\{x \in \mathbb{R} \mid f_n(x) > t\})$$

by continuity of the measure. Note that for any $n \in \mathbb{N}$,

$$\mu(\{x \in \mathbb{R} \mid f_n(x) > t\}) : \mathbb{R} \rightarrow \overline{\mathbb{R}}_{\geq 0}$$

is a monotone function of t , so it is Lebesgue measurable.

By the Monotone Convergence Theorem,

$$\lim_{n \rightarrow +\infty} \int \mu(\{x \in \mathbb{R} \mid f_n(x) > t\}) dt = \int \mu(\{x \in \mathbb{R} \mid f(x) > t\}) dt.$$

To conclude, it suffices to prove that

$$\int g d\mu = \int \mu(\{x \in \mathbb{R} \mid g(x) > t\}) dt$$

for a finite-valued and integrable function g that is a positive linear combination of characteristic functions of measurable sets.

Let $R(g) = \{a_1 < \dots < a_n\} \subset \mathbb{R}_{\geq 0}$ be the range of g , $E_i = \{x \in \mathbb{R} \mid g(x) = a_i\}$, $i = 1, \dots, n$, $E_i \cap E_j = \emptyset \forall i \neq j$. We write g in canonical form

$$g = \sum_{i=1}^n a_i \chi_{E_i},$$

i.e. g is a simple function when written in canonical form. Then

$$\int g d\mu = \sum_{i=1}^n a_i \mu(E_i)$$

and

$$\int_{(0,+\infty)} g d\mu = \int \mu(\{x \in \mathbb{R} \mid g(x) > t\}) dt = \int_{(0,a_1)} \dots + \sum_{i=1}^{n-1} \int_{[a_i, a_{i+1})} \dots + \int_{[a_{n-1}, +\infty)} \dots$$

$$\begin{aligned}
&= a_1 \mu\left(\bigcup_{j=1}^n E_j\right) + \sum_{i=1}^{n-1} (a_{i+1} - a_i) \mu\left(\bigcup_{j=i+1}^n E_j\right) + 0 \\
&= a_1 \sum_{j=1}^n \mu(E_j) + \sum_{i=1}^{n-1} \sum_{j=i+1}^n (a_{i+1} - a_i) \mu(E_j) \\
&= a_1 \sum_{j=1}^n \mu(E_j) + \sum_{j=1}^{n-1} \sum_{i=1}^j (a_{i+1} - a_i) \mu(E_{j+1}) \\
&= a_1 \sum_{j=1}^n \mu(E_j) + \sum_{j=1}^{n-1} (a_{j+1} - a_1) \mu(E_{j+1}) = \sum_{i=1}^n a_i \mu(E_i).
\end{aligned}$$

□

Remark 3.18

The sequence (f_n) constructed in the above proof is the usual sequence constructed to prove that simple functions are dense in $\mathcal{L}^1(\mu)$.

Remark 3.19

A three-line proof of this theorem can be done using Fubini Theorem.

Definition 3.10 (Weak convergence of measures)

Let $\mathfrak{M}_{\mathcal{B}}$ be the space of finite measures on $(\mathbb{R}, \mathcal{B})$, where $\mathcal{B}$ is the σ -algebra of Borel sets of $\mathbb{R}$.

We say that a sequence $(\mu_n)_{n=1}^{\infty} \subset \mathfrak{M}_{\mathcal{B}}$ *converges weakly* to a measure $\mu \in \mathfrak{M}_{\mathcal{B}}$ if

$$\lim_{n \rightarrow +\infty} \int_{\mathbb{R}} f d\mu_n = \int_{\mathbb{R}} f d\mu$$

for all $f \in C_b(\mathbb{R})$, the set of all bounded continuous functions from $\mathbb{R}$ to $\mathbb{R}$.

Proposition 3.20 (Sufficient criterion for weak convergence)

Let $\mathcal{I} \subset \mathcal{B}$ be a family of open sets that is non empty and closed by finite intersection.

Assume that any open set $U \in \mathcal{B}$ is the countable union of elements of $\mathcal{I}$.

Let $(\mu_n)_{n=1}^{\infty}$ and $\mu \in \mathfrak{M}_{\mathcal{B}}$. If

$$\lim_{n \rightarrow +\infty} \mu_n(I) = \mu(I) \quad \forall I \in \mathcal{I},$$

then (μ_n) converges weakly to μ .

Proof.

Let $I_1, I_2 \in \mathcal{I}$. Then $I_1 \cap I_2 \in \mathcal{I}$ and

$$\begin{aligned}
\mu(I_1 \cup I_2) &= \mu(I_1) + \mu(I_2) - \mu(I_1 \cap I_2) \\
&= \lim_{n \rightarrow +\infty} \mu_n(I_1) + \lim_{n \rightarrow +\infty} \mu_n(I_2) - \lim_{n \rightarrow +\infty} \mu_n(I_1 \cap I_2) \\
&= \lim_{n \rightarrow +\infty} \mu_n(I_1 \cup I_2)
\end{aligned}$$

Thus, we can assume without loss of generality that $\mathcal{I}$ is also closed by finite union.

Now, let $U = \bigcup_{n=1}^{\infty} I_n$ be an open set, $I_n \in \mathcal{I} \forall n \in \mathbb{N}$. Then $A_k = \bigcup_{n=1}^k I_n, k \in \mathbb{N}$, defines an increasing sequence of measurable sets such that $U = \bigcup_{k=1}^{\infty} A_k$. Therefore,

$$\rho(U) = \lim_{k \rightarrow +\infty} \rho(A_k)$$

for any Borel measure ρ by continuity of a measure. Equivalently, for any Borel measure ρ and for any $\epsilon > 0$, there exists $K = K(\rho, \epsilon) \in \mathbb{N}$ such that $\forall k \geq K$:

$$\rho(U) \leq \rho(A_k) + \epsilon$$

Applying this to $\rho = \mu$ and an arbitrary $\epsilon > 0$, we get

$$\mu(U) \leq \mu(A_K) + \epsilon = \lim_{n \rightarrow \infty} \mu_n(A_K) + \epsilon \leq \liminf_{n \rightarrow +\infty} \mu_n(U) + \epsilon$$

Since ϵ was arbitrary, we deduce

$$\mu(U) \leq \liminf_{n \rightarrow +\infty} \mu_n(U).$$

Let $f \in C_b(\mathbb{R})$. By applying a translation to f , we can assume that $f \geq 0$. For any $t \in \mathbb{R}$, the set $\{x \in \mathbb{R} \mid f(x) > t\}$ is open and thus Borel. Using the Layer-Cake Representation, we find

$$\begin{aligned} \int f d\mu &= \int_{(0, +\infty)} \mu(\{x \in \mathbb{R} \mid f(x) > t\}) dt \\ &\leq \int_{(0, +\infty)} \liminf_{n \rightarrow +\infty} \mu_n(\{x \in \mathbb{R} \mid f(x) > t\}) dt \\ &\leq \liminf_{n \rightarrow +\infty} \int_{(0, +\infty)} \mu_n(\{x \in \mathbb{R} \mid f(x) > t\}) dt \\ &= \liminf_{n \rightarrow +\infty} \int f d\mu_n \end{aligned}$$

where we used Fatou's Lemma for the last inequality. We get

$$\limsup_{n \rightarrow +\infty} \int f d\mu_n \leq \int f d\mu$$

by applying the same reasoning to $-f$. This allows us to conclude. $\square$

Remark 3.21

The last part of the proof (the fact that $\mu(U) \leq \liminf_{n \rightarrow +\infty} \mu_n(U)$ for any open set implies the weak convergence) is a specific case of the *Portmanteau Theorem*.

Corollary 3.22

The weak limit μ of a sequence of finite measures $(\mu_n)_{n=1}^{\infty} \subset \mathfrak{M}_{\mathcal{B}}$ is unique.

Theorem 3.23 (Helly's Selection Theorem)

Let $(\mu_n)_{n=1}^{\infty} \subset \mathfrak{M}_{\mathcal{B}}$ be a sequence of finite measures such that the sequence of cumulative distribution functions $(F_n)_{n=1}^{\infty}$ is uniformly bounded.

Then there exists a subsequence $(F_{n_k})_{k=1}^{\infty}$ of (F_n) and a non-negative, non-decreasing, bounded and right-continuous function $F : \mathbb{R} \rightarrow \mathbb{R}$ such that for all continuity points x of F , we have $\lim_{k \rightarrow +\infty} F_{n_k}(x) = F(x)$.

Proof.

Using a standard sequential compactness argument on $\mathbb{R}$ combined with a Cantor Diagonal argument, the existence of a function $G : \mathbb{Q} \rightarrow \mathbb{R}$ and a subsequence $(F_{n_k})_{k=1}^\infty$ of (F_n) such that $\lim_{k \rightarrow +\infty} F_{n_k}(x) = G(x) \forall x \in \mathbb{Q}$ is easily seen.

Now, we define $F : \mathbb{R} \rightarrow \mathbb{R}$ by $F(x) = \inf\{G(t) \mid t > x, t \in \mathbb{Q}\}$.

Clearly, G and F are non-decreasing and bounded. To check the right-continuity of F , let $x_0 \in \mathbb{R}$ be fixed, let $\epsilon > 0$ and let $t_0 > x_0$, $t_0 \in \mathbb{Q}$, such that

$$F(x_0) \leq G(t_0) \leq F(x_0) + \epsilon$$

by definition of infimum. Now, for all t such that $t_0 > t \geq x_0$ (i.e. $0 \leq t - x_0 < \delta = t_0 - x_0$), we have

$$F(x_0) \leq F(t) = \inf\{G(u) \mid u > t, u \in \mathbb{Q}\} \leq G(t_0) \leq F(x_0) + \epsilon,$$

which proves the right-continuity.

Finally, let $x_0 \in \mathbb{R}$ be a point of continuity of F . Let $\epsilon > 0$. Let $r_1 < r_2 < x_0 < r_3$, with $r_1, r_2, r_3 \in \mathbb{Q}$ and

$$F(x_0) - \epsilon/2 \leq F(r_1) \leq F(r_2) \leq F(x_0) \leq F(r_3) \leq F(x_0) + \epsilon/2$$

by continuity at x_0 and monotonicity of F .

Notice that $F(r_1) \leq G(r_2) \leq F(r_2)$ and $F(r_2) \leq G(r_3) \leq F(r_3)$ by definition of F and monotonicity of G .

Furthermore, $\lim_{k \rightarrow +\infty} F_{n_k}(r_2) = G(r_2)$ and $\lim_{k \rightarrow +\infty} F_{n_k}(r_3) = G(r_3)$ by definition of G .

Now let $K \in \mathbb{N}$ be such that $\forall k \geq K$:

$$|F_{n_k}(r_2) - G(r_2)| \leq \epsilon/2, \quad |F_{n_k}(r_3) - G(r_3)| \leq \epsilon/2.$$

Therefore, for all $k \geq K$:

$$F(x_0) - \epsilon \leq F_{n_k}(r_2) \leq F_{n_k}(x_0) \leq F_{n_k}(r_3) \leq F(r_3) \leq F(x_0) + \epsilon$$

where we used the monotonicity of F_{n_k} to include $F_{n_k}(x_0)$ in the chain. $\square$

Proposition 3.24

Let $(\mu_n)_{n=1}^\infty \subset \mathfrak{M}_{\mathcal{B}}$ be a sequence of finite measures. Assume that there exists a non-negative, non-decreasing, bounded and right-continuous function $F : \mathbb{R} \rightarrow \mathbb{R}$ for which the sequence of cumulative distribution functions $(F_n)_{n=1}^\infty$ converges pointwisely to F at all continuity points x of F , i.e. $\lim_{n \rightarrow +\infty} F_n(x) = F(x)$ for all continuity points x of F .

Then $(\mu_n)_{n=1}^\infty$ converges weakly to the Stieltjes measure induced by F .

Proof.

Since F is non-decreasing, its set of discontinuities D is countable. In particular, $D^c \subset \mathbb{R}$ is dense. Consider the family $\mathcal{I} = \{(a, b) \mid a, b \in D^c, a \neq b\}$.

Then any open set U of $\mathbb{R}$ is the countable union of elements of $\mathcal{I}$.

We apply the sufficient criterion for weak convergence, i.e. we show that

$$F_n(b-) - F_n(a) = \mu_n(I) \xrightarrow{n \rightarrow +\infty} \mu_F(I) = F(b-) - F(a) = F(b) - F(a) \forall I = (a, b) \in \mathcal{I}.$$

We already have $\lim_{n \rightarrow +\infty} F_n(a) = F(a) \forall a \in D^c$. Now, it suffices to prove $\lim_{n \rightarrow +\infty} F_n(b-) = F(b) \forall b \in D^c$.

Since F_n is non-decreasing $\forall n \in \mathbb{N}$, we have $F_n(b-) \leq F_n(b) \forall n \in \mathbb{N}$ and so

$$\limsup_{n \rightarrow +\infty} F_n(b-) \leq \lim_{n \rightarrow +\infty} F_n(b) = F(b).$$

Let $\epsilon > 0$. There exists $\delta > 0$ such that $b - \delta \in D^c$ and

$$F(b - \delta) \geq F(b) - \epsilon/2,$$

by continuity of F at b and density of D^c . Since $\lim_{n \rightarrow +\infty} F_n(b - \delta) = F(b - \delta)$, there exists $N \in \mathbb{N}$ such that $\forall n \geq N$:

$$F_n(b - \delta) \geq F(b - \delta) - \epsilon/2.$$

Therefore,

$$F_n(b-) \geq F_n(b - \delta) \geq F(b) - \epsilon$$

for all $n \geq N$, where we used the fact that F_n is non-decreasing $\forall n \in \mathbb{N}$. In particular, $\forall \epsilon > 0$:

$$\liminf_{n \rightarrow +\infty} F_n(b-) \geq F(b) - \epsilon$$

and we conclude by letting $\epsilon \rightarrow 0^+$. □

Chapter 4

The Herglotz Representation Theorem

Definition 4.1 (Bounded Variation)

Let $-\infty \leq a < b \leq +\infty$. A function $h : [a, b] \rightarrow \mathbb{C}$ is said to be *of bounded variation* if it can be written as $h = h_1 - h_2 + ih_3 - ih_4$, where each $h_i : [a, b] \rightarrow \mathbb{R}$ is a non-decreasing bounded function.

Remark 4.1

If $h : [a, b] \rightarrow \mathbb{C}$ is of bounded variation and is right-continuous, then we can choose the h_i 's to be right-continuous as well.

Definition 4.2 (Integral with respect to a function of bounded variation)

Let $h : \mathbb{R} \rightarrow \mathbb{C}$ be right-continuous, of bounded variation, with right-continuous h_i 's. Then each h_i defines a finite Stieltjes measure as in Definition 3.3.

Let $A \subset \mathbb{R}$ be measurable for each h_i . Let $f : \mathbb{R} \rightarrow \overline{\mathbb{R}}$ be a real-valued function that is integrable on A with respect to each h_i . The Lebesgue-Stieltjes integral on A with respect to h is defined as

$$\int_A f dh = \int_A f dh_1 - \int_A f dh_2 + i \int_A f dh_3 - i \int_A f dh_4.$$

The integral of an integrable complex-valued function $f : \mathbb{R} \rightarrow \mathbb{C}$ is defined similarly.

We write $\mathcal{L}_{\mathbb{K}}^p(h) = \bigcap_{i=1}^4 \mathcal{L}_{\mathbb{K}}^p(h_i) = \mathcal{L}_{\mathbb{K}}^p(\sum_{i=1}^4 h_i)$ for $1 \leq p < +\infty$.

Remark 4.2

Since each h_i defines a finite Stieltjes measure, any bounded continuous function is integrable with respect to each h_i .

Proposition 4.3

Let $h : \mathbb{R} \rightarrow \mathbb{C}$ be right-continuous, of bounded variation, and let $f \in \mathcal{L}_{\mathbb{R}}^1(h)$ be non-negative. Define

$$\begin{aligned} \Phi : \mathbb{R} &\rightarrow \mathbb{C} \\ x &\mapsto \int_{(-\infty, x]} f(t) dh(t) \end{aligned}$$

Then Φ is right-continuous and of bounded variation.

Furthermore, for any $g \in \mathcal{L}_{\mathbb{R}}^1(h)$, we have $g \in \mathcal{L}_{\mathbb{R}}^1(\Phi)$ and

$$\int_{\mathbb{R}} g(t) d\Phi(t) = \int_{\mathbb{R}} g(t) f(t) dh(t).$$

Proof.

The function Φ is right-continuous because

$$\Phi(x + \epsilon) - \Phi(x) = \int f(t) \cdot (\chi_{(-\infty, x+\epsilon]}(t) - \chi_{(-\infty, x]}(t)) dh(t) \xrightarrow{\epsilon \rightarrow 0^+} \int 0 dh(t) = 0$$

by the Dominated Convergence Theorem (write $h(t) = h_1(t) - h_2(t) + ih_3(t) - ih_4(t)$, $\int \dots dh = \int \dots dh_1 - \int \dots dh_2 + i \int \dots dh_3 - i \int \dots dh_4$, with $h_i(t)$ non-decreasing and right-continuous, and apply the theorem to each Stieltjes integral).

The function Φ is of bounded variation because

$$\Phi(x) = \int_{(-\infty, x]} f dh_1 - \int_{(-\infty, x]} f dh_2 + i \int_{(-\infty, x]} f dh_3 - i \int_{(-\infty, x]} f dh_4$$

and each $\int_{(-\infty, x]} f(t) dh_i$ is right-continuous, non-decreasing and bounded. We write $\Phi = \Phi_1 - \Phi_2 + i\Phi_3 - i\Phi_4$.

Note that the Stieltjes measure $d\Phi_i(t)$ is exactly $f dh_i(t)$ from Definition 3.8 so the identity follows. $\square$

Theorem 4.4 (Stieltjes Inversion Formula)

Let $w : \mathbb{R} \rightarrow \mathbb{R}$ be right-continuous, of bounded variation, satisfying $\lim_{t \rightarrow -\infty} w(t) = 0$.

Define

$$f(z) = \int_{\mathbb{R}} (z - t)^{-1} dw(t), \quad z \in \mathbb{H} = \{z \in \mathbb{C} \mid \Im z > 0\}.$$

Then we have the Stieltjes Inversion Formula

$$w(t) = \lim_{\delta \rightarrow 0^+} \lim_{\epsilon \rightarrow 0^+} \frac{-1}{\pi} \int_{-\infty}^{t+\delta} \Im f(s + i\epsilon) ds \quad \forall t \in \mathbb{R},$$

where this integral is to be interpreted as a standard Lebesgue integral.

In particular, if $f(z) = 0$ on $\mathbb{H}$, then $w(t) = 0$ on $\mathbb{R}$.

Proof.

Since $w(t)$ is real-valued, for $s \in \mathbb{R}$ and $\epsilon > 0$, we have

$$\Im f(s + i\epsilon) = \int_{\mathbb{R}} \Im(s + i\epsilon - t)^{-1} dw(t) = \int_{\mathbb{R}} \frac{-\epsilon}{(s - t)^2 + \epsilon^2} dw(t)$$

Using Fubini-Tonelli Theorem,

$$\begin{aligned} \int_{(-\infty, r)} ds \int_{\mathbb{R}} \frac{-\epsilon}{(s - t)^2 + \epsilon^2} dw(t) &= \int_{\mathbb{R}} dw(t) \int_{(-\infty, r)} \frac{-\epsilon}{(s - t)^2 + \epsilon^2} ds \\ &= - \int_{\mathbb{R}} \left(\arctan\left(\frac{r - t}{\epsilon}\right) + \frac{\pi}{2} \right) dw(t) \end{aligned}$$

Since $|\arctan(\frac{r-t}{\epsilon}) + \frac{\pi}{2}| \leq \pi \ \forall r, t \in \mathbb{R}, \ \forall \epsilon > 0$, and

$$\arctan(\frac{r-t}{\epsilon}) + \frac{\pi}{2} \rightarrow \begin{cases} \pi & \text{for } r > t \\ \frac{\pi}{2} & \text{for } r = t \\ 0 & \text{for } r < t \end{cases} \quad \text{as } \epsilon \rightarrow 0^+,$$

we can apply the Dominated Convergence Theorem to get

$$\begin{aligned} \lim_{\epsilon \rightarrow 0^+} \int_{(-\infty, r)} \Im f(s + i\epsilon) ds &= - \int_{(-\infty, r)} \pi dw(t) - \int_r \frac{\pi}{2} dw(t) - \int_{(r, +\infty)} 0 dw(t) \\ &= -\pi w(r-) - \frac{\pi}{2} (w(r) - w(r-)) \\ &= -\frac{\pi}{2} (w(r) + w(r-)). \end{aligned}$$

Set $r = t + \delta$, $t \in \mathbb{R}$, $\delta > 0$ and take the limit $\delta \rightarrow 0^+$. □

Theorem 4.5

Let $w : \mathbb{R} \rightarrow \mathbb{C}$ be right-continuous, of bounded variation, satisfying $\lim_{t \rightarrow -\infty} w(t) = 0$. Define

$$f(z) = \int_{\mathbb{R}} (z - t)^{-1} dw(t), \quad z \in \mathbb{C} \setminus \mathbb{R}.$$

If $f(z) = 0 \ \forall z \in \mathbb{C} \setminus \mathbb{R}$, then $w(t) = 0 \ \forall t \in \mathbb{R}$.

Theorem 4.6 (Herglotz, Nevanlinna)

Let $\mathbb{H} = \{z \in \mathbb{C} \mid \Im z > 0\}$ and let $f : \mathbb{H} \rightarrow \mathbb{C}$ be holomorphic such that $\Im f(z) \leq 0$ and $|f(z)\Im z| \leq M \ \forall z \in \mathbb{H}$. Then there exists a unique right-continuous and non-decreasing function $w(t) : \mathbb{R} \rightarrow \mathbb{R}$ for which $\lim_{t \rightarrow -\infty} w(t) = 0$ and

$$f(z) = \int_{\mathbb{R}} (z - t)^{-1} dw(t) \quad \forall z \in \mathbb{H}$$

Furthermore, $w(t)$ is bounded by M and

$$w(t) = \lim_{\delta \rightarrow 0^+} \lim_{\epsilon \rightarrow 0^+} \frac{-1}{\pi} \int_{-\infty}^{t+\delta} \Im f(s + i\epsilon) ds \quad \forall t \in \mathbb{R}$$

Proof. The last equality follows from the Stieltjes Inversion Formula and guarantees the uniqueness of w (if it exists).

Let $z \in \mathbb{H}$ be fixed. Let $0 < \epsilon < \Im z$ and let $r > |z|$.

Define the contour $\Gamma_r = \Gamma_r^1 \cup \Gamma_r^2 = \{x + iy \mid |x| \leq r, y = \epsilon\} \cup \{z \mid z = i\epsilon + re^{i\theta}, 0 \leq \theta \leq \pi\}$ (i.e. Γ is an half-circle of radius r with origin $i\epsilon$). By the Cauchy Integral Formula and the Cauchy Integral Theorem,

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \int_{\Gamma_r} \frac{f(\xi)}{\xi - z} d\xi = \frac{1}{2\pi i} \int_{\Gamma_r} \frac{f(\xi)}{\xi - z} d\xi - 0 \\ &= \frac{1}{2\pi i} \int_{\Gamma_r} \frac{f(\xi)}{\xi - z} d\xi - \int_{\Gamma_r} \frac{f(\xi)}{\xi - (\bar{z} + 2i\epsilon)} d\xi \end{aligned}$$

$$= \frac{1}{\pi} \int_{\Gamma_r} \frac{\Im z - \epsilon}{(\xi - z)(\xi - \bar{z} - 2i\epsilon)} f(\xi) d\xi$$

If $\xi \in \Gamma_r^2$, then $|f(\xi)| \leq M/\Im \xi \leq M/\epsilon$. Furthermore, there exists a constant $C = C(z, \epsilon)$ such that for all r large enough and $\xi \in \Gamma_r^2$:

$$\left| \frac{\Im z - \epsilon}{(\xi - z)(\xi - \bar{z} - 2i\epsilon)} \right| \leq C/r^2.$$

Therefore, we have

$$\left| \frac{1}{\pi} \int_{\Gamma_r^2} \frac{\Im z - \epsilon}{(\xi - z)(\xi - \bar{z} - 2i\epsilon)} f(\xi) d\xi \right| \leq (1/\pi) \cdot (M/\epsilon) \cdot (C/r^2) \cdot (2\pi r)$$

for all r large enough.

Thus, for all $z \in \mathbb{H}$ and $0 < \epsilon < \Im z$ fixed, letting $r \rightarrow +\infty$, we get

$$\begin{aligned} f(z) &= \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{\Im z - \epsilon}{(t + i\epsilon - z)(t + i\epsilon - \bar{z} - 2i\epsilon)} f(t + i\epsilon) dt \\ &= \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{\Im z - \epsilon}{(t + i\epsilon - z)(t - i\epsilon - \bar{z})} f(t + i\epsilon) dt. \end{aligned}$$

Now, write $z = x + iy$, $y > 0$, $0 < \epsilon < y$. Then

$$\begin{aligned} f(z) &= \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{\Im z - \epsilon}{(t + i\epsilon - z)(t - i\epsilon - \bar{z})} f(t + i\epsilon) dt \\ &= \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{y - \epsilon}{(x - t)^2 + (y - \epsilon)^2} f(t + i\epsilon) dt. \end{aligned}$$

Set $v(z) = \Im f(z)$, $z \in \mathbb{H}$. Let $\epsilon > 0$ be fixed. Then the function $v(t + i\epsilon)$ (which is non-positive and continuous) is Lebesgue integrable over $\mathbb{R}$. Indeed, notice that

$$v(t + i\epsilon) = \liminf_{y \rightarrow +\infty} \frac{(y - \epsilon)^2}{(x - t)^2 + (y - \epsilon)^2} v(t + i\epsilon) \quad \forall t \in \mathbb{R}$$

for any chosen $x \in \mathbb{R}$. By Fatou's Lemma,

$$\begin{aligned} 0 &\leq -\frac{1}{\pi} \int_{-\infty}^{+\infty} v(t + i\epsilon) dt \leq \liminf_{y \rightarrow +\infty} -\frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{(y - \epsilon)^2}{(x - t)^2 + (y - \epsilon)^2} v(t + i\epsilon) dt \\ &= \liminf_{y \rightarrow +\infty} -(y - \epsilon) f(x + iy) \\ &\leq \liminf_{y \rightarrow +\infty} |y f(x + iy)| \leq M, \end{aligned}$$

so $v(t + i\epsilon)$ is integrable on $\mathbb{R}$ for any $\epsilon > 0$ fixed.

For $x, t \in \mathbb{R}$ fixed, consider the function

$$y \mapsto h(y) := \frac{y}{(x - t)^2 + y^2}, \quad y \in (0, +\infty)$$

For $y \in (0, +\infty)$ and $0 < \epsilon < y$, by the Mean Value Theorem, there exists $c \in (y - \epsilon, y)$ such that

$$\begin{aligned} \left| \frac{y}{(x-t)^2 + y^2} - \frac{y-\epsilon}{(x-t)^2 + (y-\epsilon)^2} \right| &= \epsilon |h'(c)| = \epsilon \left| \frac{(x-t)^2 - c^2}{((x-t)^2 + c^2)^2} \right| \\ &\leq \epsilon \left| \frac{(x-t)^2 + c^2}{((x-t)^2 + c^2)^2} \right| \\ &= \frac{\epsilon}{(x-t)^2 + c^2} \leq \frac{\epsilon}{(y-\epsilon)^2}. \end{aligned}$$

Therefore, for any fixed $y > 0$:

$$\lim_{\epsilon \rightarrow 0^+} \int_{-\infty}^{+\infty} \left[\frac{y}{(x-t)^2 + y^2} - \frac{y-\epsilon}{(x-t)^2 + (y-\epsilon)^2} \right] v(t + i\epsilon) dt = 0$$

independently of $x \in \mathbb{R}$ using the Monotone Convergence Theorem.

In particular, for any $z = x + iy \in \mathbb{H}$, we have

$$v(z) = \lim_{\epsilon \rightarrow 0^+} \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{y}{(x-t)^2 + y^2} v(t + i\epsilon) dt$$

By Proposition 4.3, we can write

$$v(z) = \lim_{\epsilon \rightarrow 0^+} - \int_{-\infty}^{+\infty} \frac{y}{(x-t)^2 + y^2} d\theta_\epsilon(t)$$

where, for $0 < \epsilon < 1$, θ_ϵ is defined as

$$\theta_\epsilon(t) = -\frac{1}{\pi} \int_{-\infty}^t v(t + i\epsilon) dt, \quad t \in \mathbb{R},$$

which is a continuous and non-decreasing function satisfying $0 \leq \theta_\epsilon \leq M$ and $\lim_{t \rightarrow +\infty} (\theta_\epsilon(t) - \theta_\epsilon(-t)) \leq M$ for any $0 < \epsilon < 1$. By Helly's Selection Theorem and Proposition 3.24, we can find a sequence $(\epsilon_n)_{n=1}^\infty \subset (0, 1)$ converging to zero such that $d\theta_{\epsilon_n}(t)$ converges weakly to the Stieltjes measure of some non-decreasing and right-continuous function θ satisfying $0 \leq \theta \leq M$ and $\lim_{t \rightarrow +\infty} (\theta(t) - \theta(-t)) \leq M$. Therefore,

$$\begin{aligned} \Im f(z) &= v(z) = - \int_{-\infty}^{+\infty} \frac{y}{(x-t)^2 + y^2} d\theta(t) \\ &= \int_{-\infty}^{+\infty} \Im(z-t)^{-1} d\theta(t) \\ &= \Im \int_{-\infty}^{+\infty} (z-t)^{-1} d\theta(t), \quad \forall z \in \mathbb{H} \end{aligned}$$

Note that $z \mapsto \int_{-\infty}^{+\infty} (z-t)^{-1} d\theta(t)$ is holomorphic on $\mathbb{H}$. Indeed, for $z_0 \in \mathbb{H}$ fixed, let $\overline{B}_r(z_0) \subset \mathbb{H}$ be a neighborhood of z_0 . Then for $z \in \overline{B}_r(z_0)$, $t \in \mathbb{R}$, we have

$$\left| \frac{(z-t)^{-1} - (z_0-t)^{-1}}{z - z_0} \right| \leq \sqrt{2} \max\{ \left| \frac{d}{dz} (z-t)^{-1} \right| \mid z \in \overline{B}_r(z_0), t \in \mathbb{R} \} \leq C < +\infty$$

(using the Mean Value Theorem separately on the real and imaginary part), so we can apply the Dominated Convergence Theorem and get

$$\lim_{z \rightarrow z_0} \int_{-\infty}^{+\infty} \frac{(z-t)^{-1} - (z_0-t)^{-1}}{z-z_0} d\theta(t) = \int_{-\infty}^{+\infty} -(z_0-t)^{-2} d\theta(t) < +\infty$$

Therefore, $f(z)$ and $z \mapsto \int_{-\infty}^{+\infty} (z-t)^{-1} d\theta(t)$ are holomorphic functions on $\mathbb{H}$ with the same imaginary part. It follows that

$$f(z) = \int_{-\infty}^{+\infty} (z-t)^{-1} d\theta(t) + C, \quad z \in \mathbb{H}$$

for some $C \in \mathbb{R}$. Recall that $|f(z)\Im z| \leq M \quad \forall z \in \mathbb{H}$, so

$$|C\Im z| \leq |f(x)\Im z| + |\Im z| \left| \int_{-\infty}^{+\infty} (z-t)^{-1} d\theta(t) \right| \leq M + \int_{-\infty}^{+\infty} 1 d\theta(t) \leq 2M \quad \forall z \in \mathbb{H}$$

which means $C = 0$.

Finally, we conclude by defining $w(t) = \theta(t) - \lim_{s \rightarrow -\infty} \theta(s)$. □

Remark 4.7

The above theorem is, in fact, a special case of the three following theorems, which are all equivalent.

Theorem 4.8

A function $f : \mathbb{H} \rightarrow \mathbb{C}$ is holomorphic and such that $\Im f(z) \geq 0 \quad \forall z \in \mathbb{H}$ if and only if there exists $a, b \in \mathbb{R}$, $b > 0$, and a right-continuous and non-decreasing function $\mu(t) : \mathbb{R} \rightarrow \mathbb{R}$ for which

$$f(z) = a + bz + \int_{\mathbb{R}} \left(\frac{1}{t-z} - \frac{t}{t^2+1} \right) d\mu(t), \quad z \in \mathbb{H}$$

and $\int_{(0,+\infty)} (1+t^2)^{-1} d\mu(t) < +\infty$.

Theorem 4.9

A function $f : \mathbb{H} \rightarrow \mathbb{C}$ is holomorphic and such that $\Im f(z) \geq 0 \quad \forall z \in \mathbb{H}$ if and only if there exists $a, b \in \mathbb{R}$, $b > 0$, and a right-continuous, non-decreasing and bounded function $\sigma(t) : \mathbb{R} \rightarrow \mathbb{R}$ for which

$$f(z) = a + bz + \int_{\mathbb{R}} \frac{1+zt}{t-z} d\sigma(t), \quad z \in \mathbb{H}$$

Theorem 4.10

A function $f : D(0,1) \rightarrow \mathbb{C}$ is holomorphic and such that $\Im f(z) \geq 0 \quad \forall z \in \mathbb{H}$ if and only if there exists $a \in \mathbb{R}$ and a right-continuous, non-decreasing and bounded function $\omega(t) : \mathbb{R} \rightarrow \mathbb{R}$ which is constant on $\mathbb{R} \setminus [0, 2\pi]$ and for which

$$f(z) = a + i \int_{[0,2\pi]} \frac{e^{i\theta} + z}{e^{i\theta} - z} d\omega(\theta), \quad z \in D(0,1)$$

Proof.

Equivalence between Theorem 4.8 and Theorem 4.9 follows from the relation $d\sigma(t) =$

$(1+t^2)^{-1}d\mu(t)$, where the multiplication of a function with a measure is to be understood as in Definition 3.8.

Equivalence between Theorem 4.9 and Theorem 4.10 follows from a conformal transformation from $D(0, 1)$ to $\mathbb{H}$.

See [Don74] (Chapter II) or [Tam43] (Chapter II) for more details on the equivalence. A proof of (one version of) those theorems can be found in [Don74] (Chapter II) and [Akh77] (Volume 2, Section 59). The proofs rely on results from Fourier Analysis and the Helly's Selection Theorem. $\square$

Chapter 5

Integration with respect to a spectral family

In this chapter, we finally define the functional calculus for unbounded self-adjoint operators and prove some of its properties.

Definition 5.1 (Spectral family)

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space. A *spectral family* on H is a family $E(t) : \mathbb{R} \rightarrow B(H, H)$ satisfying

1. $E(t)$ is an orthogonal projection $\forall t \in \mathbb{R}$;
2. $E(s) \leq E(t) \forall s \leq t$;
3. $\forall t \in \mathbb{R}, \forall x \in H$:

$$\lim_{\epsilon \rightarrow 0^+} E(t + \epsilon)x = E(t)x;$$

4. $\forall x \in H$:

$$\lim_{t \rightarrow -\infty} E(t)x = 0, \quad \lim_{t \rightarrow +\infty} E(t)x = x.$$

Definition 5.2 (E -measurability)

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and let $(E_t)_{t \in \mathbb{R}}$ be a spectral family on H .

For $x \in H$, we define the function

$$\begin{aligned} \rho_x : \mathbb{R} &\rightarrow \mathbb{R} \\ t &\mapsto \langle x, E(t)x \rangle = \|E(t)x\|^2 \end{aligned}$$

The function ρ_x is non-negative, non-decreasing, right-continuous and bounded. Furthermore, $\lim_{t \rightarrow -\infty} \rho_x(t) = 0$ and $\lim_{t \rightarrow +\infty} \rho_x(t) = \|x\|^2$.

For $x \in H$, ρ_x defines (as in Definition 3.3) a finite Stieltjes measure such that all Borel sets of $\mathbb{R}$ are measurable.

A function $u : \mathbb{R} \rightarrow \mathbb{K}$ is said to be *E -measurable* if it is ρ_x -measurable for all $x \in H$.

Definition 5.3 (Integral with respect to a spectral family)

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and let $(E_t)_{t \in \mathbb{R}}$ be a spectral family on H . The *integral* of a step function $u : \mathbb{R} \rightarrow \mathbb{R}$, $u = \sum_{i=1}^n a_i \chi_{J_i}$, with respect to the family E is the operator defined as

$$\int u(t) dE(t) = \sum_{i=1}^n a_i E(J_i),$$

where

$$E(J) = \begin{cases} E(b) - E(a) & \text{for } J = (a, b] \\ E(b) - E(a-) & \text{for } J = [a, b] \\ E(b-) - E(a) & \text{for } J = (a, b) \\ E(b-) - E(a-) & \text{for } J = [a, b) \end{cases}$$

and $E(a-)$ is the strong limit $E(t) \xrightarrow[t \rightarrow a-]{s} E(a-)$.

For a step function $u : \mathbb{R} \rightarrow \mathbb{R}$ and an element $x \in H$, we have

$$\begin{aligned} \left\| \int u(t) dE(t)x \right\|^2 &= \left\langle \sum_{i=1}^n a_i E(J_i)x, \sum_{i=1}^n a_i E(J_i)x \right\rangle \\ &= \sum_{i=1}^n |a_i|^2 \|E(J_i)x\|^2 = \sum_{i=1}^n |a_i|^2 \langle x, E(J_i)x \rangle \\ &= \sum_{i=1}^n |a_i|^2 \rho_x(J_i) = \int |u(t)|^2 d\rho_x \end{aligned}$$

Let $u \in \mathcal{L}^2(\rho_x)$. Let $x \in H$. Since step functions are dense in $\mathcal{L}^2(\rho_x)$, we can find a sequence of step functions $(u_n)_{n=1}^\infty \subset \mathcal{L}^2(\rho_x)$ converging to u in $\mathcal{L}^2(\rho_x)$. In particular, it is a Cauchy sequence, so

$$\left\| \int u_n(t) dE(t)x - \int u_m(t) dE(t)x \right\|^2 = \int |u_n(t) - u_m(t)|^2 d\rho_x \xrightarrow{n, m \rightarrow +\infty} 0$$

and $\int u_n(t) dE(t)x$ is a Cauchy sequence in H . Thus, we define

$$\int u(t) dE(t)x := \lim_{n \rightarrow +\infty} \int u_n(t) dE(t)x$$

and we have

$$\left\| \int u(t) dE(t)x \right\|^2 = \int |u(t)|^2 d\rho_x.$$

This definition does not depend on the choice of the sequence (u_n) .

Because of this last identity, if $(u_n)_{n=1}^\infty \subset \mathcal{L}^2(\rho_x)$ is any sequence converging to u in $\mathcal{L}^2(\rho_x)$, then

$$\int u(t) dE(t)x = \lim_{n \rightarrow +\infty} \int u_n(t) dE(t)x.$$

Proposition 5.1 (Linearity of the integral)

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and let $(E_t)_{t \in \mathbb{R}}$ be a spectral family on H . Let $x \in H$, $u, v \in \mathcal{L}^2(\rho_x)$ and $a, b \in \mathbb{K}$. Then

$$\int (au(t) + bv(t)) dE(t)x = a \int u(t) dE(t)x + b \int v(t) dE(t)x$$

Remark 5.2 (About finite measures and bounded functions)

Since ρ_x is a finite measure $\forall x \in H$, if u is a bounded E -measurable function, then $u \in \mathcal{L}^2(\rho_x) \forall x \in H$.

Theorem 5.3

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and let $(E_t)_{t \in \mathbb{R}}$ be a spectral family on H . Let $u : \mathbb{R} \rightarrow \mathbb{K}$ be an E -measurable function. For any finite set $\{x_1, \dots, x_n\} \subset H$, there exists a sequence of step functions converging to u almost everywhere with respect to $\rho_{x_i} \forall i = 1, \dots, n$. Furthermore, if u is bounded, the sequence can be chosen bounded.

Proof. [Wei80] (Theorem 7.13 p.183) □

Definition 5.4

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and let $(E_t)_{t \in \mathbb{R}}$ be a spectral family on H . Let $u : \mathbb{R} \rightarrow \mathbb{K}$ be a bounded E -measurable function.

For $x, y \in H$, $t \mapsto \langle y, E(t)x \rangle$ is of bounded variation (this follows from the Polarization Identity for sesquilinear forms) and we can consider the integral

$$\int u(t) d\langle y, E(t)x \rangle = \begin{cases} \frac{1}{4} \left(\int u(t) d\rho_{x+y} - \int u(t) d\rho_{x-y} \right) \\ \frac{1}{4} \left(\int u(t) d\rho_{x+y} - \int u(t) d\rho_{x-y} + i \int u(t) d\rho_{x-iy} - i \int u(t) d\rho_{x+iy} \right) \end{cases}$$

respectively if $\mathbb{K} = \mathbb{R}$ and $\mathbb{K} = \mathbb{C}$.

Note that if $x = y$, we have

$$\int u(t) d\langle x, E(t)x \rangle = \int u(t) d\langle E(t)x, E(t)x \rangle = \int u(t) d\rho_x$$

Proposition 5.4

For any bounded E -measurable functions $u, v : \mathbb{R} \rightarrow \mathbb{K}$, the following identity holds :

$$\int \overline{v(t)} u(t) d\langle y, E(t)x \rangle = \langle \int v(t) dE(t)y, \int u(t) dE(t)x \rangle$$

Proof.

By Theorem 5.3 and the Dominated Convergence Theorem, it suffices to prove the identity for bounded step functions u, v .

We prove the case $\mathbb{K} = \mathbb{R}$. The other one is similar. Write $u = \sum_{i=1}^n a_i \chi_{J_i}$ and similarly $v = \sum_{i=1}^n b_i \chi_{J_i}$, where $J_1, \dots, J_n$ are disjoint bounded intervals (we can always take the same intervals for u and v by taking pairwise intersections of the intervals appearing in u and v 's definition). Then :

$$\begin{aligned} \frac{1}{4} \left(\int v(t) u(t) d\rho_{x+y} - \int v(t) u(t) d\rho_{x-y} \right) &= \frac{1}{4} \sum_{i=1}^n b_i a_i (\rho_{x+y}(J_i) - \rho_{x-y}(J_i)) \\ &= \frac{1}{4} \sum_{i=1}^n b_i a_i (\langle x+y, E(J_i)(x+y) \rangle - \langle x-y, E(J_i)(x-y) \rangle) \\ &= \sum_{i=1}^n b_i a_i \langle y, E(J_i)x \rangle \\ &= \sum_{i=1}^n \langle b_i y, a_i E(J_i)x \rangle = \sum_{i=1}^n \langle b_i E(J_i)y, a_i E(J_i)x \rangle \end{aligned}$$

$$\begin{aligned}
&=^2 \left\langle \sum_{i=1}^n b_i E(J_i)y, \sum_{i=1}^n a_i E(J_i)x \right\rangle \\
&= \left\langle \int v(t) dE(t)y, \int u(t) dE(t)x \right\rangle,
\end{aligned}$$

where we used the Polarization Identity for a sesquilinear form in $=^1$ and the fact that $E(J_i)^2 = E(J_i)$, $E(J_i) \perp E(J_j)$ for $i \neq j$ in $=^2$. $\square$

Theorem 5.5 (Spectral Calculus, also called Functional Calculus)

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and let $(E_t)_{t \in \mathbb{R}}$ be a spectral family on H . Let $u : \mathbb{R} \rightarrow \mathbb{K}$ be an E -measurable function.

Define the set $D(\hat{E}(u)) = \{x \in H \mid u \in \mathcal{L}^2(\rho_x)\} \subset H$. Then this set is a subspace of H and the integral operator of u

$$\begin{aligned}
\hat{E}(u) : D(\hat{E}(u)) &\rightarrow H \\
x &\mapsto \int u(t) dE(t)x
\end{aligned}$$

is well-defined and linear.

Furthermore, we have the following properties for $\hat{E}(u)$:

1. For all $x \in D(\hat{E}(u))$:

$$\|\hat{E}(u)x\|^2 = \int |u(t)|^2 d\rho_x;$$

2. If u is bounded, then $D(\hat{E}(u)) = H$, $\hat{E}(u) \in B(H, H)$ and

$$\|\hat{E}(u)\| \leq \sup\{|u(t)| \mid t \in \mathbb{R}\};$$

3. If $u(t) = 1 \ \forall t \in \mathbb{R}$, then $\hat{E}(u) = I$.

If $u, v : \mathbb{R} \rightarrow \mathbb{K}$ are E -measurable functions, define

$$\varphi_n(t) = \begin{cases} 1 & \text{if } |u(t)| \leq n \\ 0 & \text{otherwise} \end{cases}, \quad \psi_n(t) = \begin{cases} 1 & \text{if } |v(t)| \leq n \\ 0 & \text{otherwise} \end{cases}$$

We have :

4. For all $x \in D(\hat{E}(u))$, $y \in D(\hat{E}(v))$:

$$\langle \hat{E}(v)y, \hat{E}(u)x \rangle = \lim_{n \rightarrow +\infty} \int \overline{\psi_n(t)v(t)} \varphi_n(t) u(t) d\langle y, E(t)x \rangle;$$

5. For all $x \in D(\hat{E}(u))$, $y \in H$:

$$\langle y, \hat{E}(u)x \rangle = \lim_{n \rightarrow +\infty} \int \varphi_n(t) u(t) d\langle y, E(t)x \rangle;$$

In particular, if $x, y \in D(\hat{E}(u))$:

$$\langle y, \hat{E}(u)x \rangle = \int u(t) d\langle y, E(t)x \rangle;$$

6. If $u : \mathbb{R} \rightarrow \mathbb{R}$, $u(t) \geq c > 0 \forall t \in \mathbb{R}$, then for all $x \in D(\hat{E}(u))$:

$$\langle x, \hat{E}(u)x \rangle \geq c\|x\|^2;$$

7. $\hat{E}(au + bv) \supset a\hat{E}(u) + b\hat{E}(v) \forall a, b \in \mathbb{K}$, $D(\hat{E}(u) + \hat{E}(v)) = D(\hat{E}(|u| + |v|))$;

8. $\hat{E}(uv) \supset \hat{E}(u)\hat{E}(v)$, $D(\hat{E}(u)\hat{E}(v)) = D(\hat{E}(v)) \cap D(\hat{E}(uv))$;

9. $\hat{E}(\bar{u}) = \hat{E}(u)^*$ and $D(\hat{E}(\bar{u})) = D(\hat{E}(u))$.

Remark 5.6

In the following proof, for E -measurable functions $u, (u_n)_{n=1}^\infty : \mathbb{R} \rightarrow \mathbb{K}$ and an element $x \in D(\hat{E}(u)) \cap D(\hat{E}(u_n)) \forall n \in \mathbb{N}$, the notation

$$\hat{E}(u)x = \lim_{n \rightarrow +\infty} \hat{E}(u_n)x$$

means

$$\begin{aligned} \lim_{n \rightarrow +\infty} \|\hat{E}(u)(x) - \hat{E}(u_n)(x)\|^2 &= \lim_{n \rightarrow +\infty} \left\| \int (u(t) - u_n(t)) dE(t)x \right\|^2 \\ &= \lim_{n \rightarrow +\infty} \int |u(t) - u_n(t)|^2 d\rho_x \\ &= 0 \end{aligned}$$

Proof.

The operator is well-defined as seen in Definition 5.3. For a step function u , it is clear that $D(\hat{E}(u)) = H$ and $\hat{E}(u)$ is linear.

Now, for a bounded E -measurable $u : \mathbb{R} \rightarrow \mathbb{K}$, it is also clear that $D(\hat{E}(u)) = H$ (c.f. Remark 5.2). Let $x, y \in H$. By Theorem 5.3, there exists a bounded sequence of step functions $(u_n)_{n=1}^\infty$ converging to u almost everywhere with respect to ρ_x , ρ_y and ρ_{x+y} . By the Dominated Convergence Theorem, $(u_n)_{n=1}^\infty$ converges to u in $\mathcal{L}^2(\rho_x)$, $\mathcal{L}^2(\rho_y)$ and $\mathcal{L}^2(\rho_{x+y})$. Then

$$\begin{aligned} \hat{E}(u)(x + y) &= \lim_{n \rightarrow +\infty} \hat{E}(u_n)(x + y) = \lim_{n \rightarrow +\infty} \hat{E}(u_n)(x) + \lim_{n \rightarrow +\infty} \hat{E}(u_n)(y) \\ &= \hat{E}(u)(x) + \hat{E}(u)(y) \end{aligned}$$

Finally, for an arbitrary E -measurable $u : \mathbb{R} \rightarrow \mathbb{K}$, let $x, y \in D(\hat{E}(u))$. Then $(u_n)_{n=1}^\infty$, $u_n(t) = |\varphi_n(t)u(t)|$, is a non-negative monotone sequence converging to $|u(t)|$ pointwisely for all $t \in \mathbb{R}$. Furthermore, $(u_n)_{n=1}^\infty \subset \mathcal{L}^2(\rho_{x+y})$ because

$$\begin{aligned} \left(\int |\varphi_n(t)u(t)|^2 d\rho_{x+y} \right)^{1/2} &= \|\hat{E}(\varphi_n u)(x + y)\| \\ &\leq \|\hat{E}(\varphi_n u)(x)\| + \|\hat{E}(\varphi_n u)(y)\| \\ &= \left(\int |\varphi_n(t)u(t)|^2 d\rho_x \right)^{1/2} + \left(\int |\varphi_n(t)u(t)|^2 d\rho_y \right)^{1/2} \\ &\leq \left(\int |u(t)|^2 d\rho_x \right)^{1/2} + \left(\int |u(t)|^2 d\rho_y \right)^{1/2} < +\infty \end{aligned}$$

By the Monotone Convergence Theorem, $u \in \mathcal{L}^2(\rho_{x+y})$, so $x + y \in D(\hat{E}(u))$. Notice that $|\varphi_n u - u|^2 \leq 4|u|^2$, so by the Dominated Convergence Theorem, $\varphi_n u$ converges to u in $\mathcal{L}^2(\rho_x)$, $\mathcal{L}^2(\rho_y)$, $\mathcal{L}^2(\rho_{x+y})$ and

$$\begin{aligned}\hat{E}(u)(x + y) &= \lim_{n \rightarrow +\infty} \hat{E}(\varphi_n u)(x + y) = \lim_{n \rightarrow +\infty} \hat{E}(\varphi_n u)(x) + \lim_{n \rightarrow +\infty} \hat{E}(\varphi_n u)(y) \\ &= \hat{E}(u)(x) + \hat{E}(u)(y)\end{aligned}$$

The scalar multiplication part of the linearity is clear.

1. Proved in Definition 5.3;
2. $D(\hat{E}(u)) = H$ is clear. Furthermore,

$$\|\hat{E}(u)x\|^2 = \int |u(t)|^2 d\rho_x \leq \sup\{|u(t)| \mid t \in \mathbb{R}\}^2 \cdot \rho_x(\mathbb{R}) = \sup\{|u(t)| \mid t \in \mathbb{R}\}^2 \cdot \|x\|^2;$$

3. Let $x \in D(\hat{E}(u)) = H$. Notice that $u = 1$ is the limit of $(u_n)_{n=1}^\infty$ in $\mathcal{L}^2(\rho_x)$, where u_n is the bounded step function $u_n(t) = \chi_{(-n, n]}(t)$. Therefore,

$$\hat{E}(u)x = \lim_{n \rightarrow +\infty} E(\chi_{(-n, n]})x = \lim_{n \rightarrow +\infty} (E(n) - E(-n))x = x;$$

4. This follows by applying Proposition 5.4 to $\varphi_n u$ and $\psi_n v$ and then by taking the limit $n \rightarrow \infty$;
5. The first part follows from 4. for $v(t) = 1$ and the second part follows from the Dominated Convergence Theorem.
6. For all $x \in D(\hat{E}(u))$:

$$\langle x, \hat{E}(u)x \rangle = \|\hat{E}(u)x\|^2 = \int |u(t)|^2 d\rho_x \geq c^2 \cdot \rho_x(\mathbb{R}) = c^2 \cdot \|x\|^2.$$

7. It is clear that $D(a\hat{E}(u) + b\hat{E}(v)) := D(\hat{E}(u)) \cap D(\hat{E}(v)) = D(\hat{E}(|u| + |v|))$.
Let $x \in D(a\hat{E}(u) + b\hat{E}(v)) = D(\hat{E}(u)) \cap D(\hat{E}(v))$. Then $u, v \in \mathcal{L}^2(\rho_x)$, so $au + bv \in \mathcal{L}^2(\rho_x)$ since $\mathcal{L}^2(\rho_x)$ is a vector space.

Furthermore, we have

$$\hat{E}(au + bv)x = a\hat{E}(u)x + b\hat{E}(v)x$$

by Proposition 5.1, so $a\hat{E}(u) + b\hat{E}(v) \subset \hat{E}(au + bv)$;

8. If u, v are bounded E -measurable functions, then $D(\hat{E}(u)) = D(\hat{E}(v)) = D(\hat{E}(u)\hat{E}(v)) = H$ and $\forall x, y \in H$:

$$\begin{aligned}\langle y, \hat{E}(u)\hat{E}(v)x \rangle &=_{3.} \langle \hat{E}(1)y, \hat{E}(u)\hat{E}(v)x \rangle =_{4.} \int u(t) d\langle y, E(t)\hat{E}(v)x \rangle \\ &= \int \overline{u(t)} d\langle y, E(t)\hat{E}(v)x \rangle =_{4.} \langle \hat{E}(\overline{u})y, \hat{E}(1)\hat{E}(v)x \rangle \\ &=_{3.} \langle \hat{E}(\overline{u})y, \hat{E}(v)x \rangle =_{4.} \int u(t)v(t) d\langle y, E(t)x \rangle \\ &=_{4.} \langle y, \hat{E}(uv)x \rangle\end{aligned}$$

where we used points 3. and 4. extensively.

Thus, $\langle y, (\hat{E}(u)\hat{E}(v) - \hat{E}(uv))x \rangle = 0 \ \forall x, y \in H$ and we conclude by taking $x \in H$, $y = (\hat{E}(u)\hat{E}(v) - \hat{E}(uv))x \in H$.

Now, let $x \in D(\hat{E}(u)\hat{E}(v)) = \{x \in D(\hat{E}(v)) \mid \hat{E}(v)x \in D(\hat{E}(u))\}$. Then $\varphi_n u$ converges to u in $\mathcal{L}^2(\rho_{\hat{E}(v)x})$ and $\psi_m v$ converges to v in $\mathcal{L}^2(\rho_x)$. For a fixed $n \in \mathbb{N}$, $\varphi_n u$ is bounded, so $\varphi_n u \psi_m v$ converges to $\varphi_n uv$ in $\mathcal{L}^2(\rho_x)$. Consequently,

$$\begin{aligned} \hat{E}(u)\hat{E}(v)x &= \lim_{n \rightarrow +\infty} \hat{E}(\varphi_n u) \left(\lim_{m \rightarrow +\infty} \hat{E}(\psi_m v)x \right) \\ &= \lim_{n \rightarrow +\infty} \lim_{m \rightarrow +\infty} \hat{E}(\varphi_n u) \hat{E}(\psi_m v)x \\ &= \lim_{n \rightarrow +\infty} \lim_{m \rightarrow +\infty} \hat{E}(\varphi_n u \psi_m v)x = \lim_{n \rightarrow +\infty} \hat{E}(\varphi_n uv)x \end{aligned}$$

where the third equality follows from the bounded case. Therefore, $(\hat{E}(\varphi_n uv)x)_{n=1}^\infty \subset H$ is a Cauchy sequence, so $(\varphi_n uv)_{n=1}^\infty \subset \mathcal{L}^2(\rho_x)$ is a Cauchy sequence as well (because of point 1.).

Therefore, we have a converging sequence $(\varphi_n uv)_{n=1}^\infty \subset \mathcal{L}^2(\rho_x)$ that also converges pointwisely to uv almost everywhere. It follows from Theorem 3.12 that $(\varphi_n uv)_{n=1}^\infty$ converges to uv in $\mathcal{L}^2(\rho_x)$. In particular, $uv \in \mathcal{L}^2(\rho_x)$ (i.e. $x \in D(\hat{E}(uv))$) and

$$\hat{E}(u)\hat{E}(v)x = \lim_{n \rightarrow +\infty} \hat{E}(\varphi_n uv)x = \hat{E}(uv)x;$$

We have proved $D(\hat{E}(u)\hat{E}(v)) \subset D(\hat{E}(v)) \cap D(\hat{E}(uv))$. Now, let $x \in D(\hat{E}(v)) \cap D(\hat{E}(uv))$. We have to show $\hat{E}(v)x \in D(\hat{E}(u))$. We have

$$\begin{aligned} \hat{E}(uv)x &= \lim_{n \rightarrow +\infty} \hat{E}(\varphi_n uv)x = \lim_{n \rightarrow +\infty} \lim_{m \rightarrow +\infty} \hat{E}(\varphi_n u \psi_m v)x \\ &= \lim_{n \rightarrow +\infty} \lim_{m \rightarrow +\infty} \hat{E}(\varphi_n u) \hat{E}(\psi_m v)x \\ &= \lim_{n \rightarrow +\infty} \hat{E}(\varphi_n u) \left(\lim_{m \rightarrow +\infty} \hat{E}(\psi_m v)x \right) \\ &= \lim_{n \rightarrow +\infty} \hat{E}(\varphi_n u) \hat{E}(v)x \end{aligned}$$

where the third equality follows from the bounded case. Therefore $(\hat{E}(\varphi_n u)\hat{E}(v)x)_{n=1}^\infty \subset H$ is a Cauchy sequence, so $(\varphi_n u)_{n=1}^\infty \subset \mathcal{L}^2(\rho_{\hat{E}(v)x})$ is a Cauchy sequence as well. We conclude by Theorem 3.12 that $(\varphi_n u)_{n=1}^\infty$ converges to u in $\mathcal{L}^2(\rho_{\hat{E}(v)x})$ as before. Thus, $u \in \mathcal{L}^2(\rho_{\hat{E}(v)x})$, i.e. $\hat{E}(v)x \in D(\hat{E}(u))$.

9. First, we need to show that $D(\hat{E}(u))$ is dense. Let $x \in H$. Consider the sequence $(x_n)_{n=1}^\infty \subset H$, $x_n = \hat{E}(\varphi_n)x$. We have $\lim_{n \rightarrow +\infty} x_n = \lim_{n \rightarrow +\infty} \hat{E}(\varphi_n)x = \hat{E}(1)x = x$ because $(\varphi_n)_{n=1}^\infty$ converges to 1 in $\mathcal{L}^2(\rho_x)$. We show that $(x_n)_{n=1}^\infty \subset D(\hat{E}(u))$. Let $n \in \mathbb{N}$ be fixed. Since $\varphi_m u$ is bounded, we have

$$\int |\varphi_m(t)u(t)|^2 d\rho_{x_n} = \|\hat{E}(\varphi_m u)\hat{E}(\varphi_n)x\| =_8. \|\hat{E}(\varphi_n u)x\| < +\infty$$

for all $m \geq n$. By the Monotone Convergence Theorem, $u \in \mathcal{L}^2(\rho_{x_n})$, i.e. $x_n \in D(\hat{E}(u))$.

Now, $D(\hat{E}(u)) = D(\hat{E}(\bar{u}))$ because $u \in \mathcal{L}^2(\rho_x)$ if and only if $\bar{u} \in \mathcal{L}^2(\rho_x)$.
Next, we have $\hat{E}(\bar{u}) \subset \hat{E}(u)^*$ because $\hat{E}(\bar{u})$ is a formal adjoint of $\hat{E}(u)$:

$$\begin{aligned} \langle y, \hat{E}(u)x \rangle &=_{3.} \langle \hat{E}(1)y, \hat{E}(u)x \rangle =_{4.} \int u(t) d\langle y, E(t)x \rangle \\ &= \int \overline{u(t)} d\langle y, E(t)x \rangle =_{4.} \langle \hat{E}(\bar{u})y, \hat{E}(1)x \rangle \\ &=_{3.} \langle \hat{E}(\bar{u})y, x \rangle \quad \forall x, y \in D(\hat{E}(u)) \end{aligned}$$

where we used 3. and 4. extensively.

Finally, we need to show $\hat{E}(u)^* \subset \hat{E}(\bar{u})$. It suffices to show $D(\hat{E}(u)^*) \subset D(\hat{E}(\bar{u}))$.
Let $y \in D(\hat{E}(u)^*)$ and $x \in D(\hat{E}(u))$. We have

$$\begin{aligned} \langle \hat{E}(u)^*y, x \rangle &= \langle y, \hat{E}(u)x \rangle \\ &= \lim_{n \rightarrow \infty} \langle y, \hat{E}(\varphi_n u)x \rangle \\ &=_{3.,4.} \lim_{n \rightarrow \infty} \langle \hat{E}(\varphi_n \bar{u})y, x \rangle \quad \forall x \in D(\hat{E}(u)), \quad \forall y \in D(\hat{E}(u)^*) \end{aligned}$$

by definition of the adjoint.

From this last identity, observe that

$$\begin{aligned} \langle \hat{E}(\varphi_m) \hat{E}(u)^*y, x \rangle &=_{3.,4.} \langle \hat{E}(u)^*y, \hat{E}(\varphi_m)x \rangle \\ &= \lim_{n \rightarrow \infty} \langle \hat{E}(\varphi_n \bar{u})y, \hat{E}(\varphi_m)x \rangle \\ &=_{3.,4.} \lim_{n \rightarrow \infty} \langle \hat{E}(\varphi_m) \hat{E}(\varphi_n \bar{u})y, x \rangle \\ &=_{8.} \lim_{n \rightarrow \infty} \langle \hat{E}(\varphi_m \varphi_n \bar{u})y, x \rangle \\ &= \langle \hat{E}(\varphi_m \bar{u})y, x \rangle \quad \forall m \in \mathbb{N}, \quad \forall x \in H, \quad \forall y \in D(\hat{E}(u)^*). \end{aligned}$$

where we used the fact $\hat{E}(\varphi_m)x \in D(\hat{E}(u))$ as seen previously.

Therefore, we have proved

$$\langle (\hat{E}(\varphi_m) \hat{E}(u)^* - \hat{E}(\varphi_m \bar{u}))y, x \rangle = 0 \quad \forall m \in \mathbb{N}, \quad \forall x \in H, \quad \forall y \in D(\hat{E}(u)^*).$$

Take $y \in D(\hat{E}(u)^*)$, $x = (\hat{E}(\varphi_m) \hat{E}(u)^* - \hat{E}(\varphi_m \bar{u}))y \in H$ to conclude that

$$\hat{E}(\varphi_m) \hat{E}(u)^*y = \hat{E}(\varphi_m \bar{u})y \quad \forall m \in \mathbb{N}, \quad \forall y \in D(\hat{E}(u)^*)$$

By taking the limit $m \rightarrow +\infty$, we get

$$\hat{E}(u)^*y = \lim_{m \rightarrow +\infty} \hat{E}(\varphi_m \bar{u})y \quad \forall y \in D(\hat{E}(u)^*)$$

Therefore, for a fixed $y \in D(\hat{E}(u)^*)$, $(\hat{E}(\varphi_m \bar{u})y)_{m=1}^\infty \subset H$ is a Cauchy sequence, so $(\varphi_m \bar{u})_{m=1}^\infty \subset \mathcal{L}^2(\mathbb{K}, \rho_y)$ is a Cauchy sequence as well (because of point 1.).

Therefore, we have a converging sequence $(\varphi_m \bar{u})_{m=1}^\infty \subset \mathcal{L}^2(\rho_y)$ that also converges pointwisely to $\bar{u}$ almost everywhere. It follows from Theorem 3.12 that $(\varphi_m \bar{u})_{m=1}^\infty$ converges to $\bar{u}$ in $\mathcal{L}^2(\rho_y)$. In particular, $\bar{u} \in \mathcal{L}^2(\rho_y)$ (i.e. $y \in D(\hat{E}(\bar{u}))$).

□

Corollary 5.7

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and let $(E_t)_{t \in \mathbb{R}}$ be a spectral family on H . Let $u : \mathbb{R} \rightarrow \mathbb{R}$ be an E -measurable function. Then $\hat{E}(u)$ is self-adjoint.

Theorem 5.8 (Change of variables for spectral integration)

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and let $(E_t)_{t \in \mathbb{R}}$ be a spectral family on H .

Let $\phi : \mathbb{R} \rightarrow \mathbb{R}$ be an increasing function such that ϕ is a homeomorphism.

The pushforward $(F_t)_{t \in \mathbb{R}}$, $F(t) = E(\phi^{-1}(t))$, defines a spectral family on H .

Furthermore, if $u(t) : \mathbb{R} \rightarrow \mathbb{R}$ is a F -measurable function, then $u \circ \phi$ is E -measurable and

$$\int u(t) dF(t) = \int u \circ \phi(t) dE(t).$$

Proof.

It is easy to check that $(F_t)_{t \in \mathbb{R}}$ defines a spectral family on H .

Let $u(t) : \mathbb{R} \rightarrow \mathbb{R}$ be a F -measurable function. For $x \in H$, the measure $d\|F(t)x\|^2$ is the pushforward measure by ϕ of the measure $d\|E(t)x\|^2$ and conversely, $d\|E(t)x\|^2$ is the pushforward measure by ϕ^{-1} of $d\|F(t)x\|^2$. It follows from Theorem 3.9 that

$$\begin{aligned} D(\hat{F}(u)) &= \{x \in H \mid u(t) \in \mathcal{L}^2(\|F(t)x\|^2)\} \\ &= \{x \in H \mid u \circ \phi(t) \in \mathcal{L}^2(\|E(t)x\|^2)\} = D(\hat{E}(u \circ \phi)) \end{aligned}$$

and

$$\begin{aligned} \langle y, \hat{F}(u)x \rangle &= \lim_{n \rightarrow +\infty} \int \varphi_n(t) u(t) d\langle y, F(t)x \rangle \\ &= \lim_{n \rightarrow +\infty} \int \varphi_n(\phi(t)) u(\phi(t)) d\langle y, E(t)x \rangle \\ &= \lim_{n \rightarrow +\infty} \int \psi_n(t) u(\phi(t)) d\langle y, E(t)x \rangle = \langle y, \hat{E}(u \circ \phi)x \rangle \quad \forall y \in H, \forall x \in D(\hat{F}(u)) \end{aligned}$$

where we used point 4. of Theorem 5.5 and where

$$\varphi_n(t) = \begin{cases} 1 & \text{if } |u(t)| \leq n \\ 0 & \text{otherwise} \end{cases}, \quad \varphi_n(\phi(t)) = \psi_n(t) = \begin{cases} 1 & \text{if } |u(\phi(t))| \leq n \\ 0 & \text{otherwise} \end{cases}$$

Thus, $\hat{F}(u) = \hat{E}(u \circ \phi)$.

□

Chapter 6

The Spectral Theorem

Definition 6.1 (Complexification of a real Hilbert space)

Let $(H, \langle \cdot, \cdot \rangle)$ be a real Hilbert space. The complexification of H is the complex Hilbert space $H_{\mathbb{C}} = H \times H$ defined as follows :

1. Addition : $(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$
2. Scalar multiplication : $(a + ib)(x, y) = (ax - by, ay + bx)$
3. Scalar product : $\langle (x_1, y_1), (x_2, y_2) \rangle = \langle x_1, x_2 \rangle + i\langle x_1, y_2 \rangle - i\langle y_1, x_2 \rangle + \langle y_1, y_2 \rangle$
4. Induced norm : $\|(x, y)\|^2 = \langle (x, y), (x, y) \rangle = \|x\|^2 + \|y\|^2$

For a linear operator $T : D(T) \subset H \rightarrow H$, the complexification of T is the operator

$$T_{\mathbb{C}} : D(T_{\mathbb{C}}) = D(T) \times D(T) \subset H_{\mathbb{C}} \rightarrow H_{\mathbb{C}}, \quad (x, y) \mapsto (Tx, Ty)$$

It is easy to check that

1. $T_{\mathbb{C}}$ is bounded if and only if T is bounded. In that case, $\|T_{\mathbb{C}}\| = \|T\|$.
2. $T_{\mathbb{C}}$ is densely defined if and only if T is densely defined.
3. If T is densely defined, $(T_{\mathbb{C}})^* = (T^*, T^*)_{\mathbb{C}} = (T^*)_{\mathbb{C}}$.
4. $T_{\mathbb{C}}$ is self-adjoint if and only if T is self-adjoint.
5. If $D(T) = H$, $T_{\mathbb{C}}$ is symmetric if and only if T is symmetric.

Theorem 6.1 (Spectral Theorem for Self-Adjoint operators)

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space on $\mathbb{K}$. For every self-adjoint operator $T : D(T) \rightarrow H$, there exists a unique spectral family $(E_t)_{t \in \mathbb{R}}$ on H such that $T = \hat{E}(id) = \int t dE(t)$.

Furthermore, in the complex case, this spectral family satisfies

$$\langle y, (E(b) - E(a))x \rangle = \lim_{\delta \rightarrow 0^+} \lim_{\epsilon \rightarrow 0^+} \frac{-1}{2\pi i} \int_{a+\delta}^{b+\delta} \langle y, (R(t - i\epsilon, T) - R(t + i\epsilon, T))x \rangle dt \quad (\star)$$

for all $x, y \in H$ and $a \leq b \in \mathbb{R}$.

This spectral family is called the spectral resolution of T .

Proof. First, we assume $\mathbb{K} = \mathbb{C}$.

Uniqueness :

If $T = \hat{E}(id)$ for some spectral family $(E_t)_{t \in \mathbb{R}}$, then $z - T = \hat{E}(z - id)$ for all constant $z \in \mathbb{C}$ by point 7. of Theorem 5.5 (Spectral Calculus). For all $z \in \mathbb{C}$ with $\Im z \neq 0$, $u_z(t) = (z - t)^{-1}$ is bounded, so $\hat{E}(u_z)$ is defined on the whole H .

For such a z , we have $(zI - T)\hat{E}(u_z) = \hat{E}((z - t)u_z) = \hat{E}(1) = I$ on H and $\hat{E}(u_z)(zI - T) = \hat{E}(u_z(z - t)) = \hat{E}(1) = I$ on $D(T)$ by point 8. of Theorem 5.5. Thus, $(zI - T)^{-1} = R(z, T) = \hat{E}(u_z)$.

Consequently,

$$\langle x, R(z, T)x \rangle = \langle x, \hat{E}(u_z)x \rangle = \int (z - t)^{-1} d\langle x, E(t)x \rangle = \int (z - t)^{-1} d\rho_x \quad \forall x \in H$$

by point 5. of Theorem 5.5.

Since $F_x(z) = \langle x, R(z, T)x \rangle$ is holomorphic on $\{z \in \mathbb{C} \mid \Im z \neq 0\}$ (by Theorem 2.38) and $\rho_x(t) = \langle x, E(t)x \rangle = \|E(t)x\|^2$ is right-continuous and bounded, we can apply the Stieltjes Inversion Formula (Theorem 4.4) to get

$$\langle x, E(t)x \rangle = \lim_{\delta \rightarrow 0^+} \lim_{\epsilon \rightarrow 0^+} \frac{-1}{\pi} \int_{-\infty}^{t+\delta} \Im \langle x, R(s + i\epsilon, T)x \rangle ds \quad \forall t \in \mathbb{R}.$$

Since T is self-adjoint, for z with $\Im z \neq 0$, we have $(z - T)^* = \bar{z} - T$ and so $R(z, T)^* = R(\bar{z}, T)$ by Theorem 2.18. This implies

$$\langle x, R(z, T)x \rangle = \langle R(\bar{z}, T)x, x \rangle = \overline{\langle x, R(\bar{z}, T)x \rangle}.$$

Therefore,

$$\langle x, E(t)x \rangle = \lim_{\delta \rightarrow 0^+} \lim_{\epsilon \rightarrow 0^+} \frac{-1}{2\pi i} \int_{-\infty}^{t+\delta} \langle x, (R(s - i\epsilon, T) - R(s + i\epsilon, T))x \rangle ds.$$

We recover $(\star)$ using the Polarization Identity on the sesquilinear forms $\langle y, E(t)x \rangle$ and $\langle y, (R(s - i\epsilon, T) - R(s + i\epsilon, T))x \rangle$.

We have proved that any spectral family such that $T = \hat{E}(id)$ satisfies $(\star)$. Thus, $\langle y, (E_1(t) - E_2(t))x \rangle = 0 \quad \forall t \in \mathbb{R}, \quad \forall x, y \in H$ for any two spectral families such that $T = \hat{E}_1(id) = \hat{E}_2(id)$ (set $a = 0$ or $b = 0$ in $(\star)$). Consequently, $E_1 = E_2$.

Existence :

We have already proved above that if $T = \hat{E}(id)$ for some spectral family $(E_t)_{t \in \mathbb{R}}$, then the family satisfies $(\star)$. Therefore, we just need to prove the existence of a spectral family with $T = \hat{E}(id)$.

For $x \in H$, define $F_x(z) = \langle x, R(z, T)x \rangle$ on $\mathbb{H} = \{z \in \mathbb{C} \mid \Im z > 0\}$. Then

$$\Im F_x(z) = \Im \langle x, R(z, T)x \rangle = \Im \langle (zI - T)R(z, T)x, R(z, T)x \rangle = \Im(\bar{z}) \cdot \|R(z, T)\|^2 \leq 0 \quad \text{and}$$

$$|F_x(z)\Im z| \leq |\Im z| \cdot \|R(z, T)\| \cdot \|x\|^2 \leq |\Im z| \cdot \frac{1}{|\Im z|} \cdot \|x\|^2 = \|x\|^2$$

using the bound on $\|R(z, T)\|$ from Proposition 2.33.

By the Herglotz Representation Theorem (Theorem 4.6), for every $x \in H$, there exists a unique right-continuous and non-decreasing function $w(x, t) : \mathbb{R} \rightarrow \mathbb{R}$ for which $\lim_{t \rightarrow -\infty} w(x, t) = 0$ and

$$F_x(z) = \int_{-\infty}^{+\infty} (z - t)^{-1} dw(x, t) \quad \forall z \in \mathbb{H}.$$

Furthermore, $w(x, t)$ satisfies $w(x, t) \leq \|x\|^2 \forall t \in \mathbb{R}$ and

$$\begin{aligned} w(x, t) &= \lim_{\delta \rightarrow 0^+} \lim_{\epsilon \rightarrow 0^+} \frac{-1}{\pi} \int_{-\infty}^{t+\delta} \Im F_x(s + i\epsilon) ds \\ &= \lim_{\delta \rightarrow 0^+} \lim_{\epsilon \rightarrow 0^+} \frac{-1}{2\pi i} \int_{-\infty}^{t+\delta} \langle x, (R(s - i\epsilon, T) - R(s + i\epsilon, T)) x \rangle ds \forall t \in \mathbb{R}. \end{aligned}$$

Notice that if we extend the domain of $F_x(z) = \langle x, R(z, T)x \rangle$ to $\mathbb{C} \setminus \mathbb{R}$, the equality $F_x(z) = \int_{-\infty}^{+\infty} (z - t)^{-1} dw(x, t)$ still holds.

Next, for $x, y \in H$, we define

$$\begin{aligned} w(y, x, t) &= \frac{1}{4} (w(x + y, t) - w(x - y, t) + iw(x - iy, t) - iw(x + iy, t)) \\ &= \lim_{\delta \rightarrow 0^+} \lim_{\epsilon \rightarrow 0^+} \frac{-1}{2\pi i} \int_{-\infty}^{t+\delta} \langle y, (R(s - i\epsilon, T) - R(s + i\epsilon, T)) x \rangle ds \forall t \in \mathbb{R}. \end{aligned}$$

and it follows from the Polarization Identity for sesquilinear forms that for $z \in \mathbb{C} \setminus \mathbb{R}$:

$$\begin{aligned} \langle y, R(z, T)x \rangle &= \frac{1}{4} (\langle x + y, R(z, T)(x + y) \rangle - \langle x - y, R(z, T)(x - y) \rangle \\ &\quad + i\langle x - iy, R(z, T)(x - iy) \rangle - i\langle x + iy, R(z, T)(x + iy) \rangle) \\ &= \int_{-\infty}^{+\infty} (z - t)^{-1} dw(y, x, t). \end{aligned}$$

Then for all $t \in \mathbb{R}$, $w(y, x, t)$ is a semi-scalar product on H . By the Cauchy-Schwarz Inequality,

$$|w(y, x, t)| \leq w(y, y, t) \cdot w(x, x, t) \leq \|y\|^2 \cdot \|x\|^2.$$

Consequently, for all $t \in \mathbb{R}$, $x \in H$ fixed, the map $y \mapsto w(y, x, t)$ defines a continuous linear functional on H . By the Riesz Representation Theorem, there exists a unique element $\tilde{y} = \tilde{y}(x, t) = E(t)f \in H$ such that $w(y, x, t) = \langle y, \tilde{y} \rangle = \langle y, E(t)x \rangle$. The family of maps $E(t) : H \rightarrow H$ defines a spectral family on H such that $T = \hat{E}(id)$.

1. $E(t) : H \rightarrow H$ is linear by unicity of the element of Riesz Representation Theorem and linearity of $\langle y, \cdot \rangle$.
2. $E(t)$ is self-adjoint because $\langle E(t)y, x \rangle = \overline{\langle x, E(t)y \rangle} = \overline{w(x, y, t)} = w(y, x, t) = \langle y, E(t)x \rangle$ by definition of $E(t)$.
3. $E(t) \geq 0$ because $\langle x, E(t)x \rangle = w(x, x, t) \geq 0$.
4. $E(s)E(t) = E(\min\{s, t\})$: Note that $\forall z, z' \in \mathbb{C} \setminus \mathbb{R}$, $z \neq z'$, $\forall x, y \in H$, we have

$$\begin{aligned} \int_{-\infty}^{+\infty} (z - t) d\langle R(\bar{z}', T)y, E(t)x \rangle &= \langle R(\bar{z}', T)y, R(z, T)x \rangle = \langle y, R(z', T)R(z, T)x \rangle \\ &= (z' - z)^{-1} \langle y, (R(z', T) - R(z, T))x \rangle \\ &= (z' - z)^{-1} \int_{-\infty}^{+\infty} [(z' - t)^{-1} - (z - t)^{-1}] d\langle y, E(t)x \rangle \\ &= \int_{-\infty}^{+\infty} (z - t)^{-1} (z' - t)^{-1} d\langle y, E(t)x \rangle \end{aligned}$$

where we used the first Resolvent Identity. By Proposition 4.3, we have

$$\int_{-\infty}^{+\infty} (z - t)^{-1} (z' - t)^{-1} d\langle y, E(t)x \rangle = \int_{-\infty}^{+\infty} (z - t)^{-1} d\Phi(t), \quad \Phi(t) = \int_{(-\infty, t]} (z' - s)^{-1} d\langle y, E(s)x \rangle$$

and it follows from Theorem 4.5 that

$$\langle R(\bar{z}', T)y, E(t)x \rangle = \Phi(t) = \int_{(-\infty, t]} (z' - s)^{-1} d\langle y, E(s)x \rangle \quad \forall t \in \mathbb{R}.$$

But we also have

$$\langle R(\bar{z}', T)y, E(t)x \rangle = \langle y, R(z', T)E(t)x \rangle = \int_{-\infty}^{+\infty} (z' - s)^{-1} d\langle y, E(s)E(t)x \rangle.$$

By Proposition 4.3,

$$\begin{aligned} \Phi(t) &= \int_{(-\infty, t]} (z' - s)^{-1} d\langle y, E(s)x \rangle = \int_{-\infty}^{+\infty} (z' - s)^{-1} d\Psi_t(s), \text{ where} \\ \Psi_t(s) &= \int_{(-\infty, s]} \chi_{(-\infty, t]}(r) d\langle y, E(r)x \rangle = \int_{(-\infty, \min\{s, t\}]} 1 d\langle y, E(r)x \rangle = \langle y, E(\min\{s, t\})x \rangle \end{aligned}$$

where we used $\lim_{t \rightarrow -\infty} w(y, x, t) = 0 \quad \forall x, y \in H$ for the last equality.

We conclude by Theorem 4.5 that $\langle y, E(s)E(t)x \rangle = \Psi_t(s)$.

5. In particular, $E(t)^2 = E(t)$, so $E(t)$ is an orthogonal projection and $E(s) \leq E(t) \quad \forall s \leq t$ (see Section 2.3).
6. $w(x, t) = w(x, x, t) = \langle x, E(t)x \rangle = \langle x, E(t)^2x \rangle = \langle E(t)x, E(t)x \rangle = \|E(t)x\|^2$.
7. $E(t) \xrightarrow[t \rightarrow -\infty]{s} 0$:

$$\|E(t)x - 0\|^2 = w(x, x, t) = w(x, t) \xrightarrow[t \rightarrow -\infty]{} 0.$$

8. $E(t + \epsilon) \xrightarrow[\epsilon \rightarrow 0^+]{s} E(t)$:

$$\begin{aligned} \|(E(t + \epsilon) - E(t))x\|^2 &= \|E(t + \epsilon)x\|^2 + \|E(t)x\|^2 - 2\langle E(t + \epsilon)x, E(t)x \rangle \\ &= \|E(t + \epsilon)x\|^2 - \|E(t)x\|^2 \\ &= w(x, x, t + \epsilon) - w(x, x, t) = w(x, t + \epsilon) - w(x, t) \xrightarrow[\epsilon \rightarrow 0^+]{} 0 \end{aligned}$$

because $w(x, t)$ is right-continuous.

9. $E(t) \xrightarrow[t \rightarrow +\infty]{s} I$: Since $(E_t)_{t > -\infty}$ is an increasing sequence of orthogonal projections, $E(t)$ converges strongly to an orthogonal projection $E(\infty)$ as $t \rightarrow +\infty$ (by Theorem 2.28 applied to the sequence of operators $(I - E(n))_{n=1}^\infty$).

Because of properties 4. and 5. of $E(t)$, $E(\infty)^2 = E(\infty)$, $E(t) \leq E(\infty)$ and $E(t)E(\infty) = E(\infty)E(t) = E(t) \quad \forall t \in \mathbb{R}$. Finally,

$$\begin{aligned} \langle y, R(z, T)(I - E(\infty))x \rangle &= \int (z - t)^{-1} dw(y, (I - E(\infty))x, t) \\ &= \int (z - t)^{-1} d\langle y, E(t)(I - E(\infty))x \rangle \\ &= \int (z - t)^{-1} d\langle y, 0 \rangle = 0 \quad \forall x, y \in H, \quad \forall z \in \mathbb{C} \setminus \mathbb{R}. \end{aligned}$$

This implies $R(z, T)(I - E(\infty)) = 0 \quad \forall z \in \mathbb{C} \setminus \mathbb{R}$. Since $R(z, T)$ is injective for any $z \in \mathbb{C} \setminus \mathbb{R}$, it follows that $I = E(\infty)$.

This proves that $(E_t)_{t \in \mathbb{R}}$ is a spectral family.

Finally, $\langle y, R(z, T)x \rangle = \int (z - t)^{-1} d\langle y, E(t)x \rangle = \langle y, \hat{E}(u_z)x \rangle \forall z \in \mathbb{C} \setminus \mathbb{R}, \forall x, y \in H$ by point 5. of Theorem 5.5 (Spectral Calculus), so it follows that $R(z, T) = \hat{E}(u_z)$, which implies $\hat{E}(z - id) = z - T$ and $\hat{E}(id) = T$ by points 7. and 8. of Theorem 5.5.

Real Case :

Let $H_{\mathbb{C}}$ be the complexification of H and $E_{\mathbb{C}}$ be the spectral resolution of the complexification $T_{\mathbb{C}}$ of T . Let $U : H_{\mathbb{C}} \rightarrow H_{\mathbb{C}}, U(x, y) = (x, -y)$ be the complex conjugation map. Then $T_{\mathbb{C}}U(x, y) = UT_{\mathbb{C}}(x, y) \forall (x, y) \in D(T_{\mathbb{C}})$. Indeed, $U \circ U = I$ and $UT_{\mathbb{C}}U(x, y) = UT_{\mathbb{C}}(x, -y) = U(Tx, -Ty) = T_{\mathbb{C}}(x, y)$.

Define $F(t) = UE_{\mathbb{C}}(t)U$. Then $F(t)$ defines a spectral family. Most properties follow from the properties of $E_{\mathbb{C}}(t)$. The only possible problematic points are the symmetry and the property $F(t) \leq F(s) \forall t \leq s$. Note that $\langle U(x_1, y_1), (x_2, y_2) \rangle = \langle (-x_1, y_1), -(x_2, y_2) \rangle$ and $\langle (x_1, y_1), U(x_2, y_2) \rangle = \langle -(x_1, y_1), (-x_2, y_2) \rangle$, so

$$\begin{aligned} \langle F(t)(x, y), (x, y) \rangle &= \langle UE_{\mathbb{C}}(t)U(x, y), (x, y) \rangle = \langle E_{\mathbb{C}}(t)(x, y), (x, y) \rangle \text{ and} \\ \langle (x, y), F(t)(x, y) \rangle &= \langle (x, y), UE_{\mathbb{C}}(t)U(x, y) \rangle = \langle (x, y), E_{\mathbb{C}}(t)(x, y) \rangle \end{aligned}$$

and both properties follow. In particular, notice that $\rho_{(x, y)}(t) = \|F(t)(x, y)\|^2 = \|E_{\mathbb{C}}(t)(x, y)\|^2$, so $E_{\mathbb{C}}$ -measurability is equivalent to F -measurability.

For a real-valued step function $g : \mathbb{R} \rightarrow \mathbb{R}$, it is clear that

$$\int g(t) dF(t) = U \left(\int g(t) dE_{\mathbb{C}}(t) \right) U$$

and this is also true for an arbitrary real-valued F -measurable function g by taking, for a fixed $(x, y) \in \{(x, y) \in H_{\mathbb{C}} \mid g \in \mathcal{L}^2(\rho_{(x, y)})\}$, a sequence $g_n(t) \in \mathcal{L}^2(\rho_{(x, y)})$ of step functions converging to g .

In particular,

$$T_{\mathbb{C}} = UT_{\mathbb{C}}U = U \left(\int t dE_{\mathbb{C}}(t) \right) U = \int t dF(t)$$

so we must have $F(t) = E_{\mathbb{C}}(t)$ by unicity of the spectral resolution of $T_{\mathbb{C}}$, i.e. $E_{\mathbb{C}}(t)U = UE_{\mathbb{C}}(t)$.

In particular, $E_{\mathbb{C}}(t)(x, 0) = E_{\mathbb{C}}(t)U(x, 0) = UE_{\mathbb{C}}(t)(x, 0) \forall x \in H$, which implies $E_{\mathbb{C}}(t)(x, 0) = (E(t)x, 0)$ for some $E(t) : H \rightarrow H$. It follows from the properties of $E_{\mathbb{C}}(t)$ that $E(t)$ is a spectral family.

Finally, for a real-valued step function $g : \mathbb{R} \rightarrow \mathbb{R}$, we have

$$\int g(t) dE_{\mathbb{C}}(t)(x, 0) = \left(\int g(t) dE(t)x, 0 \right) \forall x \in H$$

and this is also true for an arbitrary real-valued $E_{\mathbb{C}}$ -measurable function g and $x \in H$ such that $g \in \mathcal{L}^2(\rho_x)$. Therefore,

$$(Tx, 0) = T_{\mathbb{C}}(x, 0) = \int t dE_{\mathbb{C}}(t)(x, 0) = \left(\int t dE(t)x, 0 \right) \forall x \in D(T),$$

which proves the existence of a spectral resolution in the real case.

The uniqueness follows from the fact that if $E(t)$ and $F(t)$ are two spectral resolutions of T , then their complexification $E_{\mathbb{C}}(t)$ and $F_{\mathbb{C}}(t)$ are two spectral resolutions of $T_{\mathbb{C}}$, so

they must be the same. To see that $E_{\mathbb{C}}(t)$ is a spectral resolution of $T_{\mathbb{C}}$, note that for a real-valued step function $g : \mathbb{R} \rightarrow \mathbb{R}$,

$$\int g(t) dE_{\mathbb{C}}(t)(x, y) = \left(\int g(t) dE(t)x, \int g(t) dE(t)y \right) \quad \forall (x, y) \in H_{\mathbb{C}}$$

so this must also hold in the case $g(t) = t$ and $(x, y) \in D(T_{\mathbb{C}})$, where we recover $(Tx, Ty) = T_{\mathbb{C}}(x, y)$ on the right-hand side. $\square$

Proposition 6.2 (An operator commutes with its spectral resolution)

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space, $T : D(T) \rightarrow H$ be a self-adjoint operator and let $(E_t)_{t \in \mathbb{R}}$ be its spectral resolution.

Then $E(t)(D(T)) \subset D(T) \quad \forall t \in \mathbb{R}$ and $E(t)Tx = TE(t)x \quad \forall t \in \mathbb{R}, \forall x \in D(T)$.

Proof.

Without loss of generality, we can assume we work on a complex Hilbert space (otherwise, just take the complexification of H , T and E).

By Spectral Calculus, for $z \in \rho(T)$, $z \notin \mathbb{R}$, we have

$$\begin{aligned} \langle y, R(z, T)E(t)x \rangle &= \langle y, \hat{E}((z - s)^{-1})E(t)x \rangle = \int_{-\infty}^{+\infty} (z - s)^{-1} d\langle y, E(s)E(t)x \rangle \\ &= \int_{-\infty}^{+\infty} (z - s)^{-1} d\langle E(t)y, E(s)x \rangle \\ &= \langle E(t)y, \hat{E}((z - s)^{-1})x \rangle = \langle E(t)y, R(z, T)x \rangle \\ &= \langle y, E(t)R(z, T)x \rangle \quad \forall x, y \in H, \end{aligned}$$

so $E(t)R(z, T) = R(z, T)E(t)$. In particular, $E(t)(D(T)) \subset D(T)$. Indeed, for $x \in D(T)$, we can write $x = R(z, T)y$ for some $y \in H$, so that $E(t)x = E(t)R(z, T)y = R(z, T)E(t)y \in D(T)$.

Now, since $(zI - T)R(z, T) = I$, we have $TR(z, T) = zR(z, T) - I$.

For $x \in D(T)$, we write $x = R(z, T)y$ for some $y \in H$. We get

$$E(t)Tx = E(t)TR(z, T)y = zE(t)R(z, T)y - E(t)y = zR(z, T)E(t)y - E(t)y$$

and

$$TE(t)x = TE(t)R(z, T)y = TR(z, T)E(t)y = zR(z, T)E(t)y - E(t)y.$$

$\square$

Proposition 6.3 (Spectral Calculus can be done on the spectrum)

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space, $T : D(T) \rightarrow H$ be a self-adjoint operator and let $(E_t)_{t \in \mathbb{R}}$ be its spectral resolution.

Then $E(t)$ is locally constant on $\rho(T) \cap \mathbb{R}$.

In particular, for any E -measurable function $u : \mathbb{R} \rightarrow \mathbb{K}$, we have

$$\hat{E}(u) = \int_{\mathbb{R}} u(t) dE(t) = \int_{\sigma(T)} u(t) dE(t) = \hat{E}(u \cdot \chi_{\sigma(T)}).$$

Proof.

Assume $E(t)$ is not locally constant in a neighbourhood of $t^* \in \mathbb{R}$. Then we can find a sequence $(t_n)_{n=1}^\infty \subset \mathbb{R}$ such that $t_n \neq t^* \forall n$, $\lim_{n \rightarrow \infty} t_n = t^*$ and $E(t_n) \neq E(t^*) \forall n$. In particular, since $E(t_n)$ and $E(t^*)$ are continuous and $D(T) \subset H$ is dense, $E(t_n)|_{D(T)} \neq E(t^*)|_{D(T)}$ and for all $n \in \mathbb{N}$, we can find $y_n \in D(T)$ such that $E(t_n)y_n \neq E(t^*)y_n$.

We define

$$\tilde{x}_n = (E(\max\{t_n, t^*\}) - E(\min\{t_n, t^*\}))y_n, \quad x_n = \tilde{x}_n / \|\tilde{x}_n\|, \quad n \in \mathbb{N}.$$

Note that $x_n \in D(T)$ since $E(t)(D(T)) \subset D(T) \forall t \in \mathbb{R}$. Then $t^* \in \sigma(T)$. Indeed, if t^* is an eigenvalue of T , we are done. Otherwise, $t^*I - T$ is invertible with inverse defined on the whole H . We show that its inverse is unbounded.

Note that $\lim_{n \rightarrow \infty} \|(t^*I - T)x_n\| = 0$. If we write $t^*I - T = (t^* - t_n)I + t_nI - T$, we only need to check that $\lim_{n \rightarrow \infty} \|(t_nI - T)x_n\| = 0$. This is true since

$$\begin{aligned} \|(t_nI - T)x_n\|^2 &= \int_{\mathbb{R}} |t_n - t|^2 d\|E(t)x_n\|^2 \\ &= \int_{\mathbb{R}} |t_n - t|^2 d\|E(t)(E(\max\{t_n, t^*\}) - E(\min\{t_n, t^*\}))x_n\|^2 \\ &\leq \int_{[\min\{t_n, t^*\}, \max\{t_n, t^*\}]} |t_n - t|^2 d\|E(t)x_n\|^2 \\ &\leq \max\{|t_n - t|^2 \mid t \in [\min\{t_n, t^*\}, \max\{t_n, t^*\}]\} \end{aligned}$$

Therefore, if we set $y_n = (t^*I - T)x_n$, we found a sequence in H such that $\|(t^*I - T)^{-1}y_n\| = 1 \forall n$ and $\lim_{n \rightarrow +\infty} \|y_n\| = 0$. Therefore, $(t^*I - T)^{-1}$ is not continuous at 0 and so it is unbounded. $\square$

Notation

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space on $\mathbb{K}$, $T : D(T) \rightarrow H$ be a self-adjoint operator and let $(E_t)_{t \in \mathbb{R}}$ be its spectral resolution. For a E -measurable function $u(t) : \mathbb{R} \rightarrow \mathbb{K}$, we write $u(T) = \hat{E}(u) = \int_{\sigma(T)} u(t) dE(t)$.

Corollary 6.4 (Spectrum is non-empty)

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and $T : D(T) \rightarrow H$ be a self-adjoint operator.

The spectrum of T is non-empty.

Proof.

Let $(E_t)_{t \in \mathbb{R}}$ be the spectral resolution of T . If $\sigma(T) = \emptyset$, we would have

$$T = \int_{\sigma(T)} t dE(t) = \int_{\emptyset} t dE(t) = 0.$$

But 0 is an eigenvalue for the trivial operator $T = 0$, so $\sigma(T) \neq \emptyset$. A contradiction. $\square$

Proposition 6.5 (Norm of the resolvent in the bounded case)

Let $(H, \langle \cdot, \cdot \rangle)$ be an Hilbert space and $T : H \rightarrow H$ be a bounded self-adjoint operator.

For any $z_0 \in \rho(T)$, we have $d(z_0, \sigma(T)) = \|R(z_0, T)\|^{-1}$.

Proof.

Let $(E_t)_{t \in \mathbb{R}}$ be the spectral resolution of T . Note that since $\sigma(T)$ is compact (Proposition 2.37), for any continuous function $u : \sigma(T) \rightarrow \mathbb{K}$, $\hat{E}(u) = \int_{\sigma(T)} u(t) dE(t)$ is a bounded operator defined on the whole H . Let $z_0 \in \rho(T)$. Since $z_0 \notin \sigma(T)$, the function $u(t) = (z_0 - t)^{-1}$, $t \in \sigma(T)$, is well-defined and continuous.

Furthermore, $R(z_0, T) = \hat{E}((z_0 - t)^{-1})$ and $\|R(z_0, T)\| \leq \sup\{|z_0 - t|^{-1} \mid t \in \sigma(T)\} = d(z_0, \sigma(T))^{-1}$ by Spectral Calculus.

The other inequality follows from Corollary 2.36. $\square$

Proposition 6.6

Let $(H, \langle \cdot, \cdot \rangle)$ be an Hilbert space and $T : H \rightarrow H$ be a bounded self-adjoint operator.

Then $\|T\| \leq \sup\{|\lambda| \mid \lambda \in \sigma(T)\}$.

Remark 6.7

In fact, this is an equality, but we do not need this result.

Proof.

Let $(E_t)_{t \in \mathbb{R}}$ be the spectral resolution of T .

Then $T = \hat{E}(t)$ and $\|T\| \leq \sup\{|t| \mid t \in \sigma(T)\}$. $\square$

Corollary 6.8 (Norm of the resolvent)

Let $(H, \langle \cdot, \cdot \rangle)$ be an Hilbert space and $T : D(T) \rightarrow H$ be a self-adjoint operator.

For any $z_0 \in \rho(T)$, we have $d(z_0, \sigma(T)) = \|R(z_0, T)\|^{-1}$.

Proof.

For $z_0 \in \rho(T) \cap \mathbb{C}$ (if $\mathbb{K} = \mathbb{C}$), we have that $u(t) = (z_0 - t)^{-1}$ is well-defined and bounded on $\sigma(T) \subset \mathbb{R}$, $R(z_0, T) = \hat{E}((z_0 - t)^{-1})$ and $\|R(z_0, T)\| \leq \sup\{|z_0 - t|^{-1} \mid t \in \sigma(T)\} = d(z_0, \sigma(T))^{-1}$ by Spectral Calculus.

For $z_0 \in \rho(T) \cap \mathbb{R}$, we have that $R(z_0, T)$ is a bounded self-adjoint operator, so

$$\|R(z_0, T)\| \leq \sup\{|t| \mid t \in \sigma(R(z_0, T))\} = \sup\{|z_0 - t|^{-1} \mid t \in \sigma(T)\} = d(z_0, \sigma(T))^{-1}$$

by the previous Proposition and Corollary 2.32.

The other inequality follows from Corollary 2.36. $\square$

Chapter 7

The Hamburger Moment Problem

Definition 7.1 (n -th moment)

Let $(\mathbb{R}, \mathcal{B}, \mu)$ be a measure space, where $\mathcal{B}$ denotes the Borel σ -algebra of $\mathbb{R}$. Let $n \in \mathbb{N}_{\geq 0}$. If $x^n \in \mathcal{L}^1(\mu)$, the real number

$$a_n = \int_{\mathbb{R}} x^n d\mu(x)$$

is called the n -th moment of the measure μ .

The Hamburger Moment Problem asks whether given a sequence $(a_n)_{n=0}^{\infty} \subset \mathbb{R}$, there exists a Borel measure μ on $\mathbb{R}$ for which a_n is the n -th moment of the measure μ for all n .

The above problem can be solved using the Spectral Theorem. Before this, we need to introduce the notion of (densely defined) symmetric operators and ask whether such an operator can be extended into a self-adjoint operator.

7.1 Extension of symmetric operators

Definition 7.2 (Symmetric operator)

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and let $T : D(T) \subset H \rightarrow H$ be a linear operator. We say that T is *symmetric* if it is densely defined and

$$\langle Tx, y \rangle = \langle x, Ty \rangle \quad \forall x, y \in D(T).$$

In other words, T is symmetric if it is densely defined and if it is a formal adjoint of itself. In particular, $T \subset T^*$.

Definition 7.3 (Essentially self-adjoint operator)

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and let $T : D(T) \subset H \rightarrow H$ be a symmetric operator. We say that T is *essentially self-adjoint* if T^* is self-adjoint.

Proposition 7.1

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and let $T : D(T) \subset H \rightarrow H$ be a symmetric operator. Then T is essentially self-adjoint if and only if T^* is symmetric.

Proof.

If T^* is self-adjoint, then T^* is symmetric. Conversely, if T^* is symmetric, we have

$T^* \subset T^{**}$ and since T is symmetric, $T \subset T^*$ and $T^* \supset T^{**}$ by Proposition 2.15, which proves $T^* = T^{**}$. $\square$

Proposition 7.2

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and let $T : D(T) \subset H \rightarrow H$ be a symmetric operator. Then T^{**} is symmetric.

Proof.

T^* is densely defined since $T \subset T^*$, and so T^{**} is well-defined. Furthermore, T^{**} is densely defined as $T \subset T^{**}$.

By Proposition 2.19, $\overline{G(T)} = G(T^{**})$. Let $x, y \in D(T^{**})$ and let $((x_n, Tx_n))_{n=1}^\infty, ((y_n, Ty_n))_{n=1}^\infty \subset G(T)$ be two sequences converging respectively to $(x, T^{**}x), (y, T^{**}y)$ in $H \times H$. Since T is symmetric,

$$\langle T^{**}x, y \rangle = \lim_{n \rightarrow +\infty} \langle Tx_n, y_n \rangle = \lim_{n \rightarrow +\infty} \langle x_n, Ty_n \rangle = \langle x, T^{**}y \rangle.$$

$\square$

Proposition 7.3

Let $(H, \langle \cdot, \cdot \rangle)$ be a complex Hilbert space and let $T : D(T) \rightarrow H$ be a symmetric operator. The following are equivalent :

1. T is self-adjoint;
2. For all $z \in \mathbb{C} \setminus \mathbb{R} : R(zI - T) = R(z^*I - T) = H$;
3. There exists $z \in \mathbb{C} \setminus \mathbb{R}$ such that $R(zI - T) = R(z^*I - T) = H$.

Proof.

(1) $\Rightarrow$ (2) : If T is self-adjoint, by Proposition 2.33, any $z \in \mathbb{C} \setminus \mathbb{R}$ is in $\rho(T)$ and therefore satisfies $R(zI - T) = H$.

(2) $\Rightarrow$ (3) : Trivial.

(3) $\Rightarrow$ (1) : It suffices to prove that $D(T^*) \subset D(T)$.

Let $y \in D(T^*)$, i.e. $\langle y, Tx \rangle = \langle T^*y, x \rangle \forall x \in D(T)$. By hypothesis, there exists $\gamma \in D(T)$ such that $(z^*I - T)\gamma = (z^*I - T^*)y$, where we recall that $(zI - T)^* = z^*I - T^*$ by Propositions 2.15 and 2.16. Therefore,

$$\begin{aligned} \langle y, (zI - T)x \rangle &= \langle (z^*I - T^*)y, x \rangle = \langle (z^*I - T)\gamma, x \rangle \\ &= \langle \gamma, (zI - T)x \rangle \forall x \in D(T) \end{aligned}$$

where the last equality follows from the fact that T is symmetric. Since $R(zI - T) = H$, we can choose $x \in D(T)$ such that $(zI - T)x = y - \gamma$ to deduce that $y = \gamma \in D(T)$. $\square$

Definition 7.4 (Unitary operator)

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space. A *unitary* operator is a bounded linear operator $U : H \rightarrow H$ such that $U^*U = UU^* = I$.

Proposition 7.4

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space. A linear operator $U : H \rightarrow H$ is unitary if and only if it is surjective and $\langle Ux, Uy \rangle = \langle x, y \rangle \forall x, y \in H$.

Proposition 7.5 (On the spectrum of a symmetric operator)

Let $(H, \langle \cdot, \cdot \rangle)$ be a complex Hilbert space and $T : D(T) \rightarrow H$ be a symmetric operator.

If $z \in \mathbb{C}$, $\Im z \neq 0$, we have $\|(zI - T)x\| \geq |\Im z| \cdot \|x\| > 0 \ \forall x \in H$.

In particular, $zI - T$ is injective and $\|(zI - T)^{-1}\| \leq \frac{1}{|\Im z|}$.

Proof.

Since T is symmetric, we have $\langle x, Tx \rangle = \langle Tx, x \rangle = \overline{\langle x, Tx \rangle} \in \mathbb{R} \ \forall x \in D(T)$.

By the Cauchy-Schwarz inequality, we have

$$\|(zI - T)x\| \cdot \|x\| \geq |\langle x, (zI - T)x \rangle| \geq |\Im \langle x, (zI - T)x \rangle| = |\Im z| \cdot \|x\|^2 \ \forall x \in D(T).$$

This implies $\|(zI - T)x\| \geq |\Im z| \cdot \|x\| \ \forall x \in D(T)$ and so the proposition follows. $\square$

Definition 7.5 (Cayley Transform)

Let $(H, \langle \cdot, \cdot \rangle)$ be a complex Hilbert space and let $T : D(T) \rightarrow H$ be a symmetric operator.

The *Cayley Transform* of T at $z \in \mathbb{C} \setminus \mathbb{R}$ is the surjective linear operator

$$V(z, T) : R(T - \bar{z}I) \rightarrow R(T - zI), \quad V(z, T) = (T - zI)(T - \bar{z}I)^{-1}.$$

Proposition 7.6

The Cayley transform $V(z, T)$ preserves the scalar product, i.e.

$$\langle V(z, T)x, V(z, T)y \rangle = \langle x, y \rangle \ \forall x, y \in R(T - \bar{z}I).$$

Proof.

For $u, v \in D(T)$, using the fact that T is symmetric, we have

$$\begin{aligned} \langle (T - zI)u, (T - \bar{z}I)v \rangle &= \langle Tu, Tv \rangle - 2\Re(z)\langle u, Tv \rangle + |z|^2\langle u, v \rangle \\ &= \langle Tu, Tv \rangle - 2\Re(\bar{z})\langle u, Tv \rangle + |\bar{z}|^2\langle u, v \rangle \\ &= \langle (T - \bar{z}I)u, (T - \bar{z}I)v \rangle \end{aligned}$$

Setting $u = (T - \bar{z}I)^{-1}x$, $v = (T - \bar{z}I)^{-1}y$ for $x, y \in R(T - \bar{z}I)$, we get the result. $\square$

Remark 7.7

It follows from the Polarization Identity that a linear operator $V : D(V) \subset H \rightarrow H$ on $(H, \langle \cdot, \cdot \rangle)$ preserves the scalar product of H if and only if it preserves its norm, i.e. if and only if V is isometric.

Corollary 7.8

The Cayley transform $V(z, T)$ is unitary if and only if $R(T - \bar{z}I) = R(T - zI) = H$ if and only if T is self-adjoint.

Proposition 7.9

The Cayley transform $V(z, T)$ is an operator such that $R(I - V(z, T))$ is dense. More precisely, $R(I - V(z, T)) = D(T)$.

Proof.

Notice that

$$\begin{aligned} I - V(z, T) &= I - (T - zI)(T - \bar{z}I)^{-1} \\ &= (T - \bar{z}I)(T - \bar{z}I)^{-1} - (T - zI)(T - \bar{z}I)^{-1} \\ &= (z - \bar{z})(T - \bar{z}I)^{-1}. \end{aligned}$$

Thus, $R(I - V(z, T)) = D(T)$ and $R(I - V(z, T))$ is dense. $\square$

Lemma 7.10

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and $V : D(V) \subset H \rightarrow H$ be an isometric linear operator such that $R(I - V)$ is dense in H . Then $N(I - V) = \{0\}$.

Proof.

Let $x \in N(I - V)$. We show that $x \in R(I - V)^\perp = \{0\}$ (by density). We have

$$\langle x, (I - V)y \rangle = \langle Vx, (I - V)y \rangle = \langle x, y \rangle - \langle Vx, Vy \rangle = \langle x, y \rangle - \langle x, y \rangle = 0 \quad \forall y \in D(V)$$

$\square$

Definition 7.6 (Inverse Cayley Transform)

Let $(H, \langle \cdot, \cdot \rangle)$ be a complex Hilbert space and $V : D(V) \subset H \rightarrow H$ be an isometric linear operator such that $R(I - V)$ is dense in H . By the previous lemma, $I - V$ is injective.

The *Inverse Cayley Transform* of V at $z \in \mathbb{C} \setminus \mathbb{R}$ is the operator $T(z, V) : R(I - V) \rightarrow H$, $T(z, V) = (zI - \bar{z}V)(I - V)^{-1}$.

Proposition 7.11

The inverse Cayley transform $T(z, V)$ is a densely defined symmetric operator with Cayley transform $V(z, T(z, V)) = V$.

Proof.

By definition, $T(z, V)$ is densely defined. For $x_1, x_2 \in D(T(z, V)) = R(I - V)$, we write $x_i = (I - V)y_i$ for some $y_i \in D(I - V)$, $i = 1, 2$. Using the fact that $V(z, T)$ preserves the scalar product, we have

$$\langle T(z, V)x_1, x_2 \rangle = \langle (zI - \bar{z}V)y_1, (I - V)y_2 \rangle = \bar{z}\langle y_1, y_2 \rangle - \bar{z}\langle y_1, Vy_2 \rangle - z\langle Vy_1, y_2 \rangle + z\langle y_1, y_2 \rangle$$

and

$$\langle x_1, T(z, V)x_2 \rangle = \langle (I - V)y_1, (zI - \bar{z}V)y_2 \rangle = z\langle y_1, y_2 \rangle - \bar{z}\langle y_1, Vy_2 \rangle - z\langle Vy_1, y_2 \rangle + \bar{z}\langle y_1, y_2 \rangle,$$

which proves that $T(z, V)$ is symmetric.

Finally, $D(V(z, T(z, V))) \stackrel{\text{def}}{=} R(T(z, V) - \bar{z}I) = D(V)$ since

$$\begin{aligned} T(z, V) - \bar{z}I &= (zI - \bar{z}V)(I - V)^{-1} - \bar{z}I \\ &= (zI - \bar{z}V)(I - V)^{-1} - \bar{z}(I - V)(I - V)^{-1} \\ &= (z - \bar{z})(I - V)^{-1}. \end{aligned}$$

Furthermore,

$$\begin{aligned}
V(z, T(z, V))(T(z, V) - \bar{z}I) &= (T(z, V) - zI)(T(z, V) - \bar{z}I)^{-1}(T(z, V) - \bar{z}I) \\
&= T(z, V) - zI \\
&= (zI - \bar{z}V)(I - V)^{-1} - z(I - V)(I - V)^{-1} \\
&= V(z - \bar{z})(I - V)^{-1} \\
&= V(T(z, V) - \bar{z}I)
\end{aligned}$$

i.e. $V(z, T(z, V)) = V$ on $R(T(z, V) - \bar{z}I) = D(V) = D(V(z, T(z, V)))$. $\square$

Proposition 7.12

The Cayley transform $V(z, T)$ has inverse Cayley transform $T(z, V(z, T)) = T$.

Proof.

By Proposition 7.9, $D(T(z, V(z, T))) \stackrel{\text{def}}{=} R(I - V(z, T)) = D(T)$.

Furthermore, we have

$$\begin{aligned}
T(z, V(z, T))(I - V(z, T)) &= (zI - \bar{z}V(z, T)) \\
&= zI - \bar{z}(T - zI)(T - \bar{z}I)^{-1} \\
&= z(T - \bar{z}I)(T - \bar{z}I)^{-1} - \bar{z}(T - zI)(T - \bar{z}I)^{-1} \\
&= T(z - \bar{z})(T - \bar{z}I)^{-1} \\
&= T(I - V(z, T))
\end{aligned}$$

i.e. $T(z, V(z, T)) = T$ on $R(I - V(z, T)) = D(T)$.

Note that we used the fact that $I - V(z, T) = (z - \bar{z})(T - \bar{z}I)^{-1}$ for the last equality, as seen in the proof of Proposition 7.9. $\square$

Theorem 7.13

Let $(H, \langle \cdot, \cdot \rangle)$ be a complex Hilbert space and let $z \in \mathbb{C} \setminus \mathbb{R}$ be fixed.

The Cayley transform $T \mapsto V(z, T)$ is a bijective mapping between symmetric operators $T : D(T) \subset H \rightarrow H$ and isometric linear operators $V : D(V) \subset H \rightarrow H$ for which $R(I - V)$ is dense in H . Its inverse is given by the inverse Cayley transform $V \mapsto T(z, V)$.

In the following, the Cayley transform of T will always be taken at $z = i$ and will be denoted by V .

Lemma 7.14

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and let $X, Y \subset H$ be two closed subspaces. There exists a linear isometry between X and Y if and only if they have the same dimension.

Remark 7.15

Here, the dimension of a Hilbert space H is defined as the cardinality of a maximal orthonormal basis of H and the dimension of a subspace M is defined as the dimension of the associated Hilbert space $\overline{M}$. See [Akh77] (Volume 1, pp.27-32) for more details.

Definition 7.7 (Deficiency spaces, Defect indices)

Let $(H, \langle \cdot, \cdot \rangle)$ be a complex Hilbert space and let $T : D(T) \rightarrow H$ be a symmetric operator.

The spaces $K_{\pm} = N(T^* \mp iI) = R(T \pm iI)^{\perp}$ are called the *deficiency spaces* of T and their dimension $d_{\pm}$ the *defect indices*.

Theorem 7.16

Let $(H, \langle \cdot, \cdot \rangle)$ be a complex Hilbert space. Let $T : D(T) \rightarrow H$ be a symmetric operator. The following are equivalent :

1. T admits a self-adjoint extension;
2. The deficiency spaces K_+ and K_- are linearly isometric;
3. The defect indices are equal.

Proof.

First, we observe that we always have a particular decomposition for H . Since T^* is a closed operator, $T^* \mp iI$ is a closed operator as well, and so $N(T^* \mp iI)$ must be closed. By Theorem 2.2, Proposition 2.13 and Proposition 2.24, we have

$$\begin{aligned} H &= N(T^* \mp iI)^{\perp} \oplus N(T^* \mp iI) = \overline{R(T \pm iI)} \oplus N(T^* \mp iI) \\ &= R(T^{**} \pm iI) \oplus N(T^* \mp iI) \quad (\star) \end{aligned}$$

(1) $\Rightarrow$ (2) : Assume that T has a self-adjoint extension $T \subset T_1$. Observe that since T_1 is a self-adjoint extension of T , we have the tower of extensions $T \subset T^{**} \subset T_1 \subset T^*$. Let V and V_1 be the Cayley transform of T^{**} and T_1 respectively. Since $T^{**} \subset T_1$, V_1 is an extension of V . In particular, $V_1(R(T^{**} + iI)) = R(T^{**} - iI)$.

As T_1 is self-adjoint, V_1 is unitary. Therefore, V_1 restricts to an isometry between $N(T^* - iI)$ and $N(T^* + iI)$ using the decomposition $(\star)$ of H .

(2) $\Rightarrow$ (1) : By replacing T with T^{**} , we can assume without loss of generality that T is closed. Indeed, we have $T \subset T^{**}$, T^{**} is closed and symmetric, and by Corollary 2.23, $T^* = (T^{**})^*$, so the deficiency spaces of T^{**} are equal to the deficiency spaces of T .

Let V be the Cayley transform of T . Let $U : N(T^* - iI) \rightarrow N(T^* + iI)$ be a linear isometry. Then the linear operator

$$\begin{aligned} V_1 : D(V) \oplus N(T^* - iI) &\longrightarrow R(V) \oplus N(T^* + iI) \\ x + y &\mapsto Vx + Uy \end{aligned}$$

is a well-defined isometric extension of V , which is unitary. The sums are direct since $D(V) = R(T + iI) = R(T^{**} + iI)$, $R(V) = R(T - iI) = R(T^{**} - iI)$ (which are closed by Proposition 2.24) and we have the decomposition $(\star)$ of H . In particular, $H = D(V_1) = R(V_1)$. It remains to prove that V is isometric. This follows from the fact that U and V are isometric and the direct sum in the decomposition $(\star)$ of H is orthogonal.

The inverse Cayley transform T_1 of V_1 is therefore a self-adjoint extension of T .

(2) $\Longleftrightarrow$ (3) : Follows from Lemma 7.14. □

Definition 7.8 (Conjugation)

Let $(H, \langle \cdot, \cdot \rangle)$ be a complex Hilbert space. An anti-linear map $C : H \rightarrow H$, $C(\lambda x + \mu y) = \bar{\lambda}C(x) + \bar{\mu}C(y)$, is called a *conjugation* if it is isometric and $C^2 = I$.

Remark 7.17

It follows from the Polarization Identity that an anti-linear operator $C : D(C) \subset H \rightarrow H$ on $(H, \langle \cdot, \cdot \rangle)$ is isometric if and only if $\langle Cx, Cy \rangle = \overline{\langle x, y \rangle} \forall x, y \in D(C)$.

Theorem 7.18 (von Neumann's Theorem)

Let $(H, \langle \cdot, \cdot \rangle)$ be a complex Hilbert space and let $T : D(T) \rightarrow H$ be a symmetric operator. Assume there exists a conjugation $C : H \rightarrow H$ such that $C(D(T)) \subset D(T)$ and $CT = TC$. Then the deficiency spaces of T are (anti-linearly) isometric and T admits a self-adjoint extension.

Proof.

Since $C^2 = I$, we have $C(D(T)) = D(T)$. Let $x \in N(T^* - iI)$.

We show that $Cx \in N(T^* + iI) = R(T - iI)^\perp$, i.e. $C(N(T^* - iI)) \subset N(T^* + iI)$. Indeed,

$$\begin{aligned} \langle Cx, (T - iI)Cy \rangle &= \langle Cx, C(T + iI)y \rangle \\ &= \overline{\langle x, (T + iI)y \rangle} = 0 \quad \forall y' = Cy \in D(T) = D(T - iI) \end{aligned}$$

Similarly, $C(N(T^* + iI)) \subset N(T^* - iI)$ and since $C^2 = I$, the inclusions are equality. Therefore, C restricts to a bijective, isometric, anti-linear map $C : N(T^* - iI) \rightarrow N(T^* + iI)$, which ensures that the dimensions of $N(T^* - iI)$ and $N(T^* + iI)$ are the same. $\square$

7.2 The Hamburger Moment Problem

Theorem 7.19

Given a sequence $(a_n)_{n=0}^\infty \subset \mathbb{R}$, there exists a Borel measure μ on $\mathbb{R}$ for which $x^n \in \mathcal{L}^1(\mu)$ and

$$a_n = \int_{\mathbb{R}} x^n d\mu(x) \quad \forall n \in \mathbb{N}_{\geq 0}$$

if and only if for all $N \in \mathbb{N}_{\geq 0}$ and all $\beta_0, \dots, \beta_N \in \mathbb{C}$,

$$\sum_{n,m=0}^N \overline{\beta_n} \beta_m a_{n+m} \geq 0 \quad (\star)$$

Proof.

$\Rightarrow$: Assume that μ is a Borel measure with n -th moment a_n . Then

$$\begin{aligned} \sum_{n,m=0}^N \overline{\beta_n} \beta_m a_{n+m} &= \int_{\mathbb{R}} \sum_{n,m=0}^N \overline{\beta_n} \beta_m x^{n+m} d\mu(x) \\ &= \int_{\mathbb{R}} \left| \sum_{n=0}^N \beta_n x^n \right|^2 d\mu(x) \geq 0 \end{aligned}$$

for all $N \in \mathbb{N}_{\geq 0}$ and all $\beta_0, \dots, \beta_N \in \mathbb{C}$.

$\Leftarrow$: Conversely, assume that $(\star)$ holds. Let $\mathbb{C}[x]$ be the set of polynomial with complex coefficients. We define the following semi-scalar product on $\mathbb{C}[x]$:

$$\left\langle \sum_{n=0}^N \beta_n x^n, \sum_{m=0}^M \gamma_m x^m \right\rangle_{\mathbb{C}[x]} = \sum_{n=0}^N \sum_{m=0}^M \overline{\beta_n} \gamma_m a_{n+m}$$

Let $Q = \{p(x) \in \mathbb{C}[x] \mid \langle p(x), p(x) \rangle_{\mathbb{C}[x]} = 0\}$. It follows from the Parallelogram law that Q is a subspace of $\mathbb{C}[x]$ and it follows from the Polarization identity that $\langle p(x), q(x) \rangle_{\mathbb{C}[x]} = 0 \forall p, q \in Q$.

Consider the quotient space $P = \mathbb{C}[x]/Q$ (i.e. $p(x)$ and $q(x) \in \mathbb{C}[x]$ are identified if $p(x) - q(x) \in Q$) with the following well-defined scalar product :

$$\langle \sum_{n=0}^N \beta_n x^n + Q, \sum_{m=0}^M \gamma_m x^m + Q \rangle_P = \langle \sum_{n=0}^N \beta_n x^n, \sum_{m=0}^M \gamma_m x^m \rangle_{\mathbb{C}[x]} = \sum_{n=0}^N \sum_{m=0}^M \overline{\beta_n} \gamma_m a_{n+m}$$

Consider the completion $(H, \langle \cdot, \cdot \rangle)$ of the Pre-Hilbert space $(P, \langle \cdot, \cdot \rangle_P)$. The space P can be seen as a dense subset of H and the scalar product $\langle \cdot, \cdot \rangle_P$ as the restriction of $\langle \cdot, \cdot \rangle$ to $P \times P$.

Consider the shift operator S on $\mathbb{C}[x]$:

$$\begin{aligned} S : \mathbb{C}[x] &\rightarrow \mathbb{C}[x] \\ \sum_{n=0}^N \beta_n x^n &\mapsto \sum_{n=0}^N \beta_n x^{n+1} \end{aligned}$$

Then T is a symmetric operator on $(\mathbb{C}[x], \langle \cdot, \cdot \rangle_{\mathbb{C}[x]})$. Indeed,

$$\begin{aligned} \langle q(x), Sp(x) \rangle_{\mathbb{C}[x]} &= \langle \sum_{n=0}^N \gamma_n x^n + Q, 0 + \sum_{m=1}^{N+1} \beta_{m-1} x^m + Q \rangle \\ &= \sum_{n=0}^N \sum_{m=1}^{N+1} \overline{\gamma_n} \beta_{m-1} a_{n+m} \\ &= \sum_{m=0}^N \sum_{n=1}^{N+1} \overline{\gamma_m} \beta_{n-1} a_{m+n} \\ &= \sum_{m=1}^{N+1} \sum_{n=0}^N \overline{\gamma_{m-1}} \beta_n a_{m+n} \\ &= \langle Sq(x), p(x) \rangle_{\mathbb{C}[x]} \quad \forall p(x) \in \mathbb{C}[x] \end{aligned}$$

Furthermore, it follows from the Cauchy-Schwarz inequality that $S(Q) \subset Q$:

$$\begin{aligned} \langle Sp(x), Sp(x) \rangle_{\mathbb{C}[x]} &= \langle S^2 p(x), p(x) \rangle_{\mathbb{C}[x]} \\ &\leq \langle S^2 p(x), S^2 p(x) \rangle_{\mathbb{C}[x]}^{1/2} \cdot \langle p(x), p(x) \rangle_{\mathbb{C}[x]}^{1/2} \quad \forall p(x) \in \mathbb{C}[x] \end{aligned}$$

Therefore, the shift operator

$$\begin{aligned} T : P \subset H &\rightarrow P \subset H \\ \sum_{n=0}^N \beta_n x^n + Q &\mapsto \sum_{n=0}^N \beta_n x^{n+1} + Q \end{aligned}$$

is a well-defined symmetric operator on a dense subset of H .

Consider the conjugation operator $C : P \rightarrow P$, $C(p(x) + Q) = \overline{p(x)} + Q$, which extends to a conjugation $\tilde{C}$ on the whole H (it is defined as $\tilde{C} : H \rightarrow H$, $\tilde{C}([(p_n(x) + Q)_{n=1}^\infty]) = [(\overline{p_n(x)} + Q)_{n=1}^\infty]$ for an equivalence class of Cauchy sequences in P).

Clearly, $CT = TC$. Therefore, T admits a self-adjoint extension $\tilde{T} : H \rightarrow H$.

Let $(E_t)_{t \in \mathbb{R}}$ be its spectral resolution. Let μ be the Stieltjes measure associated to $t \mapsto \|E(t)(1 + Q)\|^2$, where 1 is to be interpreted as the constant polynomial $p(x) \equiv 1$. By Spectral Calculus, we have

$$\begin{aligned} a_n &= \langle 1 + Q, x^n + Q \rangle = \langle (1 + Q), T^n(1 + Q) \rangle \\ &= \langle (1 + Q), \hat{E}(x^n)(1 + Q) \rangle = \int_{\mathbb{R}} x^n d\mu. \end{aligned}$$

□

Chapter 8

Operator Monotone Functions

In this chapter, our goal will be to characterize functions $f : (0, +\infty) \rightarrow [0, +\infty)$ that are monotone on symmetric positive definite matrices, i.e. functions $f : (0, +\infty) \rightarrow [0, +\infty)$ such that if $0 \leq A \leq B$, then $0 \leq f(A) \leq f(B)$ (where $A \leq B$ is to be understood as " $B - A$ is positive definite" and $f(A)$ is defined via the functional/spectral calculus of A). This will be made more precise a bit later.

8.1 Dissipative operators

Definition 8.1 (Complete Bernstein Function)

A monotone function $f : (0, +\infty) \rightarrow [0, +\infty)$ is called complete Bernstein if f can be written as

$$f(\lambda) = a + b\lambda + \int_{(0, +\infty)} \frac{\lambda}{\lambda + t} d\sigma(t), \quad \lambda > 0$$

where $a, b \geq 0$ and σ is a sigma-finite Borel measure on $(0, +\infty)$ such that

$$\int_{(0, +\infty)} \frac{1}{1 + t} d\sigma(t) < +\infty$$

Remark 8.1

1. The last condition on the measure ensures that σ is finite on bounded measurable sets because

$$\sigma((0, n)) = \int_{(0, n)} 1 d\sigma(t) \leq \text{cst} \int_{(0, n)} \frac{1}{1 + t} d\sigma(t) \leq \text{cst} \int_{(0, +\infty)} \frac{1}{1 + t} d\sigma(t) < +\infty$$

2. Since for a fixed $t > 0$, the function $\lambda \mapsto \frac{\lambda}{\lambda + t}$ is monotone in $\lambda \geq 0$, it follows from the Monotone Convergence Theorem that a complete Bernstein function is continuous.
3. In what follows, a monotone function f on $(0, +\infty)$ is always implicitly prolonged by continuity at 0 by setting $f(0) = \lim_{\lambda \rightarrow 0^+} f(\lambda)$.

For the sake of completeness, we state without proof the following theorem about complete Bernstein functions. We will not make use of it in this chapter.

Theorem 8.2

A function $f : (0, +\infty) \rightarrow [0, +\infty)$ is complete Bernstein if and only if f is continuous on $(0, +\infty)$, f has a continuous continuation to $\mathbb{H} \cup (0, +\infty)$ such that f is analytic on $\mathbb{H}$, $\Im f(z) \geq 0 \ \forall z \in \mathbb{H}$, and the limit $\lim_{\substack{\lambda \in \mathbb{R} \\ \lambda \rightarrow 0^+}} f(\lambda)$ exists and is real.

Proof.

See [Son10] (Theorem 6.2 and Remark 6.4). The direction $\Leftarrow$ is similar to the proof of Theorem 4.6 (Herglotz) : integrate on a contour and use Helly's Selection Theorem. $\square$

Definition 8.2 (Dissipative Operator)

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space on $\mathbb{K}$ and let $T : D(T) \rightarrow H$ be a self-adjoint operator. T is said to be *dissipative* if

$$\langle x, Tx \rangle \leq 0 \ \forall x \in D(T).$$

We say that a self-adjoint operator T is *positive* if $-T$ is dissipative.

Proposition 8.3

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and let $T : D(T) \rightarrow H$ be a self-adjoint operator. The following are equivalent :

1. T is dissipative;
2. $\forall \lambda > 0 : \|(\lambda I - T)x\| \geq \lambda \cdot \|x\| \ \forall x \in H$;
3. $\forall \lambda > 0 : \lambda \in \rho(T)$ and $\|R(\lambda, T)x\| \leq \lambda^{-1} \cdot \|x\| \ \forall x \in H$;
4. $\sigma(T) \subset (-\infty, 0]$.

Proof.

It is clear that (2) $\iff$ (3) and (3) $\Rightarrow$ (4).

(4) $\Rightarrow$ (3) : If $\sigma(T) \subset (-\infty, 0]$, then for all fixed $\lambda > 0$, $u(t) = (\lambda - t)^{-1}$ is well-defined and bounded on $\sigma(T)$, so we get $R(\lambda, T) = (\lambda I - T)^{-1} = \hat{E}((\lambda - t)^{-1})$ by usual spectral calculus.

(1) $\Rightarrow$ (2) : Since T is self-adjoint, we have

$$\|(\lambda I - T)x\|^2 = \lambda^2 \|x\|^2 - 2\lambda \langle x, Tx \rangle + \|Tx\|^2 \ \forall x \in D(T), \ \forall \lambda > 0$$

with $\langle Tx, x \rangle \leq 0$. Therefore, $\|(\lambda I - T)x\|^2 \geq \lambda^2 \|x\|^2 \ \forall x \in D(T), \ \forall \lambda > 0$.

(2) $\Rightarrow$ (1) : Since T is self-adjoint, we have

$$\|(\lambda I - T)x\|^2 = \lambda^2 \|x\|^2 - 2\lambda \langle x, Tx \rangle + \|Tx\|^2 \ \forall x \in D(T), \ \forall \lambda > 0$$

with $\|(\lambda I - T)x\|^2 \geq \lambda^2 \|x\|^2 \ \forall x \in D(T), \ \forall \lambda > 0$. Therefore, $-2\lambda \langle x, Tx \rangle + \|Tx\|^2 \geq 0 \ \forall x \in D(T), \ \forall \lambda > 0$. This is only possible for $\langle x, Tx \rangle \leq 0$, since λ can be chosen arbitrarily large. $\square$

Proposition 8.4 (Square root of a positive operator)

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and let $T : D(T) \rightarrow H$ be a positive operator. Then there exists a unique positive self-adjoint operator S_T , $D(T) \subset D(S_T)$, $D((S_T)^2) = D(T)$, such that $(S_T)^2 = T$. The operator S_T is called the square root of T .

Proof.

Let $(E_t)_{t \in \mathbb{R}}$ be the spectral resolution of T . Since $\sigma(T) \subset [0, +\infty)$, $u(t) = \sqrt{t} : \sigma(T) \rightarrow \mathbb{R}$ is a well-defined E -measurable function.

Define $S_T = \hat{E}(\sqrt{t}) = \int_{\sigma(T)} \sqrt{t} dE(t)$. The inclusion

$$\begin{aligned} D(T) &= D(\hat{E}(t)) = \{x \in H \mid t \in \mathcal{L}^2(\mathbb{K}, \rho_x)\} \\ &\subset \{x \in H \mid \sqrt{t} \in \mathcal{L}^2(\mathbb{K}, \rho_x)\} = D(\hat{E}(\sqrt{t})) = D(S_T) \end{aligned}$$

follows from the elementary inequality $\sqrt{t} \leq t$ for $t \geq 1$ and the fact that the Stieltjes measures ρ_x are Borel and finite. We also have $\{x \in D(S_T) \mid S_T x \in D(S_T)\} = D(S_T S_T) = D(S_T) \cap D(T) = D(T)$ by point 8. of Theorem 5.5.

By Spectral Calculus, $(S_T)^2 = \hat{E}(\sqrt{t})\hat{E}(\sqrt{t}) = \hat{E}(t) = T$ on $D(T)$ and S_T is self-adjoint since $\sqrt{t}$ is real-valued.

The operator S_T is positive since

$$\langle x, Sx \rangle = \int_{[0, +\infty)} \sqrt{t} d\|E(t)x\|^2 \geq 0 \quad \forall x \in D(S_T).$$

Finally, we verify unicity. Assume that S is a positive self-adjoint square root of T , $D(T) \subset D(S)$, $D(S^2) = D(T)$. Let $F(t)$ be its spectral resolution.

By Spectral Calculus, $T = S^2 = \hat{F}(t)\hat{F}(t) \subset \hat{F}(t^2)$, so $T = \int_{[0, +\infty)} t^2 dF(t)$ on $D(T)$. Therefore, by Theorem 5.8

$$T = \int_{[0, +\infty)} t^2 dF(t) = \int_{[0, +\infty)} t^2 dF(t) + \int_{(-\infty, 0)} -t^2 dF(t) = \int_{\mathbb{R}} t dG(t) \text{ on } D(T),$$

where $G(t) = F(\phi^{-1}(t))$, $\phi(t) = t^2$ for $t \geq 0$, $\phi(t) = -t^2$ for $t \leq 0$.

By unicity of the spectral resolution of T , $E(t) = G(t) \forall t$. Therefore, $F(t) = E(t^2)$ for $t \geq 0$, $F(t) = E(-t^2)$ for $t \leq 0$, i.e. F is precisely determined by E . $\square$

Proposition 8.5

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and let $T : D(T) \rightarrow H$ be a dissipative operator. Then for any $x \in H$, $f(z) = \langle x, zR(z, T)x \rangle$, $z \in \mathbb{K} \setminus (\infty, 0]$ is (the extension) of a complete Bernstein function.

Furthermore, $\lim_{\lambda \rightarrow +\infty} \langle x, \lambda R(\lambda, T)y \rangle = \langle x, y \rangle \quad \forall x, y \in H$.

Proof.

Let $(E_t)_{t \in \mathbb{R}}$ be the spectral resolution of $-T$.

For $\lambda \in (0, +\infty)$, we have

$$\begin{aligned} \langle x, \lambda R(\lambda, T)x \rangle &= \langle x, \hat{E}[\lambda \cdot (\lambda + (-t))^{-1}]x \rangle = \int_{[0, +\infty)} \frac{\lambda}{\lambda + t} d\|E(t)x\|^2 \\ &= \|(E(0) - E(0-))x\|^2 + \int_{(0, +\infty)} \frac{\lambda}{\lambda + t} d\|E(t)x\|^2 \\ &= a + \int_{(0, +\infty)} \frac{\lambda}{\lambda + t} d\|E(t)x\|^2 \end{aligned}$$

$$= a + \int_{(0,+\infty)} \frac{\lambda}{\lambda+t} d\sigma_x(t),$$

where σ_x is the bounded Stieltjes measure induced by the function

$$\sigma_x(t) = \begin{cases} \|E(t)x\|^2 & \text{if } t > 0 \\ \|E(0)x\|^2 & \text{otherwise} \end{cases}$$

Finally,

$$\lim_{\lambda \rightarrow +\infty} \langle x, \lambda R(\lambda, T)x \rangle = \lim_{\lambda \rightarrow +\infty} \int_{[0,+\infty)} \frac{\lambda}{\lambda+t} d\|E(t)x\|^2 = \int_{[0,+\infty)} 1 d\|E(t)x\|^2 = \|x\|^2 \quad \forall x \in H$$

using the Dominated Convergence Theorem (the measure is finite) and we conclude using the Polarization Identity for sesquilinear forms. $\square$

8.2 Operator Monotone Functions

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and let $T, S : H \rightarrow H$ be two bounded self-adjoint operators. Recall that we write $T \leq S$ if $S - T$ is positive, i.e. if

$$\langle x, Tx \rangle \leq \langle x, Sx \rangle \quad \forall x \in H$$

We wish to extend this notion to dissipative self-adjoint operators T, S . Since $D(S - T) = D(S) \cap D(T)$ is not necessarily dense, we cannot talk about $S - T$ being self-adjoint or even dissipative.

Definition 8.3 (Quadratic form associated to a dissipative operator)

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and let $T : D(T) \rightarrow H$ be a dissipative operator. We define the following quadratic form on H :

$$Q_T(x, x) = \begin{cases} \langle (-T)^{1/2}x, (-T)^{1/2}x \rangle & \text{if } x \in D(Q_T) = D((-T)^{1/2}) \\ +\infty & \text{otherwise} \end{cases}$$

If $(E_t)_{t \in \mathbb{R}}$ is the spectral resolution of T , this can be written as

$$Q_T(x, x) = \int_{(-\infty, 0]} -t d\|E(t)x\|^2 \quad \forall x \in H.$$

Finally, notice that if $x \in D(T) \subset D((-T)^{1/2})$, we have $Q_T(x, x) = -\langle x, Tx \rangle$.

Definition 8.4 (Domination in quadratic form)

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and let $T : D(T) \rightarrow H, S : D(S) \rightarrow H$ be two dissipative operators. We say that $-T$ *dominates* $-S$ in quadratic form, and write $-S \leq_{qf} -T$, if

$$Q_S(x, x) \leq Q_T(x, x) \quad \forall x \in H$$

Notice that if $-S \leq_{qf} -T$, we necessarily have $D(Q_T) \subset D(Q_S)$.

Definition 8.5 (Matrix Monotone Operator of order n)

A monotone function $f : (0, +\infty) \rightarrow [0, +\infty)$ is called *matrix monotone of order n* if for any symmetric matrices $A, B \in \mathbb{R}^{n \times n}$, we have

$$0 \leq B \leq A \implies 0 \leq f(B) \leq f(A).$$

We denote by P_n the set of functions which are matrix monotone of order n .

A monotone function $f : (0, +\infty) \rightarrow [0, +\infty)$ is called *matrix monotone* (or *operator monotone*) if it is matrix monotone of order n for all $n \in \mathbb{N}$.

Definition 8.6 (Monotone Operator in quadratic sense)

A monotone function $f : (0, +\infty) \rightarrow [0, +\infty)$ is called *monotone in the quadratic sense* if for any pair of dissipative operators $T : D(T) \rightarrow H$, $S : D(S) \rightarrow H$ on a Hilbert space $(H, \langle \cdot, \cdot \rangle)$, we have :

$$0 \leq_{qf} -S \leq_{qf} -T \implies 0 \leq_{qf} f(-S) \leq_{qf} f(-T).$$

Remark 8.6

Clearly, $\{f : \text{monotone in the quadratic sense}\} \subset \{f : \text{matrix monotone}\}$. Our goal will be to prove that those two notions are equivalent and that we, in fact, have

$$\{f : \text{complete Bernstein}\} = \{f : \text{monotone in the quadratic sense}\} = \{f : \text{matrix monotone}\}.$$

This result is known as *Löwner's Theorem*.

Lemma 8.7

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and let $T : D(T) \rightarrow H$ be a dissipative operator. Then for all $\lambda \in \mathbb{R} \cap \rho(T)$, $TR(\lambda, T)$ is a bounded operator.

Proof.

Let $\lambda \in \mathbb{R} \cap \rho(T)$ and $x \in H$. Then

$$\|TR(\lambda, T)x\|^2 = \int_{(-\infty, 0]} \left| \frac{t}{\lambda - t} \right| d\|E(t)x\|^2 = \int_{(-\infty, 0]} \frac{-t}{\lambda - t} d\|E(t)x\|^2 \leq \|x\|^2.$$

□

Proposition 8.8

Let $(H, \langle \cdot, \cdot \rangle)$ be a Hilbert space and let $T : D(T) \rightarrow H$, $S : D(S) \rightarrow H$ be two dissipative operators. The following are equivalent :

1. $0 \leq_{qf} -S \leq_{qf} -T$;
2. $0 \leq R(\lambda, T) \leq R(\lambda, S) \forall \lambda > 0$;
3. $0 \leq (-S)R(\lambda, S) \leq (-T)R(\lambda, T) \forall \lambda > 0$.

Proof.

(i) $\Rightarrow$ (ii) : Let $\lambda > 0$ be fixed. The operators $(\lambda I - T) = \hat{E}^T(\lambda - t)$ and $R(\lambda, T) = \hat{E}^T((\lambda - t)^{-1})$ are positive, since they can be defined via the Spectral Calculus of T with some real-valued and non-negative functions on $(-\infty, 0]$. In particular, the square root

$(\lambda I - T)^{1/2}$, $D(T) \subset D((\lambda I - T)^{1/2}) = D((-T)^{1/2}) = D(Q_T)$, is well-defined. This remains unchanged if we replace T by S .

For $x \in H$, define

$$\begin{aligned}\phi &= R(\lambda, S)x = (\lambda I - S)^{-1}x \in D(T) \subset D(Q_T) \\ \psi &= R(\lambda, T)x = (\lambda I - T)^{-1}x \in D(S) \subset D(Q_S)\end{aligned}$$

We then have

$$\begin{aligned}\langle x, R(\lambda, T)x \rangle^2 &= \langle (\lambda I - S)\phi, R(\lambda, T)x \rangle^2 \\ &= \langle (\lambda I - S)^{1/2}\phi, (\lambda I - S)^{1/2}R(\lambda, T)x \rangle^2 \\ &\leq \langle (\lambda I - S)^{1/2}\phi, (\lambda I - S)^{1/2}\phi \rangle \cdot \langle (\lambda I - S)^{1/2}R(\lambda, T)x, (\lambda I - S)^{1/2}R(\lambda, T)x \rangle \\ &= \langle (\lambda I - S)\phi, \phi \rangle \cdot \langle (\lambda I - S)^{1/2}\psi, (\lambda I - S)^{1/2}\psi \rangle \\ &\leq \langle (\lambda I - S)\phi, \phi \rangle \cdot \langle (\lambda I - T)^{1/2}\psi, (\lambda I - T)^{1/2}\psi \rangle \\ &= \langle (\lambda I - S)\phi, \phi \rangle \cdot \langle (\lambda I - T)\psi, \psi \rangle \\ &= \langle x, R(\lambda, S)x \rangle \cdot \langle x, R(\lambda, T)x \rangle\end{aligned}$$

where we used the Cauchy-Schwarz Inequality for the first inequality and the hypothesis $(\lambda I - S) \leq_{qf} (\lambda I - T)$ for the last inequality.

(ii) $\Rightarrow$ (iii) : Let $\lambda > 0$ be fixed. Let $x \in H$. We have

$$\begin{aligned}\langle x, -SR(\lambda, S)x \rangle &= \langle x, (\lambda I - S)R(\lambda, S)x - \lambda R(\lambda, S)x \rangle \\ &= \langle x, x \rangle - \langle x, \lambda R(\lambda, S)x \rangle \\ &\leq \langle x, x \rangle - \langle x, \lambda R(\lambda, T)x \rangle \\ &= \langle x, (\lambda I - T)R(\lambda, T)x - \lambda R(\lambda, T)x \rangle \\ &= \langle x, -TR(\lambda, T)x \rangle\end{aligned}$$

(iii) $\Rightarrow$ (i) : By Proposition 8.5, we have

$$\langle (-S)^{1/2}x, (-S)^{1/2}x \rangle = \lim_{\lambda \rightarrow +\infty} \langle (-S)^{1/2}x, \lambda R(\lambda, S)(-S)^{1/2}x \rangle \quad \forall x \in D(Q_S)$$

The right-hand side is equal to

$$\lim_{\lambda \rightarrow +\infty} \lambda \langle x, (-S)R(\lambda, S)x \rangle$$

using the Spectral Calculus. We get the result by doing the same with T . $\square$

Example 8.1

Consider the function $f : (0, +\infty) \rightarrow [0, +\infty)$, $f(\lambda) = \frac{\lambda}{\lambda + 1}$.

Then f is a complete Bernstein function. Indeed, take $a = b = 0$ and σ the Stieltjes measure given by $\sigma(t) = \begin{cases} 1 & \text{if } t \geq 1 \\ 0 & \text{otherwise} \end{cases}$.

We check that f is matrix monotone. Let $A, B \in \mathbb{R}^{n \times n}$ be symmetric matrices for which $0 \leq B \leq A$ (i.e. A , B , and $A - B$ are semi-positive definite).

We have

$$f(A) = \int_{[0,+\infty)} \frac{\lambda}{\lambda+t} dE^A(t) = \hat{E}^A(f) = \hat{E}^A(t)(\hat{E}^A(1)+\hat{E}^A(t))^{-1} = A(I+A)^{-1} = (-A)R(-1, A)$$

and similarly

$$f(B) = B(I+B)^{-1} = (-B)R(-1, B).$$

It follows directly from Proposition 8.8 that $0 \leq f(A) \leq f(B)$. We have the more general theorem :

Theorem 8.9

Let $f : (0, +\infty) \rightarrow [0, +\infty)$ be a complete Bernstein function. Then f is monotone in the quadratic sense.

Proof.

Write

$$f(\lambda) = a + b\lambda + \int_{(0,+\infty)} \frac{\lambda}{\lambda+t} d\sigma(t), \quad \lambda > 0$$

as in Definition 8.1.

Let $T : D(T) \rightarrow H$, $S : D(S) \rightarrow H$ be a pair of dissipative operators on a Hilbert space $(H, \langle \cdot, \cdot \rangle)$ and let $(E_t^T)_{t \in \mathbb{R}}$, $(E_t^S)_{t \in \mathbb{R}}$ be their respective spectral resolution.

Observe that for all $x \in H$, we have

$$\begin{aligned} Q_{f(S)}(x, x) &= \int_{(-\infty, 0]} f(-\lambda) d\|E^S(\lambda)x\|^2 \\ &= \int_{(-\infty, 0]} \left(a + b(-\lambda) + \int_{(0,+\infty)} \frac{-\lambda}{-\lambda+t} d\sigma(t) \right) d\|E^S(\lambda)x\|^2 \\ &= a\|x\|^2 + bQ_S(x, x) + \int_{(-\infty, 0]} \left(\int_{(0,+\infty)} \frac{-\lambda}{t-\lambda} d\sigma(t) \right) d\|E^S(\lambda)x\|^2 \\ &= a\|x\|^2 + bQ_S(x, x) + \int_{(0,+\infty)} \left(\int_{(-\infty, 0]} \frac{-\lambda}{t-\lambda} d\|E^S(\lambda)x\|^2 \right) d\sigma(t) \\ &= a\|x\|^2 + bQ_S(x, x) + \int_{(0,+\infty)} \langle (-S)R(t, S)x, x \rangle \sigma(t) \quad (\star) \end{aligned}$$

where we can separate the first integral into three parts since all the integrands are non-negative and where we can switch the order of integration of the third integral by Tonelli's Theorem since, again, the integrand is non-negative.

But $0 \leq_{qf} -S \leq_{qf} -T$ is equivalent to $Q_S(x, x) \leq Q_T(x, x) \forall x \in H$ and $\langle (-S)R(t, S)x, x \rangle \leq \langle (-T)R(t, T)x, x \rangle \forall x \in H, \forall t > 0$ by Proposition 8.8. Since integration is monotone, we have

$$0 \leq Q_{f(-S)}(x, x) \leq Q_{f(-T)}(x, x) \quad \forall x \in H$$

by comparison of each term in $(\star)$ for S and T . □

Proposition 8.10

Let $f : (0, +\infty) \rightarrow [0, +\infty)$ be a monotone function. Then f is matrix monotone of order

n if and only if for all $\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_n \geq 0$, for all orthogonal matrices $Q \in \mathbb{R}^{n \times n}$:

$$\sum_{j=1}^n (Qx)_j^2 \beta_j \leq \sum_{j=1}^n x_j^2 \alpha_j \quad \forall x \in \mathbb{R}^n \implies \sum_{j=1}^n (Qx)_j^2 f(\beta_j) \leq \sum_{j=1}^n x_j^2 f(\alpha_j) \quad \forall x \in \mathbb{R}^n$$

Proof.

The above condition can be rewritten as : f is matrix monotone of order n if and only if for all $\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_n \geq 0$, for all orthogonal matrices $Q \in \mathbb{R}^{n \times n}$:

$$QD(\beta_i)Q^T \leq D(\alpha_i) \quad \forall x \in \mathbb{R}^n \implies QD(f(\beta_i))Q^T \leq D(f(\alpha_i)) \quad \forall x \in \mathbb{R}^n$$

where $D(a_i)$ denotes a diagonal matrix with coefficient $D_{i,i} = a_i$.

Recall that f is matrix monotone of order n if and only if for all symmetric matrices $A, B \in \mathbb{R}^{n \times n}$:

$$0 \leq B \leq A \implies 0 \leq f(B) \leq f(A)$$

A symmetric positive semi-definite matrix $M \in \mathbb{R}^{n \times n}$ has real non-negative eigenvalues $m_1, \dots, m_n \geq 0$ and is diagonalizable over $\mathbb{R}$ with an orthogonal change of basis matrix, i.e. there exists an orthogonal matrix $P \in \mathbb{R}^{n \times n}$ such that $P^T M P = D(m_i)$ is diagonal. Then it is easy to check that $f(M) = P D(f(m_i)) P^T$ by approximating $f(x)$ in $\sigma(M) \subset [0, \max_i \{m_i\} + 1]$ with a sequence of polynomials (using Weierstrass Approximation Theorem). Therefore, the direction $\implies$ is clear.

For the other direction, fix $A, B \in \mathbb{R}^{n \times n}$ two symmetric positive semi-definite matrices and write $P_1^T A P_1 = D(\alpha_i)$, $P_2^T B P_2 = D(\beta_i)$, $\alpha_i, \beta_i \geq 0$, $P_1, P_2 \in \mathbb{R}^{n \times n}$ orthogonal. Then

$$\begin{aligned} 0 \leq B \leq A &\iff 0 \leq x^T (B - A) x \quad \forall x \in \mathbb{R}^n \iff 0 \leq x^T (B - A) x \quad \forall x \in \mathbb{R}^n \\ &\iff 0 \leq y^T P_1^T (B - A) P_1 y \quad \forall y \in \mathbb{R}^n \iff 0 \leq Q D(\beta_i) Q^T \leq D(\alpha_i) \end{aligned}$$

where $Q^T = P_2^T P_1$.

□

Notation

Let R be a ring. We denote by $R[x]$ the ring of polynomials with coefficients in R and by $R_k[x] \subset R[x]$ the subset of polynomials of degree less than or equal to k .

Lemma 8.11

Let $p(x) \in \mathbb{R}_n[x]$, $n \geq 0$. Then $p(x)$ is uniquely determined by its values at $n+1$ distinct points $\lambda_1, \dots, \lambda_{n+1} \in \mathbb{R}$.

Furthermore, $p(x)$ can be recovered in the following way :

$$p(x) = \sum_{j=1}^{n+1} \left(p(\lambda_j) \prod_{\substack{1 \leq i \leq n+1 \\ i \neq j}} \frac{x - \lambda_i}{\lambda_j - \lambda_i} \right).$$

Lemma 8.12

Let $n \geq 1$. Let $\lambda_1, \dots, \lambda_{2n} \in \mathbb{R} \setminus \{0\}$ be fixed distinct points. Define $\pi(x) = \prod_{j=1}^{2n} (x + \lambda_j)$.

Then the map

$$\begin{aligned}\phi : \mathbb{R}_{2n-1}[x] &\rightarrow \mathbb{R}^{2n} \\ p(x) &\mapsto \left(\frac{p(-\lambda_j)}{\pi'(-\lambda_j)\lambda_j} \right)_{1 \leq j \leq 2n}\end{aligned}$$

is a linear bijection from $\mathbb{R}_{2n-1}[x]$ to $\mathbb{R}^{2n}$ such that

$$\frac{p(x)}{\pi(x)} = \sum_{j=1}^{2n} \phi(p)_j \left(\frac{\lambda_j}{x + \lambda_j} \right) \quad \forall x \in \mathbb{R}.$$

Furthermore, $p(x) \in \mathbb{R}_{2n-1}[x]$, $p(0) = 0$, if and only if $\sum_{j=1}^{2n} \phi(p)_j = 0$.

Proof.

The linearity of ϕ is clear. The injectivity follows from the fact that a polynomial $p(x) \in \mathbb{R}_{2n-1}[x]$ is uniquely determined by its values at $2n$ distinct points. Since ϕ is a linear map between two $\mathbb{R}$ -vector spaces of the same dimension, it follows from the Rank-Nullity Theorem that ϕ is surjective.

Now, notice that

$$\pi'(-\lambda_j) = \prod_{\substack{1 \leq i \leq 2n \\ i \neq j}} (\lambda_i - \lambda_j).$$

By Lemma 8.11, for a given $p(x) \in \mathbb{R}_{2n-1}[x]$, we have

$$\begin{aligned}p(x) &= \sum_{j=1}^{2n} \left(p(-\lambda_j) \prod_{\substack{1 \leq i \leq 2n \\ i \neq j}} \frac{x + \lambda_i}{-\lambda_j + \lambda_i} \right) = \sum_{j=1}^{2n} p(-\lambda_j) \left(\prod_{\substack{1 \leq i \leq 2n \\ i \neq j}} (x + \lambda_i) \right) \pi'(-\lambda_j)^{-1} \\ &= \sum_{j=1}^{2n} p(-\lambda_j) \frac{\pi(x)}{\pi'(-\lambda_j)(x + \lambda_j)} = \pi(x) \sum_{j=1}^{2n} \phi(p)_j \left(\frac{\lambda_j}{x + \lambda_j} \right)\end{aligned}$$

□

Definition 8.7

Let $n \geq 1$. We say that a function $f : (0, +\infty) \rightarrow [0, +\infty)$ is in the set M_n if for all distinct $\lambda_1, \dots, \lambda_{2n} > 0$, for all $a_1, \dots, a_{2n} \in \mathbb{R}$ such that $\sum_{i=1}^{2n} a_i = 0$:

$$\sum_{j=1}^{2n} a_j \frac{\lambda_j}{x + \lambda_j} \geq 0 \quad \forall x > 0 \implies \sum_{j=1}^{2n} a_j f(\lambda_j) \geq 0.$$

Observe that $M_{n+1} \subset M_n \quad \forall n \geq 1$.

Proposition 8.13

A function $f : (0, +\infty) \rightarrow [0, +\infty)$ is in the set M_n if and only if for all distinct $\lambda_1, \dots, \lambda_{2n} > 0$, for all $a_1, \dots, a_{2n} \in \mathbb{R}$ such that $\sum_{i=1}^{2n} a_i = 0$:

$$\sum_{j=1}^{2n} a_j \frac{\lambda_j x - 1}{x + \lambda_j} \geq 0 \quad \forall x > 0 \implies \sum_{j=1}^{2n} a_j f(\lambda_j) \geq 0.$$

Proof.

Notice that

$$\begin{aligned} \sum_{j=1}^{2n} a_j \frac{\lambda_j x - 1}{x + \lambda_j} \geq 0 \quad \forall x > 0 &\iff \sum_{j=1}^{2n} a_j \left(\left(x + \frac{1}{x} \right) \frac{\lambda_j}{x + \lambda_j} - \frac{1}{x} \right) \geq 0 \quad \forall x > 0 \\ &\iff \left(x + \frac{1}{x} \right) \sum_{j=1}^{2n} a_j \frac{\lambda_j}{x + \lambda_j} \geq 0 \quad \forall x > 0 \\ &\iff \sum_{j=1}^{2n} a_j \frac{\lambda_j}{x + \lambda_j} \geq 0 \quad \forall x > 0 \end{aligned}$$

where we used the fact that $\sum_{j=1}^{2n} a_j = 0$ to get the second equivalence. $\square$

Proposition 8.14

A function $f : (0, +\infty) \rightarrow [0, +\infty)$ is in the set M_n if and only if for all distinct $\lambda_1, \dots, \lambda_{2n} > 0$, for all $p(x) \in \mathbb{R}_{2n-1}[x]$ with $p(0) = 0$:

$$p(x) \geq 0 \quad \forall x > 0 \implies \sum_{j=1}^{2n} \phi(p)_j f(\lambda_j) \geq 0 \quad (\star)$$

where ϕ is defined as in Lemma 8.12.

Proof.

This follows from Lemma 8.12 and its proof. $\square$

Remark 8.15

Let $f : (0, +\infty) \rightarrow [0, +\infty)$ be any function and let $\lambda_1, \dots, \lambda_{2n} > 0$ be distinct. If $(\star)$ holds for some $p(x), q(x) \in \mathbb{R}_{2n-1}[x]$ such that $p(0) = q(0) = 0$, $p(x), q(x) > 0 \quad \forall x > 0$, then by linearity of ϕ , $(\star)$ also holds for $p(x) + q(x)$.

Theorem 8.16

Let $m \geq 0$. Any polynomial $p(x) \in \mathbb{R}_m[x]$ such that $p(x) \geq 0 \quad \forall x > 0$ can be written as $p(x) = xq_1^2(x) + q_2^2(x)$ for some $q_1(x), q_2(x) \in \mathbb{R}[x]$.

Remark 8.17

It follows from degree comparison that $q_1(x), q_2(x) \in \mathbb{R}_{\lfloor m/2 \rfloor}[x]$.

Proof.

Observe that for all $\forall a, b, c, d \in \mathbb{R}[x]$, the following identity holds :

$$[xa^2(x) + b^2(x)][xc^2(x) + d^2(x)] = x[a(x)d(x) - b(x)c(x)]^2 + [a(x)c(x)x + b(x)d(x)]^2$$

Therefore, the set $\mathcal{P} = \{xa^2(x) + b^2(x) \mid a, b \in \mathbb{R}[x]\}$ is closed by multiplication. It is also closed by positive scalar multiplication. Thus, it suffices to prove that $p(x)$ is a product of a positive constant and elements of $\mathcal{P}$.

Let

$$p(x) = A \cdot \prod (x + a_i)^{\alpha_i} \cdot \prod (x + b_i x + c_i)^{\beta_i}$$

be the decomposition into irreducible factors (over $\mathbb{R}$) of $p(x)$. We study each factor independently :

1. Since $p(x) \geq 0 \forall x > 0$, by taking $x \rightarrow +\infty$, it follows that $A \geq 0$.
2. If $a_i \geq 0$, then $x + a_i = x \cdot 1^2 + (\sqrt{a_i})^2$ is an element of $\mathcal{P}$.
3. Since $(x^2 + b_i x + c_i)$ is irreducible, we have

$$(x^2 + b_i x + c_i) = (x - \omega_i)(x - \bar{\omega}_i) = x^2 - 2\Re(\omega_i)x + |\omega_i|^2 = (x - |\omega_i|)^2 + 2(|\omega_i|^2 - \Re(\omega_i)^2)x$$

for some $\omega_i \in \mathbb{C} \setminus \mathbb{R}$. Since $|\omega|^2 - \Re(\omega_i)^2 \geq 0$, this polynomial is in $\mathcal{P}$.

4. Finally, if $a_i < 0$, then $(x + a_i)$ must have even multiplicity. Assume not. Order the negative roots in increasing order $a_k < \dots < a_1 < a_0 = 0$. Consider the smallest $a_i < 0$ with odd multiplicity. Then for $x \in (-a_{i-1}, -a_i)$, we have $p(x) < 0$, which is a contradiction. Therefore, $(x + a_i)^{\alpha_i}$ is in $\mathcal{P}$.

□

Theorem 8.18

$P_{n+1} \subset M_n \forall n \geq 1$.

Proof.

From Remark 8.15 and Theorem 8.16, it suffices to prove that :

If $f \in P_{n+1}$, then for all distinct $\lambda_1, \dots, \lambda_{2n} > 0$, for all $p(x) \in \mathbb{R}_{2n-1}[x]$ with $p(0) = 0$ of the form $xq^2(x)$ or $q^2(x)$, $q(x) \in \mathbb{R}_{n-1}[x]$:

$$\sum_{j=1}^{2n} \phi(p)_j f(\lambda_j) \geq 0 \quad (\star)$$

1. Write $\{\lambda_1, \dots, \lambda_{2n}\} = \{0 < \beta_1 < \alpha_1 < \dots < \beta_n < \alpha_n\}$ and $\pi(x) = \prod_{j=1}^{2n} (x + \lambda_j)$, $\pi'(-\beta_j) > 0$, $\pi'(-\alpha_j) < 0$ for all j . For any $q(x) \in \mathbb{R}_{n-1}[x]$, we have

$$\begin{aligned} \frac{xq^2(x)}{\pi(x)} &= \sum_{j=1}^{2n} \phi(xq^2(x))_j \left(\frac{\lambda_j}{x + \lambda_j} \right) = \sum_{j=1}^{2n} \left(\frac{(-\lambda_j)q^2(-\lambda_j)}{\pi'(-\lambda_j)\lambda_j} \right) \left(\frac{\lambda_j}{x + \lambda_j} \right) \\ &= - \sum_{j=1}^n \left(\frac{q(-\beta_j)}{\sqrt{\pi'(-\beta_j)}} \right)^2 \left(\frac{\beta_j}{x + \beta_j} \right) + \sum_{j=1}^n \left(\frac{q(-\alpha_j)}{\sqrt{-\pi'(-\alpha_j)}} \right)^2 \left(\frac{\alpha_j}{x + \alpha_j} \right) \geq 0 \quad \forall x > 0 \end{aligned}$$

The mappings

$$\begin{aligned} \varphi : \mathbb{R}_{n-1}[x] &\rightarrow \mathbb{R}^n, & \psi : \mathbb{R}_{n-1}[x] &\rightarrow \mathbb{R}^n \\ q(x) &\mapsto \left(\frac{q(-\beta_j)}{\sqrt{\pi'(-\beta_j)}} \right)_{1 \leq j \leq n}, & q(x) &\mapsto \left(\frac{q(-\alpha_j)}{\sqrt{-\pi'(-\alpha_j)}} \right)_{1 \leq j \leq n} \end{aligned}$$

are linear bijections (since a polynomial of degree $\leq n-1$ is uniquely determined by its values at n distinct points and $\mathbb{R}_{n-1}[x]$, $\mathbb{R}^n$ have the same dimension). Therefore,

$$\begin{aligned} \frac{xq^2(x)}{\pi(x)} &= - \sum_{j=1}^n \varphi(q)_j^2 \left(\frac{\beta_j}{x + \beta_j} \right) + \sum_{j=1}^n \psi(q)_j^2 \left(\frac{\alpha_j}{x + \alpha_j} \right) \\ &= - \sum_{j=1}^n (Q\psi(q))_j^2 \left(\frac{\beta_j}{x + \beta_j} \right) + \sum_{j=1}^n \psi(q)_j^2 \left(\frac{\alpha_j}{x + \alpha_j} \right) \geq 0 \quad \forall x > 0 \end{aligned}$$

for some matrix $Q \in \mathbb{R}^{n \times n}$. Evaluating at $x = 0$, we get

$$-\sum_{j=1}^n (Q\psi(q))_j^2 + \sum_{j=1}^n \psi(q)_j^2 = 0$$

Since $q(x) \in \mathbb{R}_{n-1}[x]$ was arbitrary and ψ is a linear bijection, we have

$$-\sum_{j=1}^n (Qz)_j^2 + \sum_{j=1}^n z_j^2 = 0 \quad \forall z \in \mathbb{R}^n$$

and

$$-\sum_{j=1}^n (Qz)_j^2 \left(\frac{\beta_j}{x + \beta_j} \right) + \sum_{j=1}^n z_j^2 \left(\frac{\alpha_j}{x + \alpha_j} \right) \geq 0 \quad \forall x > 0, \quad \forall z \in \mathbb{R}^n.$$

The left-hand side of the first equation is exactly $z^T(I - Q^T Q)z$, so the matrix Q must be orthogonal.

Multiplying by x the second equation and letting $x \rightarrow +\infty$, we get

$$-\sum_{j=1}^n (Qz)_j^2 \beta_j + \sum_{j=1}^n z_j^2 \alpha_j \geq 0 \quad \forall z \in \mathbb{R}^n.$$

If $f \in P_n$, we then have

$$-\sum_{j=1}^n (Qz)_j^2 f(\beta_j) + \sum_{j=1}^n z_j^2 f(\alpha_j) \geq 0 \quad \forall z \in \mathbb{R}^n,$$

i.e.

$$-\sum_{j=1}^n \varphi(q)_j^2 f(\beta_j) + \sum_{j=1}^n \psi(q)_j^2 f(\alpha_j) = \sum_{j=1}^{2n} \phi(xq^2(x))_j f(\lambda_j) \geq 0 \quad \forall q(x) \in \mathbb{R}_{n-1}[x],$$

which is precisely $(\star)$ for polynomials of the form $xq^2(x)$, $q(x) \in \mathbb{R}_{n-1}[x]$. Since $P_{n+1} \subset P_n$, this also holds if $f \in P_{n+1}$.

Remark : We have proved that for any $m \in \mathbb{N}$, if $f \in P_m$, then $(\star)$ holds for all distinct $\lambda_1, \dots, \lambda_{2m} > 0$ and all $p(x) = xq^2(x) \in \mathbb{R}_{2m-1}[x]$. To prove the case where $p(x) = q^2(x) \in \mathbb{R}_{2n-1}[x]$, $p(0) = 0$, we will apply this for $m = n + 1$.

Observation : If $f \in P_2$, it follows from $(\star)$ applied to $n = 2$, $\beta_1 = 1$, $\alpha_1 = 2$, $\beta_2 = \beta$, $\alpha_2 = 2\beta$, $p(x) = x(x + 2\beta)^2 = xq^2(x)$, $\beta > 2$, that

$$-\sum_{j=1}^2 \left(\frac{q(-\beta_j)}{\sqrt{\pi'(-\beta_j)}} \right)^2 f(\beta_j) + \sum_{j=1}^2 \left(\frac{q(-\alpha_j)}{\sqrt{-\pi'(-\alpha_j)}} \right)^2 f(\alpha_j) \geq 0$$

i.e.

$$\begin{aligned} & -\frac{(-1 + 2\beta)^2}{(2-1)(\beta-1)(2\beta-1)} f(1) - \frac{(-\beta + 2\beta)^2}{(1-\beta)(2-\beta)(2\beta-\beta)} f(\beta) \\ & + \frac{(-2 + 2\beta)^2}{-(1-2)(\beta-2)(2\beta-2)} f(2) + \frac{(-2\beta + 2\beta)^2}{-(1-2\beta)(2-2\beta)(\beta-2\beta)} f(2\beta) \end{aligned}$$

$$= -\frac{2\beta-1}{\beta-1}f(1) - \frac{\beta}{(\beta-1)(\beta-2)}f(\beta) + \frac{2\beta-2}{\beta-2}f(2) \geq 0$$

Multiplying this last inequality by $(\beta-1)(\beta-2)/\beta^3$, we get

$$-\frac{(2\beta-1)(\beta-2)}{\beta^3}f(1) - \frac{1}{\beta^2}f(\beta) + \frac{2(\beta-1)^2}{\beta^3}f(2) \geq 0$$

i.e.

$$0 \leq \frac{f(\beta)}{\beta^2} \leq \frac{(2\beta-1)(\beta-2)}{\beta^3}f(1) + \frac{2(\beta-1)^2}{\beta^3}f(2) \xrightarrow{\beta \rightarrow +\infty} 0.$$

2. Write $\{\lambda_1, \dots, \lambda_{2n}\} = \{0 < \beta_1 < \alpha_1 < \dots < \beta_n < \alpha_n\}$ and $\pi(x) = \prod_{j=1}^{2n}(x + \lambda_j)$, $\pi'(-\beta_j) > 0$, $\pi'(-\alpha_j) < 0$ for all j . For any $q(x) \in \mathbb{R}_{n-1}[x]$ with $q(0) = 0$, we have

$$\begin{aligned} \frac{q^2(x)}{\pi(x)} &= \sum_{j=1}^{2n} \phi(q^2(x))_j \left(\frac{\lambda_j}{x + \lambda_j} \right) = \sum_{j=1}^{2n} \left(\frac{q^2(-\lambda_j)}{\pi'(-\lambda_j)\lambda_j} \right) \left(\frac{\lambda_j}{x + \lambda_j} \right) \\ &= + \sum_{j=1}^n \left(\frac{q(-\beta_j)}{\sqrt{\pi'(-\beta_j)\beta_j}} \right)^2 \left(\frac{\beta_j}{x + \beta_j} \right) - \sum_{j=1}^n \left(\frac{q(-\alpha_j)}{\sqrt{-\pi'(-\alpha_j)\alpha_j}} \right)^2 \left(\frac{\alpha_j}{x + \alpha_j} \right) \\ &= + \sum_{j=1}^n y_j^2 \left(\frac{\beta_j}{x + \beta_j} \right) - \sum_{j=1}^n z_j^2 \left(\frac{\alpha_j}{x + \alpha_j} \right) \end{aligned}$$

Let $0 < \alpha_0 < \beta_1$ and $\alpha_n < \beta_{n+1}$. Define $\tilde{\pi}(x) = (x + \alpha_0)\pi(x)(x + \beta_{n+1})$, $\tilde{\pi}'(\beta_j) < 0$, $\tilde{\pi}'(\alpha_j) > 0$ for all j . For any $q(x) \in \mathbb{R}_{n-1}[x]$ with $q(0) = 0$, we consider

$$\begin{aligned} \frac{x}{x + \alpha_0} \frac{q^2(x)}{\pi(x)} \frac{\beta_{n+1}}{x + \beta_{n+1}} &= \frac{\beta_{n+1}xq^2(x)}{\tilde{\pi}(x)} = \sum_{j=0}^{2n+1} \tilde{\phi}(x\beta_{n+1}q^2(x))_j \left(\frac{\lambda_j}{x + \lambda_j} \right) \\ &= + \sum_{j=1}^{n+1} \left(\frac{q(-\beta_j)\sqrt{\beta_{n+1}}}{\sqrt{-\tilde{\pi}'(-\beta_j)}} \right)^2 \left(\frac{\beta_j}{x + \beta_j} \right) \\ &\quad - \sum_{j=0}^n \left(\frac{q(-\alpha_j)\sqrt{\beta_{n+1}}}{\sqrt{\tilde{\pi}'(-\alpha_j)}} \right)^2 \left(\frac{\alpha_j}{x + \alpha_j} \right) \\ &= + \sum_{j=1}^{n+1} \tilde{y}_j^2 \left(\frac{\beta_j}{x + \beta_j} \right) - \sum_{j=0}^n \tilde{z}_j^2 \left(\frac{\alpha_j}{x + \alpha_j} \right) \end{aligned}$$

It follows from the first part that if $f \in P_{n+1}$, then

$$+ \sum_{j=1}^{n+1} \tilde{y}_j^2 f(\beta_j) - \sum_{j=0}^n \tilde{z}_j^2 f(\alpha_j) \geq 0$$

For $j = 1, \dots, n$, we have $\tilde{\pi}'(-\alpha_j) = (-\alpha_j + \alpha_0)\pi'(-\alpha_j)(-\alpha_j + \beta_{n+1})$ and $\tilde{\pi}'(-\beta_j) = (-\beta_j + \alpha_0)\pi'(-\beta_j)(-\beta_j + \beta_{n+1})$. Therefore,

$$\lim_{\substack{\alpha_0 \rightarrow 0^+ \\ \beta_{n+1} \rightarrow +\infty}} \tilde{y}_j = y_j, \quad \lim_{\substack{\alpha_0 \rightarrow 0^+ \\ \beta_{n+1} \rightarrow +\infty}} \tilde{z}_j = z_j$$

where the order of the double limit does not matter.

It remains to study what happens to $\tilde{y}_{n+1}^2 f(\beta_{n+1})$ and $\tilde{z}_0^2 f(\alpha_0)$ when $\alpha_0 \rightarrow 0^+$ and $\beta_{n+1} \rightarrow +\infty$. If both limits are zero, we recover $(\star)$ and we are done.

Since any $f \in P_{n+1}$ is a non-negative and non-decreasing function, f is bounded in a neighbourhood of 0 and

$$\begin{aligned} \lim_{\substack{\alpha_0 \rightarrow 0^+ \\ \beta_{n+1} \rightarrow +\infty}} \tilde{z}_0^2 f(\alpha_0) &= \lim_{\substack{\alpha_0 \rightarrow 0^+ \\ \beta_{n+1} \rightarrow +\infty}} \left(\frac{q(-\alpha_0)\sqrt{\beta_{n+1}}}{\sqrt{\tilde{\pi}'(-\alpha_0)}} \right)^2 f(\alpha_0) \\ &= \lim_{\substack{\alpha_0 \rightarrow 0^+ \\ \beta_{n+1} \rightarrow +\infty}} \frac{q^2(-\alpha_0)\beta_{n+1}}{\pi(-\alpha_0)(-\alpha_0 + \beta_{n+1})} f(\alpha_0) = 0 \end{aligned}$$

since $q(0) = 0$, $\pi(0) \neq 0$. Again, the order of the double limit does not matter.

Since $f \in P_{n+1} \subset P_2$, it follows from the first part $\lim_{\beta_{n+1} \rightarrow +\infty} f(\beta_{n+1})/\beta_{n+1}^2 = 0$ and

$$\begin{aligned} \lim_{\substack{\alpha_0 \rightarrow 0^+ \\ \beta_{n+1} \rightarrow +\infty}} \tilde{y}_{n+1}^2 f(\beta_{n+1}) &= \lim_{\substack{\alpha_0 \rightarrow 0^+ \\ \beta_{n+1} \rightarrow +\infty}} \left(\frac{q(-\beta_{n+1})\sqrt{\beta_{n+1}}}{\sqrt{-\tilde{\pi}'(-\beta_{n+1})}} \right)^2 f(\beta_{n+1}) \\ &= \lim_{\substack{\alpha_0 \rightarrow 0^+ \\ \beta_{n+1} \rightarrow +\infty}} \frac{q^2(-\beta_{n+1})\beta_{n+1}}{\pi(-\beta_{n+1})(\beta_{n+1} - \alpha_0)} f(\beta_{n+1}) = 0 \end{aligned}$$

since $\deg(q^2) \leq 2n - 2$ and $\deg(\pi) = 2n$.

□

Remark 8.19

We have proved that $f : (0, +\infty) \rightarrow [0, +\infty)$ is complete Bernstein $\Rightarrow f$ is monotone in the quadratic sense $\Rightarrow f \in \bigcap_{n \in \mathbb{N}} P_n \Rightarrow f \in \bigcap_{n \in \mathbb{N}} M_n$. It remains to prove that if $f \in \bigcap_{n \in \mathbb{N}} M_n$, then f is Complete Bernstein.

Definition 8.8

Let X be a topological space and $M \subset (C_b(X), \|\cdot\|_\infty)$ be a subspace. A linear functional $\Lambda : M \rightarrow \mathbb{R}$ will be called positive if $\Lambda(f) \geq 0$ whenever $f(X) \subset [0, +\infty)$, $f \in M$.

Lemma 8.20

Let X be a topological space and consider the uniform closure $C_0(X)$ of $C_c(X) \subset C_b(X)$. Any positive linear functional $\Lambda : C_0(X) \rightarrow \mathbb{R}$ is bounded.

Proof.

Notice that by positivity, if $f, g \in C_0(X)$ and $f(x) \leq g(x) \forall x \in X$, then $\Lambda(f) - \Lambda(g) = \Lambda(f - g) \geq 0$, i.e. $\Lambda(f) \geq \Lambda(g)$.

Assume for a contradiction that there exists a sequence $(f_n)_{n=1}^\infty \subset C_0(X)$ such that $\|f_n\| \leq 1 \forall n$ and $\Lambda(f_n) \xrightarrow{n \rightarrow +\infty} +\infty$. By passing to a subsequence, we can assume $\Lambda(f_n) \geq 2^n \forall n$. Furthermore, it follows from the reverse triangular inequality that if $f_k \in C_0(X)$, then $|f_k| \in C_0(X)$ as well, so we can assume that $f_k(X) \subset [0, +\infty)$ by

replacing f_k with its absolute value.

Define

$$f(x) = \sum_{k=1}^{\infty} 2^{-k} f_k(x).$$

Then $f(x) \in C_0(X)$. Indeed, let $\epsilon > 0$. Let $N \in \mathbb{N}$ be large enough so that $\sum_{k=N+1}^{\infty} 2^{-k} \leq \epsilon/2$. For $k = 1, \dots, N$, let $g_k \in C_c(X)$ be such that $\|g_k - 2^{-k} f_k\|_{\infty} \leq \epsilon/(2N)$. Finally, define $g(x) = \sum_{k=1}^N g_k(x) \in C_c(X)$. Then

$$\|f - g\|_{\infty} \leq \sum_{k=1}^N \|2^{-k} f_k - g_k\|_{\infty} + \sum_{k=N+1}^{\infty} \|2^{-k} f_k\|_{\infty} \leq \epsilon.$$

Now, observe that $f(x) \geq \sum_{k=1}^n 2^{-k} f_k(x) \forall n \in \mathbb{N}$. By positivity and linearity of Λ :

$$\Lambda(f) \geq \sum_{k=1}^n 2^{-k} \Lambda(f_k) \geq n \forall n.$$

Therefore, $\Lambda(f) = +\infty$, which is a contradiction. $\square$

Corollary 8.21

Let X be a compact Hausdorff space and consider the uniform closure $C_0(X)$ of $C_c(X) \subset C_b(X)$. Let $\Lambda : C_0(X) \rightarrow \mathbb{R}$ be a positive linear functional.

Then there exists a complete Borel measure space $(X, \mathcal{M}, \mu)$, where μ is a finite Radon measure, such that

$$\Lambda(f) = \int_X f d\mu, \quad f \in C_0(X).$$

Proof.

By the Riesz-Markov-Kakutani Representation Theorem, there exists a complete Borel measure space $(X, \mathcal{M}, \mu)$, where μ is a finite Radon measure, such that

$$\Lambda(f) = \int_X f d\mu, \quad f \in C_c(X).$$

Therefore, $f \mapsto \Lambda(f)$ and $f \mapsto \int_X f d\mu$, $f \in C_0(X)$, are two bounded linear functional that coincide on a dense subset (i.e. $C_c(X)$) of $C_0(X)$. Thus, they are equal on the whole $C_0(X)$. $\square$

Theorem 8.22

Let $f : (0, +\infty) \rightarrow [0, +\infty)$, $f \in \bigcap_{n \in \mathbb{N}} M_n$. Then f is a Complete Bernstein function.

Proof.

Let $X = [0, +\infty]$ be the one-point compactification of the half-line $[0, +\infty)$.

Define the set

$$G = \{g(x), x \in [0, +\infty] \mid g(x) = \sum_{j=1}^n a_j \frac{\lambda_j x - 1}{x + \lambda_j} = \sum_{j=1}^n a_j \phi_{\lambda_j}(x), n \in \mathbb{N}, a_j \in \mathbb{R}, \lambda_j > 0, \sum_{j=1}^n a_j = 0\}$$

which is a subspace of $(C_0([0, +\infty]), \|\cdot\|_{\infty})$, the set of continuous functions from $[0, +\infty]$ to $\mathbb{R}$ vanishing at infinity endowed with the sup norm. Let $f \in \bigcap_{n \in \mathbb{N}} M_n$. Define the following linear functional on G :

$$\Lambda_f : G \rightarrow \mathbb{R}$$

$$\sum_{j=1}^n a_j \phi_{\lambda_j} \mapsto \sum_{j=1}^n a_j f(\lambda_j)$$

Since $f \in \bigcap_{n \in \mathbb{N}} M_n$, $\Lambda_f(g) \geq 0$ whenever $g(x) \geq 0 \forall x > 0$ (see Proposition 8.13). Let

$$g_0(x) = 5(\phi_2(x) - \phi_1(x)) = 5 \left(\frac{2x-1}{x+2} - \frac{x-1}{x+1} \right) = \frac{5(x^2+1)}{x^2+3x+2} \in G$$

Then $g_0(x) \geq 1 \forall x > 0$ since

$$\frac{x^2+1}{x^2+3x+2} \geq \frac{1}{5} \forall x > 0 \iff \frac{4}{5}x^2 - \frac{3}{5}x + \frac{3}{5} \geq 0 \forall x > 0,$$

which is true since $\frac{4}{5}x^2 - \frac{3}{5}x + \frac{3}{5}$ has no real roots and is strictly positive when $|x|$ is large.

Therefore, for any $g \in G$, we have $g(x) \leq \max\{\sup g, 0\} \cdot g_0(x) \forall x > 0$. By positivity of Λ_f ,

$$\Lambda_f(g) \leq \max\{\sup g, 0\} \cdot \Lambda_f(g_0) \forall g \in G.$$

Observe that $g \mapsto \max\{\sup g, 0\} \cdot \Lambda_f(g_0)$ is a sublinear functional. Therefore by the Hahn-Banach Extension Theorem, the linear functional Λ_f can be extended to a linear functional $\tilde{\Lambda}_f : C_0([0, +\infty]) \rightarrow \mathbb{R}$ in such a way that $\tilde{\Lambda}_f(g) \leq \max\{\sup g, 0\} \cdot \Lambda_f(g_0) \forall g \in C_0([0, +\infty])$.

For any positive $h \in C_0([0, +\infty])$, we have

$$\tilde{\Lambda}_f(-h) \leq \max\{\sup -h, 0\} \cdot \Lambda_f(g_0) = 0$$

i.e. $\tilde{\Lambda}_f$ is a positive linear functional on $C_0([0, +\infty])$. In particular, it is bounded. By the Riesz-Markov Representation Theorem, there exists a finite Radon measure μ such that

$$\tilde{\Lambda}_f(h) = \int_{[0, +\infty]} h(t) d\mu(t) \forall h \in C_0([0, +\infty]).$$

For $g(t) = \phi_\lambda(t) - \phi_1(t)$, $\lambda > 0$, we get $\tilde{\Lambda}_f(g) = \Lambda_f(g) = f(\lambda) - f(1)$, i.e.

$$\begin{aligned} f(\lambda) &= \tilde{\Lambda}_f(g) + f(1) = \tilde{\Lambda}_f(\phi_\lambda) - \tilde{\Lambda}_f(\phi_1) + f(1) = \int_{[0, +\infty]} \frac{\lambda t - 1}{t + \lambda} d\mu(t) - \Lambda(\phi_1) + f(1) \\ &= (-\Lambda(\phi_1) + f(1)) + \lambda \mu(\{\infty\}) + \int_{[0, +\infty]} \frac{\lambda t - 1}{t + \lambda} d\mu(t) \end{aligned}$$

Observe that for $t \geq 0$, $\lambda \mapsto \frac{\lambda t - 1}{t + \lambda}$ is an increasing function (the derivative is given by $(1 + t^2)/(\lambda + t)^2$). By the Monotone Convergence Theorem :

$$f(0+) = \lim_{\lambda \rightarrow 0^+} f(\lambda) = (-\Lambda(\phi_1) + f(1)) - \int_{[0, +\infty]} \frac{1}{t} d\mu(t)$$

In particular, $\int_{[0, +\infty]} t^{-1} d\mu(t)$ is finite. Yet, we have

$$+\infty > \int_{[0, +\infty]} \frac{1}{t} d\mu(t) > \int_{[0, x)} \frac{1}{t} d\mu(t) > \mu([0, x)) \cdot \frac{1}{x} > 0 \forall x > 0$$

Taking the limit $x \rightarrow 0^+$, by continuity of the measure, we get that $\mu(\{0\}) = 0$, so

$$f(\lambda) = (-\Lambda(\phi_1) + f(1)) + \lambda\mu(\{\infty\}) + \int_{(0,+\infty)} \frac{\lambda t - 1}{t + \lambda} d\mu(t)$$

Finally, notice that

$$\begin{aligned} \int_{(0,+\infty)} \frac{\lambda t - 1}{t + \lambda} d\mu(t) &= \int_{(0,+\infty)} \frac{\lambda}{t + \lambda} t d\mu(t) - \left(\int_{(0,+\infty)} \frac{1}{t + \lambda} - \frac{1}{t} d\mu(t) \right) - \int_{(0,+\infty)} \frac{1}{t} d\mu(t) \\ &= \int_{(0,+\infty)} \frac{\lambda}{t + \lambda} \left(t + \frac{1}{t} \right) d\mu(t) - \int_{(0,+\infty)} \frac{1}{t} d\mu(t) \\ &= \int_{(0,+\infty)} \frac{\lambda}{t + \lambda} d\sigma(t) - \int_{(0,+\infty)} \frac{1}{t} d\mu(t) \end{aligned}$$

where $d\sigma(t) = \left(t + \frac{1}{t} \right) d\mu(t)$ is defined as in Definition 3.8.

Therefore,

$$f(\lambda) = f(0+) + \lambda\mu(\{\infty\}) + \int_{(0,+\infty)} \frac{\lambda}{t + \lambda} d\sigma(t).$$

□

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